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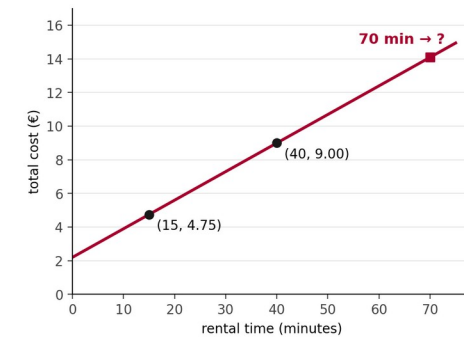
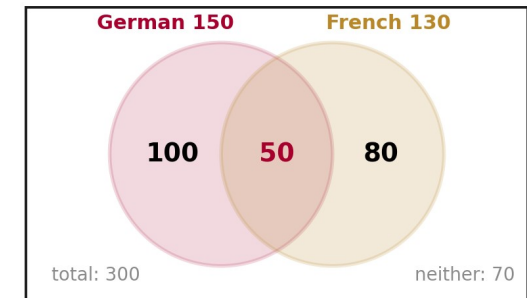
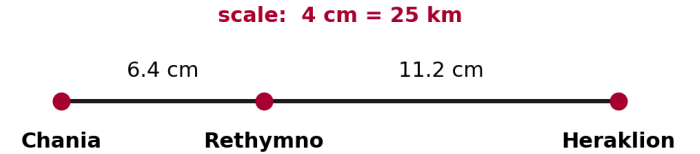
DIGITAL SAT — MATH

Problem-Solving & Data Analysis

Theory Companion — Explained Rules, Worked Problems & Cheat Sheet

Ratios • Probability • Units & Rates • Statistics • Data Displays • Scatterplots • Sampling

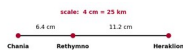
Prepared by the CLC Digital SAT Team



7 Skills in This Deck

Each skill opens with key terms in English and Greek, explained theory with examples, and a Watch Out slide — then 8 Hard / Very Hard worked problems. A cheat sheet closes the deck.

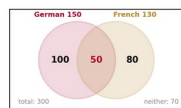
01



Ratios & Proportions

Part-to-part ratios, proportions, and direct and inverse variation.

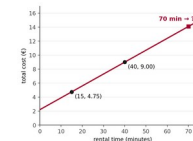
02



Probability

Two-way tables, conditional probability, and compound events.

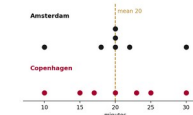
03



Unit Conversions & Rates

Dimensional analysis, rates, and converting area and volume units.

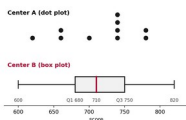
04



Statistics — Center & Spread

Mean, median, weighted averages, outliers, and standard deviation.

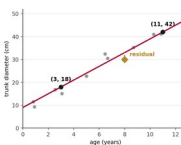
05



Data Displays — Histograms, Dot Plots & Box Plots

Reading frequency displays and five-number summaries.

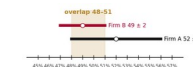
06



Scatterplots

Lines of best fit, residuals, model types, and extrapolation.

07



Sampling, Bias & Margin of Error

Random samples, bias, generalizing, and confidence intervals.

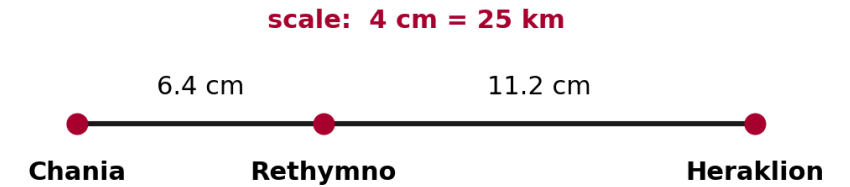
01

Ratios & Proportions

Part-to-part ratios, proportions, and direct and inverse variation.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

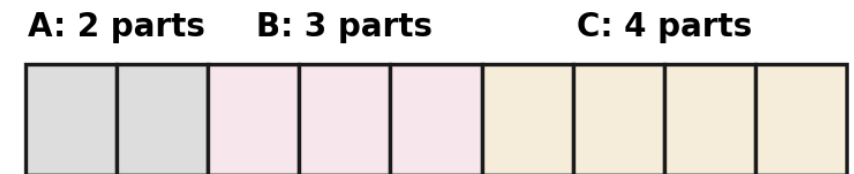


Key Terminology

Term (English)	Ελληνικά
Ratio	Λόγος
Proportion	Αναλογία
Cross-multiplication	Εξίσωση γινομένων (χιαστί πολλαπλασιασμός)
Scale factor	Λόγος κλίμακας
Unit rate	Μοναδιαία τιμή

A ㄣ The big picture

- A ratio compares parts; a proportion sets two ratios equal.
- Most questions reduce to one unknown multiplier k .
- Hard versions chain ratios or change one quantity.



9 equal parts $\rightarrow$ one part = total $\div$ 9

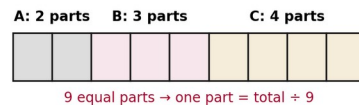
Theory Explained (1/3)

Ratio Parts

$$a : b : c \rightarrow a \cdot k, b \cdot k, c \cdot k$$

WHAT IT MEANS

- Total = $(a + b + c) \cdot k$ — solve for k first.
- Every group is its ratio number times k .
- Then answer the exact comparison asked.



Example

2:3:4 of 108 stamps $\rightarrow 9k = 108 \rightarrow k = 12$.

Proportions

$$a/b = c/d \rightarrow ad = bc$$

WHAT IT MEANS

- Keep the same units in the same positions.
- Cross-multiply to solve.
- Scale drawings and maps are proportions.



Example

4 cm = 25 km $\rightarrow$ 17.6 cm = 110 km.

Theory Explained (2/3)

Chaining Ratios

match the shared term using the LCM

WHAT IT MEANS

- $F:S = 5:3$ and $S:B = 2:5$ share S .
- Scale both so S matches (LCM of 3 and 2 = 6).
- Result: $F:S:B = 10:6:15$.



Example

Butter 300 g = 15 parts $\rightarrow$ 1 part = 20 g $\rightarrow$ flour 200 g.

Direct vs Inverse Variation

direct: $y = kx$ inverse: $xy = k$

WHAT IT MEANS

- Direct: both grow together (paint vs wall area).
- Inverse: one grows, the other shrinks (workers vs days).
- Find k from the known pair first.



Example

8 workers $\times$ 15 days = 120 $\rightarrow$ $120/6 = 20$ workers.

Theory Explained (3/3)

Changing Ratios

write each quantity as a multiple of k

WHAT IT MEANS

- Only the changed quantity gets the extra amount.
- Set up the new ratio as an equation in k.
- The unchanged quantity is your anchor.



Example

$$3k + 12 = 7k \text{ (new 1:1)} \rightarrow k = 3.$$

Watch Out



Ratio $\neq$ actual count

- Always multiply back by k .
- $4 - 3 = 1$ part, not 1 stamp.



Identify what is fixed

- In mixing problems, the ingredient NOT added stays constant.
- Scale from that fixed quantity.

Problem 1

Hard

A philatelist's collection contains Greek, Italian, and French stamps in the ratio 2:3:4. Each Greek stamp in the collection is individually insured for €0.50, and the collection contains 108 stamps in total. How many more French stamps are there than Italian stamps?

Formula / Rule

Three-part ratio 'x method': let the parts be $2x$, $3x$, $4x$ and use the total to solve for x .

First move

Turn a three-part ratio into $2x$, $3x$, $4x$, set the sum equal to the given total, and solve for x before answering the actual question asked.

Solution

1. Let the numbers of Greek, Italian, and French stamps be $2x$, $3x$, and $4x$, matching the ratio 2:3:4.
2. The insurance value of €0.50 per Greek stamp describes cost, not a count of stamps, so it plays no role in this counting problem — set it aside.
3. Add the parts and set equal to the total: $2x + 3x + 4x = 108$, so $9x = 108$.
4. Solve for x : $x = 108/9 = 12$.
5. Find each count: Greek = $2x = 24$, Italian = $3x = 36$, French = $4x = 48$.
6. Subtract to find the difference: French – Italian = $48 - 36 = 12$.

 **Answer:** 12 more French stamps than Italian stamps.



Problem 1 — Strategy & Elimination



Tip

Turn a three-part ratio into $2x$, $3x$, $4x$, set the sum equal to the given total, and solve for x before answering the actual question asked.



Trap

A rushed student, seeing €0.50 as the only 'extra' number in the problem, might try to work it into the ratio calculation (e.g., dividing 108 by 0.50) instead of recognizing it is irrelevant to the stamp count, producing a nonsensical result.



Shortcut

Since only the difference between two of the three groups is needed, you can skip listing all three counts and compute $(4 - 3)x = x = 12$ directly once x is known.



Desmos Tip

Type 108/9 directly into the calculator to instantly confirm $x = 12$ before multiplying out each stamp count.



Process of Elimination

Correct: **12 more French stamps than Italian stamps.**

Tempting wrong answer	Exactly what went wrong
24	Computed French – Greek ($48 - 24 = 24$) instead of French – Italian, answering with the wrong pair of groups.
1	Found the difference in ratio parts ($4 - 3 = 1$) but never multiplied back by $x = 12$, leaving the answer in ratio units instead of actual stamp counts.
48	Reported the number of French stamps (4×12) instead of the difference.

Problem 2

Very Hard

At a bakery in Lyon, the ratio of flour to sugar used in a recipe is 5:3, and the ratio of sugar to butter used is 2:5. If the bakery uses 300 grams of butter for one batch, how many grams of flour does it use?

Formula / Rule

Combine two ratios sharing a term by scaling each to a common value for that shared term (using the LCM).

First move

When two ratios share a term, scale each ratio (using the LCM of the shared term's two values) so that term matches in both, then merge them into one three-part ratio before using any given total.

Solution

1. Write the two given ratios: flour:sugar = 5:3 and sugar:butter = 2:5.
2. The shared term is sugar, which appears as 3 in the first ratio and 2 in the second — find the LCM of 3 and 2, which is 6.
3. Scale flour:sugar = 5:3 by 2 (so sugar becomes 6): flour:sugar = 10:6.
4. Scale sugar:butter = 2:5 by 3 (so sugar becomes 6): sugar:butter = 6:15.
5. Since both ratios now share the same sugar value, combine them into one three-part ratio: flour:sugar:butter = 10:6:15.
6. The butter amount, 300 g, corresponds to the 15 parts, so one part = $300/15 = 20$ g.
7. Find the flour amount: flour = $10 \times 20 = 200$ grams.

 **Answer:** 200 grams of flour.



Problem 2 — Strategy & Elimination



Tip

When two ratios share a term, scale each ratio (using the LCM of the shared term's two values) so that term matches in both, then merge them into one three-part ratio before using any given total.



Trap

Because sugar is '3' in the first ratio and '2' in the second, a rushed student might just concatenate the numbers as flour:sugar:butter = 5:3:5 (reusing the sugar:butter figures unscaled) instead of finding the LCM of 3 and 2 and scaling both ratios to a common sugar value of 6, producing an incorrect combined ratio and a wrong flour amount.



Shortcut

Since only the flour amount is needed, you can isolate the flour:butter portion of the combined ratio (10:15) and solve $300 \times 10/15 = 200$ directly, skipping the sugar amount entirely.



Desmos Tip

Type $300 \cdot (10/15)$ into the calculator to instantly confirm the flour amount is 200 grams.



Process of Elimination

Correct: 200 grams of flour.

Tempting wrong answer	Exactly what went wrong
300 grams	Concatenated the two ratios directly as flour:sugar:butter = 5:3:5 (reusing butter's raw ratio number without scaling to match sugar's LCM value of 6), overstating the flour amount.
500 grams	Applied the flour:sugar ratio (5:3) directly to the butter amount as if sugar and butter were interchangeable, computing flour = $300 \times 5/3$ instead of combining through the properly scaled three-part ratio.
120 grams	Correctly found one part = $300/15 = 20$ g but then multiplied by sugar's part count (6) instead of flour's (10), reporting the sugar amount instead of the flour amount.

Problem 3

Very Hard

A construction crew of 8 workers can build a section of highway retaining wall near Thessaloniki in 15 days, with the number of days needed inversely proportional to the number of workers (assuming every worker works at the same steady rate). The company now wants to complete an identical wall section in only 6 days. How many additional workers, beyond the original 8, must be added to the crew to meet this deadline?

Formula / Rule

Inverse proportion: $\text{workers} \times \text{days} = k$ (a constant); find k first, then use it to solve the new scenario.

First move

For inverse proportion word problems, first find the constant of variation $k = (\text{quantity 1}) \times (\text{quantity 2})$ from the given pair, then reuse k to solve for the missing quantity in the new scenario.

Solution

1. Since days needed is inversely proportional to the number of workers, $\text{workers} \times \text{days} = k$ for some constant k .
2. Use the original data to find k : $k = 8 \times 15 = 120$.
3. For the new deadline, set up the equation with the new number of workers, w , and 6 days: $w \times 6 = 120$.
4. Solve for w : $w = 120/6 = 20$.
5. This means 20 workers total are needed to finish in 6 days.
6. The question asks for additional workers beyond the original 8, so subtract: $20 - 8 = 12$.



Answer: 12 additional workers (20 workers total).



Problem 3 — Strategy & Elimination



Tip

For inverse proportion word problems, first find the constant of variation $k = (\text{quantity 1}) \times (\text{quantity 2})$ from the given pair, then reuse k to solve for the missing quantity in the new scenario.



Shortcut

There is no shortcut past finding k and reapplying it — and since the question asks for additional workers, not the total, the final subtraction step cannot be skipped.



Trap

A student might set up a direct proportion instead ($8/15 = w/6$), treating workers and days as increasing together, and/or might stop at $w = 20$ without subtracting the original 8 workers, answering a question that actually asks how many MORE workers are needed.



Desmos Tip

Graph $y = 120/x$ on the calculator and read the y -value at $x = 6$ to instantly confirm 20 total workers are needed.



Process of Elimination

Correct: **12 additional workers (20 workers total).**

Tempting wrong answer	Exactly what went wrong
20 additional workers	Correctly solved for the new total number of workers ($w = 120/6 = 20$) but forgot the question asks for additional workers beyond the original 8, submitting the total instead of the difference.
$w = 3.2$	Set up a direct proportion ($8/15 = w/6$) instead of using the inverse-proportion relationship $\text{workers} \times \text{days} = k$, incorrectly assuming fewer days would require fewer workers rather than more.
5 additional workers	Correctly found $w = 20$ total workers needed but subtracted 15 (the original number of days) instead of 8 (the original number of workers), mixing up which given number represents workers.

Problem 4

Hard

Sail X and Sail Y are similar triangular sails, and their areas are in the ratio 4:9 (Sail X to Sail Y). If the perimeter of Sail X is 20 meters, what is the perimeter of Sail Y?

Formula / Rule

For similar figures, the ratio of corresponding lengths squared equals the ratio of areas: $(\text{length ratio})^2 = \text{area ratio}$.



First move

When similar figures give you an area ratio but ask about a length (a side or perimeter), take the square root of the area ratio first to get the correct length ratio — never use the area ratio as the length ratio directly.

Solution

1. Sail X and Sail Y are similar, so the ratio of their perimeters (a length ratio) equals the ratio r of their corresponding side lengths.
2. For similar figures, area ratio $= r^2$, so $r^2 = 4/9$.
3. Solve for r by taking the square root of both sides: $r = \sqrt{4/9} = 2/3$.
4. This means perimeter of X : perimeter of Y = 2:3.
5. Set up the proportion: $20/(\text{perimeter of Y}) = 2/3$.
6. Cross-multiply: $20 \times 3 = 2 \times (\text{perimeter of Y})$, so $60 = 2 \times (\text{perimeter of Y})$.
7. Solve: perimeter of Y = $60/2 = 30$ meters.



Answer: Perimeter of Sail Y = 30 meters.



Problem 4 — Strategy & Elimination



Tip

When similar figures give you an area ratio but ask about a length (a side or perimeter), take the square root of the area ratio first to get the correct length ratio — never use the area ratio as the length ratio directly.



Trap

A student who forgets that area scales with the square of the length ratio will use 4:9 directly as the perimeter ratio, computing perimeter of Y = $20 \times 9/4 = 45$ meters — exactly the wrong answer this problem is built to catch.



Shortcut

Once you know the length ratio is 2:3, you can skip formal cross-multiplication and just scale 20 by $3/2$ directly to get 30.



Process of Elimination

Correct: **Perimeter of Sail Y = 30 meters.**

Tempting wrong answer	Exactly what went wrong
45 meters	Used the area ratio 4:9 directly as the perimeter ratio without taking the square root, computing $20 \times 9/4 = 45$ instead of first finding the length ratio 2:3.
40/3 meters (≈ 13.33)	Found the correct length ratio 2:3 but inverted it when setting up the proportion (used $3/2$ instead of $2/3$), computing $20 \times 2/3$ instead of $20 \times 3/2$.
90 meters	Took the square root of only the numerator of the area ratio ($\sqrt{4} = 2$, leaving $2/9$) instead of the square root of the entire fraction ($\sqrt{4/9} = 2/3$), producing an inflated length ratio.

Problem 5

Very Hard

A jar contains only red and blue marbles in the ratio red:blue = 3:7. After 12 more red marbles are added to the jar (the number of blue marbles stays the same), the new ratio of red to blue becomes 1:1. What is the total number of marbles in the jar now?

Formula / Rule

Reverse ratio setup: let the original quantities be $3k$ and $7k$, write an equation from the described change, and solve for k .

First move

When a final ratio is given after a stated change, assign the original quantities as multiples of a variable ($3k$, $7k$), write an equation reflecting exactly what changed, and solve for the variable.

Solution

1. Let the original numbers of red and blue marbles be $3k$ and $7k$, matching the given ratio 3:7.
2. After adding 12 red marbles, red becomes $3k + 12$, while blue stays at $7k$.
3. The new ratio is 1:1, meaning the new red count equals the blue count: $3k + 12 = 7k$.
4. Solve for k : $12 = 7k - 3k = 4k$, so $k = 3$.
5. Find the blue count (unchanged): $\text{blue} = 7k = 7(3) = 21$.
6. Find the new red count: $\text{red}_{\text{new}} = 3k + 12 = 9 + 12 = 21$.
7. Add both counts to get the current total: $21 + 21 = 42$.

 **Answer:** 42 marbles total (21 red and 21 blue).



Problem 5 — Strategy & Elimination



Tip

When a final ratio is given after a stated change, assign the original quantities as multiples of a variable ($3k$, $7k$), write an equation reflecting exactly what changed, and solve for the variable.



Shortcut

Since the new ratio is exactly 1:1, you can skip a full cross-multiplied proportion and just recognize red_new must equal blue ($7k$), giving the simpler equation $3k + 12 = 7k$ directly.



Trap

A student might assume the 1:1 ratio means equal numbers were added to both colors, or might set $3k:7k = 1:1$ directly and ignore that only the red count changed by +12, leading to a wrong value of k or a wrong total.



Desmos Tip

Graph $y = 3x + 12$ and $y = 7x$ and find their intersection point to instantly confirm $k = 3$.



Process of Elimination

Correct: **42 marbles total (21 red and 21 blue).**

Tempting wrong answer	Exactly what went wrong
21	Correctly found $k = 3$ and computed blue = 21 (which also equals the new red count) but forgot to add red and blue together for the total, reporting only one color's count.
52	Solved $4k = 12$ but divided incorrectly, getting $k = 4$ instead of $k = 3$, then added the resulting mismatched red and blue counts, which no longer actually satisfy the 1:1 condition.
30	Used the original total before the change ($3k + 7k = 10k = 30$) instead of the total after the 12 red marbles were added, answering the wrong point in time.

Problem 6

Hard

A map of Crete uses a scale where 4 centimeters represents 25 kilometers of actual distance. On this map, the distance from Chania to Rethymno measures 6.4 centimeters, and the distance from Rethymno to Heraklion measures 11.2 centimeters. Assuming the three cities lie along a straight coastal road in that order, what is the actual distance, in kilometers, from Chania to Heraklion?

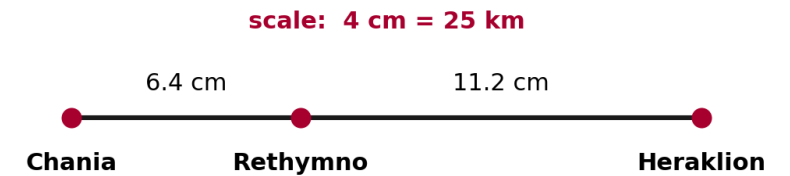
Formula / Rule

Translate a scale into a unit rate (actual distance per cm), then apply that rate as a proportion to each map measurement.

Solution

1. Translate the scale into a unit rate: 4 cm represents 25 km, so 1 cm represents $25/4 = 6.25$ km of actual distance.
2. Convert the Chania–Rethymno map distance: $6.4 \text{ cm} \times 6.25 \text{ km/cm} = 40 \text{ km}$.
3. Convert the Rethymno–Heraklion map distance: $11.2 \text{ cm} \times 6.25 \text{ km/cm} = 70 \text{ km}$.
4. Since the three cities lie on a straight road in the order Chania, Rethymno, Heraklion, the actual Chania–Heraklion distance equals the sum of the two segments.
5. Add the two actual distances: $40 \text{ km} + 70 \text{ km} = 110 \text{ km}$.

 **Answer:** 110 kilometers.





Problem 6 — Strategy & Elimination



Tip

Convert a map or drawing scale into a single unit rate (actual distance per centimeter) first, then multiply every given map measurement by that same rate.



Shortcut

Since both segments use the same scale, you can add the map distances first ($6.4 + 11.2 = 17.6$ cm) and multiply once by 6.25 km/cm to get 110 km in a single multiplication instead of two.



Trap

A student might invert the scale (using 4/25 cm per km instead of 25/4 km per cm) or might convert only one segment and forget that the straight-road condition requires adding both actual distances to get the full Chania-to-Heraklion distance.



Desmos Tip

Type $17.6 \cdot 6.25$ into the calculator to instantly verify the total actual distance is 110 km.



Process of Elimination

Correct: **110 kilometers.**

Tempting wrong answer	Exactly what went wrong
≈2.82 km	Inverted the scale, using 4/25 km per cm instead of 25/4 km per cm (6.25 km/cm), producing a drastically undersized distance.
40 km	Converted only the Chania–Rethymno segment ($6.4 \text{ cm} \times 6.25 = 40 \text{ km}$) and forgot to add the Rethymno–Heraklion segment, treating the first conversion as the final answer.
30 km	Subtracted the two map distances ($11.2 - 6.4 = 4.8 \text{ cm}$) instead of adding them, misunderstanding that both segments lie along the same road and must be summed for the total distance.

Problem 7

Hard

The amount of paint needed to cover a wall varies directly with the wall's area. A painter in Porto used 3 liters of paint to cover a 15 square-meter wall. For a new job, the painter must cover three walls with areas 20, 35, and 25 square meters. The client wants two coats of paint on only the largest of the three walls, and one coat on each of the other two. How many liters of paint does the painter need in total?

Formula / Rule

Direct variation: $\text{liters} = k \times \text{area}$; find k from the given pair, then apply it to each new area, adjusting for any wall that needs an extra coat.

First move

For direct variation, find $k = \text{value}/\text{other value}$ from the given pair first, then apply that same k to every new quantity — treat any extra requirement (like a second coat) as a separate addition after the base amounts are found.

Solution

1. Direct variation means $\text{liters} = k \times \text{area}$; use the given pair to find k : $3 = k \times 15$, so $k = 3/15 = 0.2$ liters per square meter.
2. Find the one-coat paint amount for each wall: $20 \times 0.2 = 4$ L, $35 \times 0.2 = 7$ L, $25 \times 0.2 = 5$ L.
3. Add these one-coat amounts: $4 + 7 + 5 = 16$ liters.
4. Identify the largest wall (35 m^2) — this is the one that needs a second coat, requiring an extra 7 liters.
5. Add the extra coat to the one-coat total: $16 + 7 = 23$ liters.

 **Answer:** 23 liters of paint.



Problem 7 — Strategy & Elimination



Tip

For direct variation, find k = value/other value from the given pair first, then apply that same k to every new quantity — treat any extra requirement (like a second coat) as a separate addition after the base amounts are found.



Shortcut

Since only one wall needs a second coat, you can skip recomputing all three walls under a doubled scenario and just add one extra single-coat amount (7 L) for the 35 m² wall to the one-coat total.



Trap

A student who computes only the one-coat amounts for each wall and reports 16 liters forgets that the problem explicitly requires a second coat on the largest wall, an amount that must be added separately.



Desmos Tip

Type $0.2 \cdot (20 + 35 + 25) + 0.2 \cdot 35$ directly into the calculator to verify the total in one line.



Process of Elimination

Correct: **23 liters of paint.**

Tempting wrong answer	Exactly what went wrong
16 liters	Correctly found the one-coat total ($4 + 7 + 5 = 16$) but forgot the largest wall needs a second coat, never adding the extra 7 liters.
21 liters	Applied the second coat to the 25 m ² wall instead of the actual largest wall (35 m ²), misreading which wall the client wants double-coated.
575 liters	Inverted the direct-variation constant, using $k = 15/3 = 5$ L per square meter instead of $k = 3/15 = 0.2$ L per square meter, inflating every computed amount.

Problem 8

Very Hard

A dough recipe requires flour and water in the ratio 8:3. While preparing a large batch, a baker accidentally reverses the ratio, mixing flour and water in the ratio 3:8 instead, using 9 kilograms of flour for the batch. Without discarding any of the mixture, the baker plans to add only more flour until the ratio of flour to water matches the correct 8:3 ratio. How many kilograms of additional flour must the baker add?

Formula / Rule

Use the mistaken ratio to find the quantity that stays fixed (water), then use the correct ratio with that fixed quantity to solve for the target amount, subtracting what's already there.



First move

When fixing a ratio by adding only one ingredient, first identify which quantity stays constant (here, water) and use it as the fixed anchor to solve for the target amount of the other ingredient.

Solution

1. Under the mistaken ratio, flour:water = 3:8, and the flour used is 9 kg, so 9 kg corresponds to 3 parts: one part = $9/3 = 3$ kg.
2. Find the water used (8 parts): water = $8 \times 3 = 24$ kg. This amount stays fixed, since only flour will be added.
3. Let F be the total flour needed so the ratio becomes the correct 8:3: $F/24 = 8/3$.
4. Cross-multiply: $3F = 8 \times 24 = 192$.
5. Solve: $F = 192/3 = 64$ kg of flour needed in total.
6. Subtract the flour already used: additional flour = $64 - 9 = 55$ kg.



Answer: 55 kilograms of additional flour.



Problem 8 — Strategy & Elimination



Tip

When fixing a ratio by adding only one ingredient, first identify which quantity stays constant (here, water) and use it as the fixed anchor to solve for the target amount of the other ingredient.



Shortcut

There's no shortcut past anchoring on the fixed quantity (water) first — but once you have it, you can combine the last three steps into one line: $24 \times (8/3) - 9$.



Trap

A student who assumes the given 9 kg of flour was already correctly proportioned and simply rescales it by $8/3$ ($9 \times 8/3 = 24$) never recognizes that water — not flour — is the fixed quantity that must anchor the new proportion, and reports 24 as if it were the answer.



Desmos Tip

Type $24 \cdot 8/3 - 9$ into the calculator to confirm the additional flour needed is 55 kg in one line.



Process of Elimination

Correct: **55 kilograms of additional flour.**

Tempting wrong answer	Exactly what went wrong
64 kg	Correctly solved for the total flour needed (64 kg) but forgot to subtract the 9 kg of flour already used, reporting the total instead of the additional amount.
24 kg	Assumed the given 9 kg of flour was already in the correct 8:3 ratio and scaled it directly by $8/3$ ($9 \times 8/3 = 24$), never recognizing that water is the fixed quantity needed to solve for the new flour amount.
15 kg	Assumed the total mixture weight ($9 + 24 = 33$ kg) stays fixed and re-split it into the new 8:3 ratio (getting 24 kg flour), instead of recognizing that only flour is added, so water must remain fixed at 24 kg.

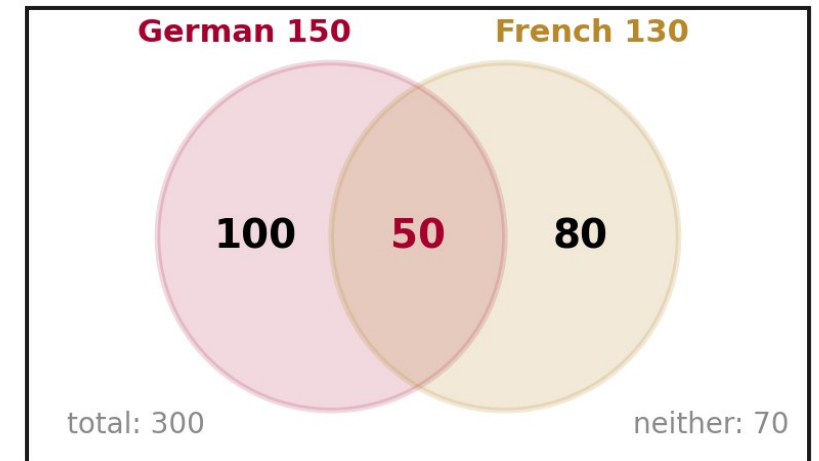
02

Probability

Two-way tables, conditional probability, and compound events.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



Key Terminology

Term (English)	Ελληνικά
Probability	Πιθανότητα
Independent events	Ανεξάρτητα ενδεχόμενα
Dependent events	Εξαρτημένα ενδεχόμενα
Mutually exclusive events	Ασυμβίβαστα ενδεχόμενα
Sample space	Δειγματικός χώρος

A ㄱ The big picture

- Probability = favorable outcomes $\div$ total outcomes.
- The key skill is choosing the right denominator.
- Two-way tables and 'without replacement' appear constantly.

	Yes	No	Total
Group 1	a	b	a+b
Group 2	c	d	c+d
Total	a+c	b+d	N

Theory Explained (1/3)

Basic & Complement

$$P(A) = \text{favorable} / \text{total} \quad P(\text{not } A) = 1 - P(A)$$

WHAT IT MEANS

- Always between 0 and 1.
- 'At least one' → use $1 - P(\text{none})$.
- The complement is often the shortest path.



Example

$$P(\text{at least 1 of 3 fails}) = 1 - (0.9)^3 = 0.271.$$

Conditional Probability

$$P(A | B) = P(A \text{ and } B) / P(B)$$

WHAT IT MEANS

- 'Given' or 'of those who...' shrinks the denominator.
- The denominator is the condition's row or column total.
- Complete the two-way table before dividing.

	Yes	No	Total
Group 1	a	b	a+b
Group 2	c	d	c+d
Total	a+c	b+d	N



Example

$$P(\text{South} | \text{owns bike}) = 60/130 = 6/13.$$

Theory Explained (2/3)

Independent Events

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

WHAT IT MEANS

- Independent: one outcome doesn't change the other.
- With replacement → independent.
- 'Exactly one' needs every order: win-lose AND lose-win.



Example

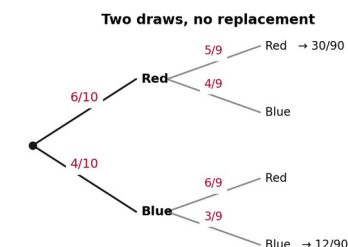
$$P(\text{exactly 1 of 2 wins}) = 2 \cdot (3/20)(17/20).$$

Without Replacement

second draw: numerator and denominator both drop

WHAT IT MEANS

- The first draw changes what remains.
- Multiply along a branch; add different branches.
- 'Same color' = both red OR both blue.



Example

$$6R, 4B: P(\text{same}) = (6/10)(5/9) + (4/10)(3/9) = 7/15.$$

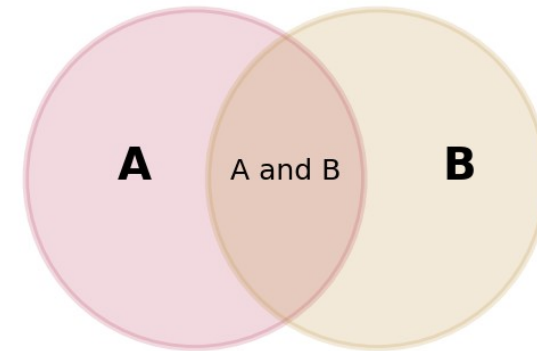
Theory Explained (3/3)

Addition Rule & Odds

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

WHAT IT MEANS

- Subtract the overlap once — it was counted twice.
- Odds a to b in favor $\rightarrow P = a/(a + b)$.
- Odds are NOT a probability.



$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$



Example

Odds 3 to 17 $\rightarrow P(\text{win}) = 3/20$.

Watch Out



Pick the correct denominator

- 'Of all students' vs 'of bike owners' changes everything.
- Underline the condition in the question.



Order matters for 'exactly'

- Count every arrangement that fits.
- 'First defective on the 3rd test' allows only one order.

Problem 1

Hard

CLC Educational Institute surveyed 200 students across its Athens (North) and Piraeus (South) branches about whether they own a bicycle. Of the 120 students at the North branch, 70 own a bicycle. Of the 80 students at the South branch, 20 do NOT own a bicycle. If a student who owns a bicycle is selected at random from the entire survey, what is the probability that this student is from the South branch?

Formula / Rule

Conditional probability from a two-way table: $P(\text{South} \mid \text{owns bike}) = (\text{South students who own bikes}) / (\text{total students who own bikes})$.

Solution

- Find the number of North branch students who do NOT own a bicycle: $120 - 70 = 50$.
- Find the number of South branch students who own a bicycle: $80 - 20 = 60$ (since 20 of the 80 South students do not own one).
- Find the total number of students, from both branches, who own a bicycle: 70 (North) + 60 (South) = 130 .
- Conditional probability restricts the denominator to the 'given' group — here the given condition is 'owns a bicycle,' so the denominator is 130, not the grand total of 200.
- Divide the South owners by the total owners: $P(\text{South} \mid \text{owns bike}) = 60/130$.
- Simplify the fraction: $60/130 = 6/13$.

✓ **Answer: 6/13**

Two-way table (completed)

	Owns bike	No bike	Total
North	70	50	120
South	60	20	80
Total	130	70	200



Problem 1 — Strategy & Elimination



Tip

In a two-way table conditional probability question, first fill in any missing cell using its row or column total, then restrict your denominator to exactly the 'given' condition — never the grand total.



Trap

A rushed student divides by the grand total of 200 students instead of the 130 students who own bikes, getting $60/200 = 3/10$, which actually answers a different question ('South AND owns a bike') rather than the conditional probability that was asked.



Shortcut

There's no way to skip filling in the missing cell — that step is required — but once the table is complete you only need to track two numbers: the South-owner count and the total-owner count, not the whole table.



Process of Elimination

Correct: **6/13**

Tempting wrong answer	Exactly what went wrong
3/10	Divided the South-owner count by the grand total of 200 students instead of the 130 total bike owners, answering 'P(South and owns a bike)' instead of the requested conditional probability.
7/13	Put the North branch's bike-owner count (70) in the numerator instead of the South branch's (60), effectively finding $P(\text{North} \mid \text{owns bike})$ instead of $P(\text{South} \mid \text{owns bike})$.
3/4	Divided the South bike owners by the South branch total (80) instead of the total bike owners across both branches (130), finding $P(\text{owns bike} \mid \text{South})$ instead of $P(\text{South} \mid \text{owns bike})$ — the condition was reversed.

Problem 2

Very Hard

At the Port of Piraeus, random safety inspections are performed on shipping containers. Historical records show that, independent of any other container, 1 out of every 10 containers fails its safety check. The port handles about 4,000 containers per month. If an inspector randomly selects 3 containers to check, what is the probability that at least one of the 3 fails its safety check?

Formula / Rule

Complement rule for 'at least one': $P(\text{at least one}) = 1 - P(\text{none})$.



First move

Whenever a problem asks for the probability of 'at least one' success among repeated independent trials, immediately switch to finding the probability of 'none' and subtract from 1 — it's almost always faster than adding up every possible case.

Solution

1. Translate the rate into a probability: 1 out of every 10 containers fails, so $P(\text{fail}) = 1/10$ and $P(\text{pass}) = 1 - 1/10 = 9/10$.
2. The monthly total of 4,000 containers is not needed — the question only concerns the 3 selected containers.
3. 'At least one fails' is the complement of 'none of the 3 fail,' so find $P(\text{none fail})$ first.
4. Since the containers are independent, multiply: $P(\text{none fail}) = 9/10 \times 9/10 \times 9/10 = 729/1000$.
5. Apply the complement rule: $P(\text{at least one fails}) = 1 - 729/1000$.
6. Subtract: $1000/1000 - 729/1000 = 271/1000$.



Answer: $271/1000 (= 0.271)$



Problem 2 — Strategy & Elimination



Tip

Whenever a problem asks for the probability of 'at least one' success among repeated independent trials, immediately switch to finding the probability of 'none' and subtract from 1 — it's almost always faster than adding up every possible case.



Shortcut

The complement approach IS the shortcut here — directly listing every combination of 1, 2, or 3 failing containers would take three separate calculations instead of one subtraction.



Trap

A rushed student multiplies the single-event probability by the number of trials — $3 \times 1/10 = 3/10$ — treating the probabilities as if they simply add, but that overcounts cases where more than one container fails and ignores how the complement actually compounds.



Desmos Tip

Type $1 - (9/10)^3$ directly into the calculator to get 0.271 instantly, avoiding any arithmetic slip when cubing $9/10$ by hand.



Process of Elimination

Correct: **271/1000 (= 0.271)**

Tempting wrong answer	Exactly what went wrong
3/10	Multiplied the single-container failure probability by the number of trials ($3 \times 1/10$) instead of applying the complement rule, treating probabilities as additive across independent trials.
729/1000	Correctly computed $P(\text{none fail}) = 9/10 \times 9/10 \times 9/10$ but forgot the final step of subtracting from 1, submitting the complement of the requested event instead of the event itself.
243/1000	Computed $P(\text{exactly one fails}) = 3 \times (1/10) \times (9/10)^2 = 243/1000$, mistaking 'at least one' for 'exactly one' fails and only counting the single-failure case.

Problem 3

Very Hard

A drawer at a stationery shop in Thessaloniki contains 6 red pens and 4 blue pens, with no other colors. Two pens are drawn at random, one after another, without putting the first pen back. What is the probability that both pens drawn are the same color?

Formula / Rule

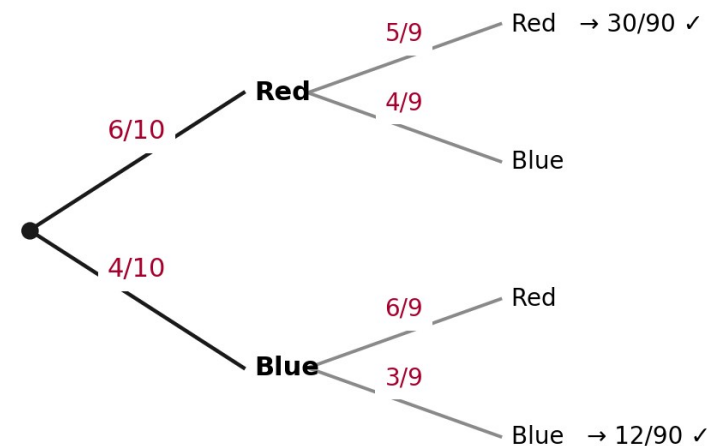
Dependent 'and': $P(A \text{ and } B) = P(A) \times P(B|A)$, combined across colors with the mutually exclusive 'or' rule.

Solution

1. There are 10 pens total (6 red + 4 blue), and the draws are without replacement, so the second draw's probability depends on the first.
2. Find $P(\text{both red})$: $P(\text{red 1st}) = 6/10$, then $P(\text{red 2nd} | \text{red 1st}) = 5/9$, since one red pen is gone and 9 pens remain.
3. Multiply: $P(\text{both red}) = 6/10 \times 5/9 = 30/90 = 1/3$.
4. Find $P(\text{both blue})$: $P(\text{blue 1st}) = 4/10$, then $P(\text{blue 2nd} | \text{blue 1st}) = 3/9$, since one blue pen is gone and 9 pens remain.
5. Multiply: $P(\text{both blue}) = 4/10 \times 3/9 = 12/90 = 2/15$.
6. 'Both red' and 'both blue' cannot happen at the same time, so they are mutually exclusive — add the two probabilities: $P(\text{same color}) = 1/3 + 2/15$.
7. Find a common denominator (15) and add: $5/15 + 2/15 = 7/15$.

✓ **Answer: 7/15**

Two draws, no replacement





Problem 3 — Strategy & Elimination



Tip

When a 'same color' or 'same type' question involves more than one category, compute one dependent-draw probability per category, then combine the category results with the mutually exclusive OR rule.



Trap

A student who correctly computes $P(\text{both red}) = 1/3$ often stops there, forgetting that 'both blue' is a separate valid way to get 'same color,' which gives an incomplete answer of $1/3$ instead of the full $7/15$.



Shortcut

There isn't a faster route than computing each color's dependent probability separately, but noticing both cases share the same fraction structure ($\text{count}/10 \times (\text{count}-1)/9$) lets you set up both products side-by-side before simplifying, saving rework.



Process of Elimination

Correct: **7/15**

Tempting wrong answer	Exactly what went wrong
1/3	Computed only $P(\text{both red})$ and stopped, forgetting that 'both blue' is a second valid way to satisfy 'same color' and must be added.
13/25	Treated the two draws as independent with replacement, computing $6/10 \times 6/10 + 4/10 \times 4/10 = 52/100$ instead of reducing the denominator to 9 and the count by 1 for the second, dependent draw.
8/15	Computed $P(\text{different colors}) = 1 - 7/15 = 8/15$, answering the complement of the question that was actually asked.

Problem 4

Hard

A drawer of laboratory samples at a CLC science lab contains 8 clearly labeled 'Group X' vials and an unknown number of unlabeled 'Group Y' vials, with no other types. The drawer can hold up to 50 vials in total. If the probability of randomly selecting a Group Y vial is $\frac{5}{13}$, how many Group Y vials are in the drawer?

Formula / Rule

Reverse probability setup: $P(Y) = (\text{number of Y}) / (\text{total})$, then solve the resulting equation for the unknown count.



First move

In a reverse-setup probability problem, name the unknown count with a variable, write the total as a sum that includes it, and solve the resulting probability equation like any other algebraic fraction equation.

Solution

1. Let y = the number of Group Y vials. The drawer's 50-vial capacity is irrelevant here — it's a maximum, not the actual count.
2. The total number of vials in the drawer is $8 + y$, the 8 Group X vials plus the unknown Group Y vials.
3. Set up the probability equation: $P(Y) = y / (8 + y) = 5/13$.
4. Cross-multiply: $13 \times y = 5 \times (8 + y)$.
5. Expand the right side: $13y = 40 + 5y$.
6. Subtract $5y$ from both sides: $8y = 40$.
7. Divide by 8: $y = 5$.

**Answer: 5 Group Y vials**



Problem 4 — Strategy & Elimination



Tip

In a reverse-setup probability problem, name the unknown count with a variable, write the total as a sum that includes it, and solve the resulting probability equation like any other algebraic fraction equation.



Trap

A rushed student sets up $y/8 = 5/13$, dividing by the known count instead of the full total, which gives $y = 40/13$ — not a whole number of vials, a clear sign the wrong quantity was used as the denominator.



Shortcut

Since the given probability $5/13$ already has denominator 13, you can suspect the total is 13 and check $y = 13 - 8 = 5$ directly; solving the equation algebraically is just as fast here and confirms it without guessing.



Desmos Tip

In Desmos, graph $f(y) = y/(8+y)$ alongside the horizontal line $y = 5/13$ and find their intersection at $y = 5$ to confirm the algebra.



Process of Elimination

Correct: 5 Group Y vials

Tempting wrong answer	Exactly what went wrong
$y = 40/13$	Set up $y/8 = 5/13$, dividing by the known Group X count instead of the total $8 + y$, so the denominator never included the unknown quantity being solved for.
$y = 64/5$ (12.8)	Set up the equation for $P(X)$ instead of $P(Y)$ — solved $8/(8+y) = 5/13$, using the labeled count as the numerator instead of the unlabeled one.
$y = 250/13$	Used the drawer's maximum capacity (50) as the actual total instead of $8 + y$, solving $y/50 = 5/13$ with an irrelevant number pulled from the problem.

Problem 5

Hard

CLC Multinational employs 300 people across two offices — 180 in Athens and 120 in Vienna. Across all 300 employees, 150 speak German, 130 speak French, and 50 speak both languages fluently. If an employee is selected at random from the full company, what is the probability that they speak German or French?

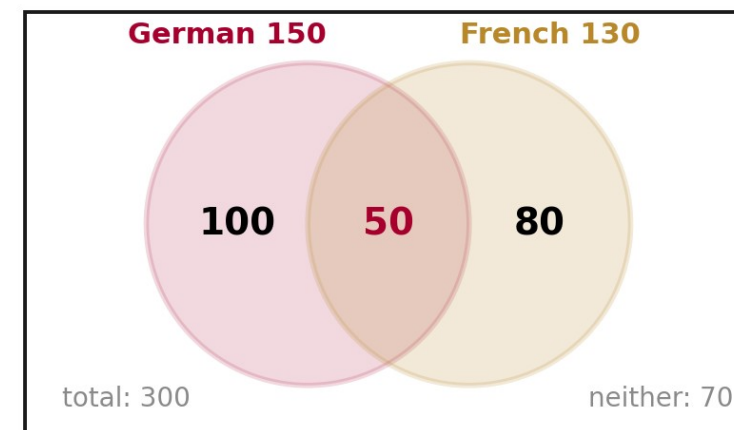
Formula / Rule

General 'or' rule for non-mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Solution

1. The Athens/Vienna office split (180 and 120) has nothing to do with which languages employees speak — it's extra information to ignore.
2. Speaking German and speaking French are NOT mutually exclusive, since 50 employees speak both, so the basic addition rule alone would double-count them.
3. Find each individual probability: $P(\text{German}) = 150/300$ and $P(\text{French}) = 130/300$.
4. Find the overlap probability: $P(\text{both}) = 50/300$.
5. Apply the general OR rule: $P(\text{German or French}) = 150/300 + 130/300 - 50/300$.
6. Combine the numerators: $150 + 130 - 50 = 230$, so $P(\text{German or French}) = 230/300$.
7. Simplify: $230/300 = 23/30$.

✓ **Answer:** 23/30





Problem 5 — Strategy & Elimination



Tip

Whenever an 'or' problem gives you an overlap number (people who do BOTH things), treat that as your signal the events are not mutually exclusive and you must subtract the overlap.



Trap

A student who ignores the phrase 'both languages' simply adds $150/300 + 130/300 = 280/300 = 14/15$, double-counting the 50 employees who speak both languages.



Shortcut

No shortcut beats the direct formula here, but working the whole calculation over the common denominator of 300 from the start, rather than simplifying each fraction first, avoids extra fraction work.



Process of Elimination

Correct: **23/30**

Tempting wrong answer	Exactly what went wrong
14/15	Added $150/300 + 130/300$ without subtracting the overlap of $50/300$, double-counting the employees who speak both languages.
1/6	Computed only the overlap probability $50/300 = 1/6$ and submitted it directly, mistaking the 'and' (both languages) value for the requested 'or' (German or French) value.
3/5	Subtracted the overlap twice — $150/300 + 130/300 - 50/300 - 50/300 = 180/300$ — instead of the single subtraction the general OR rule requires.

Problem 6

Very Hard

A jar at a school fair in Rotterdam contains only red marbles and blue marbles, in the ratio 2:3. A volunteer adds 10 more blue marbles to the jar, which changes the ratio of red to blue marbles to 2:5. Using the ORIGINAL numbers of marbles (before the blue ones were added), if two marbles are drawn at random without replacement, what is the probability that both are red?

Formula / Rule

Ratio-to-count setup (red = $2k$, blue = $3k$) to solve for k , then dependent 'and' probability using the original counts.



First move

When a ratio equation is hidden inside a probability question, solve completely for the ratio's constant (k) first, then plug the resulting counts into the probability calculation as a separate second step — don't try to do both at once.

Solution

1. Let the original counts be red = $2k$ and blue = $3k$, using the given ratio 2:3.
2. After adding 10 blue marbles, the new ratio is 2:5, so $\text{red}/(\text{blue} + 10) = 2/5$, giving $2k/(3k + 10) = 2/5$.
3. Cross-multiply: $5 \times 2k = 2 \times (3k + 10)$, so $10k = 6k + 20$.
4. Subtract $6k$ from both sides: $4k = 20$, so $k = 5$.
5. Find the original counts: red = $2k = 10$, blue = $3k = 15$, for an original total of 25 marbles.
6. Use these ORIGINAL counts for the probability question: $P(\text{1st red}) = 10/25$.
7. Since the draw is without replacement, $P(\text{2nd red} \mid \text{1st red}) = 9/24$, since one red marble is gone and 24 remain.
8. Multiply: $P(\text{both red}) = 10/25 \times 9/24 = 90/600$.
9. Simplify: $90/600 = 3/20$.



Answer: 3/20



Problem 6 — Strategy & Elimination



Tip

When a ratio equation is hidden inside a probability question, solve completely for the ratio's constant (k) first, then plug the resulting counts into the probability calculation as a separate second step — don't try to do both at once.



Shortcut

There's no way to skip solving for k first — the probability cannot be computed until the actual counts are known — but once $k = 5$ is found, the probability part is just two quick multiplications.



Trap

A student in a hurry uses the NEW post-addition counts (10 red and 25 blue, out of 35) instead of the ORIGINAL counts (10 red and 15 blue, out of 25) that the question specifically asks for, giving the wrong probability $10/35 \times 9/34 = 90/1190 = 9/119$.



Desmos Tip

Graph $f(k) = 2k/(3k+10)$ alongside the line $y = 2/5$ in Desmos and find their intersection at $k = 5$ to confirm the ratio equation before computing the final probability.



Process of Elimination

Correct: 3/20

Tempting wrong answer	Exactly what went wrong
9/119	Used the NEW post-addition counts (10 red, 25 blue, 35 total) instead of the ORIGINAL counts the question specifically asked for (10 red, 15 blue, 25 total).
1/10	Used the raw ratio numbers 2 and 3 directly as if they were the actual marble counts ($P = 2/5 \times 1/4$) instead of solving for k first to find the true counts of 10 and 15.
4/25	Treated the two draws as independent with replacement, computing $10/25 \times 10/25$ instead of reducing both the numerator and denominator for the second, dependent draw.

Problem 7

Hard

At a CLC Educational Institute charity raffle in Vienna, the odds in favor of winning a prize with a single ticket are stated as 3 to 17 (that is, for every 3 winning tickets sold, there are 17 non-winning tickets). A student buys 2 raffle tickets, and the outcome of each ticket is independent of the other. What is the probability that the student wins with exactly one of the two tickets?

Formula / Rule

Convert odds to probability (parts of a whole), then combine independent 'and' with mutually exclusive 'or' for an 'exactly one of two' event.

First move

When a problem gives odds instead of a probability, first convert by adding the two parts to get the total, then treat the rest as a normal probability problem — and for 'exactly one of two independent trials,' always account for both possible orders.

Solution

1. Convert the odds 3 to 17 into a probability: the total parts are $3 + 17 = 20$, so $P(\text{win}) = 3/20$ and $P(\text{lose}) = 17/20$.
2. 'Exactly one of the two tickets wins' can happen two different ways: (win, lose) or (lose, win).
3. Since the tickets are independent, compute $P(\text{win then lose}) = 3/20 \times 17/20 = 51/400$.
4. Compute the other order: $P(\text{lose then win}) = 17/20 \times 3/20 = 51/400$.
5. These two orders are mutually exclusive (they can't both occur), so add them: $P(\text{exactly one win}) = 51/400 + 51/400 = 102/400$.
6. Simplify: $102/400 = 51/200$.

 **Answer:** 51/200



Problem 7 — Strategy & Elimination



Tip

When a problem gives odds instead of a probability, first convert by adding the two parts to get the total, then treat the rest as a normal probability problem — and for 'exactly one of two independent trials,' always account for both possible orders.



Shortcut

Once $P(\text{win}) = 3/20$ is found, 'exactly one of two independent trials' can be computed in one line as $2 \times (3/20) \times (17/20) = 102/400 = 51/200$, without writing out both orders separately.



Trap

A rushed student treats the odds '3 to 17' as if they were the probability itself, using $3/17$ as $P(\text{win})$ instead of correctly converting it to $3/20$ by adding $3 + 17$ to get the total number of parts.



Desmos Tip

Type $2 \cdot (3/20) \cdot (17/20)$ directly into the calculator to get $51/200$ in one step, avoiding a separate addition of two equal fractions.



Process of Elimination

Correct: **51/200**

Tempting wrong answer	Exactly what went wrong
84/289	Misread the odds '3 to 17' as the probability itself, using $3/17$ as $P(\text{win})$ and $14/17$ as $P(\text{lose})$ instead of converting to parts-of-20 ($3/20$ and $17/20$).
51/400	Computed only $P(\text{win then lose}) = 3/20 \times 17/20$ and forgot to also add $P(\text{lose then win})$, missing the second order in which exactly one ticket wins.
111/400	Confused 'exactly one wins' with 'at least one wins' and applied the complement rule instead: $1 - P(\text{lose, lose}) = 1 - (17/20)^2 = 111/400$, which answers a different question.

Problem 8

Very Hard

A shipment of 15 lightbulbs bound for a hardware store in Bergen contains 4 defective bulbs, mixed in with 11 working bulbs. A technician tests the bulbs one at a time, without replacement, stopping as soon as a defective bulb is found. What is the probability that the very first defective bulb found is the third bulb tested?

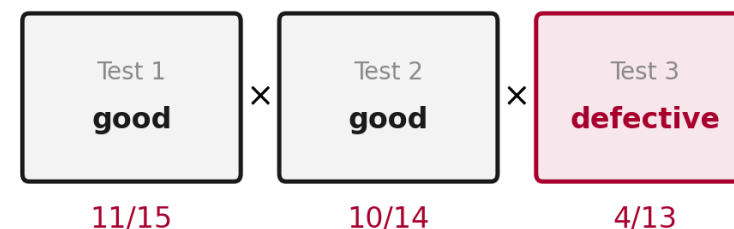
Formula / Rule

Sequential dependent 'and': the required order is (good, good, defective), multiplying probabilities that update as bulbs are removed.

Solution

1. For the third bulb to be the first defective one found, the first two bulbs tested must both be working, and the third must be defective — this is one specific order, not any arrangement.
2. The shipment size of 500 units to the store is not used — only the 15 bulbs in this particular batch matter.
3. Find $P(\text{1st bulb good})$: 11 working bulbs out of 15 total, so $P = 11/15$.
4. Find $P(\text{2nd bulb good} \mid \text{1st was good})$: 10 working bulbs remain out of 14 total, so $P = 10/14$.
5. Find $P(\text{3rd bulb defective} \mid \text{first two were good})$: 4 defective bulbs remain out of 13 total, so $P = 4/13$.
6. Multiply the three probabilities in this exact order: $11/15 \times 10/14 \times 4/13 = 440/2730$.
7. Simplify: $440/2730 = 44/273$ (dividing numerator and denominator by 10 gives $44/273$, which has no further common factors).

✓ **Answer: 44/273**





Problem 8 — Strategy & Elimination



Tip

When a problem asks for a specific position where an event first occurs, translate it into one exact ordered sequence (e.g., 'not yet, not yet, then yes') and multiply the dependent probabilities for that sequence only — don't average over all possible positions.



Shortcut

No shortcut beyond direct sequential multiplication is available here, since only one specific order (good, good, defective) is being asked for — unlike an 'any position' version of this problem, which would require multiplying the final product by 3.



Trap

A rushed student computes the probability that exactly one defective bulb appears anywhere among the first three tested (multiplying by 3 for the three possible positions), which answers a broader question than 'the third bulb specifically is the first defective one.'



Desmos Tip

Type $(11/15) \cdot (10/14) \cdot (4/13)$ into the calculator to get 44/273 directly, avoiding an arithmetic slip across the three-step multiplication.



Process of Elimination

Correct: 44/273

Tempting wrong answer	Exactly what went wrong
44/91	Computed P(exactly one defective among the first three bulbs, in any order) by multiplying by 3 for the three possible positions, instead of the single required order good-good-defective.
4/15	Computed only the plain, unconditioned probability that the third bulb drawn is defective, ignoring the requirement that the first two bulbs must also both be good for it to be the FIRST defective found.
58/91	Computed P(at least one defective among the first three bulbs) via the complement $1 - (11/15 \times 10/14 \times 9/13)$, answering whether any defective appears within three draws rather than whether the third draw specifically is the first defective.

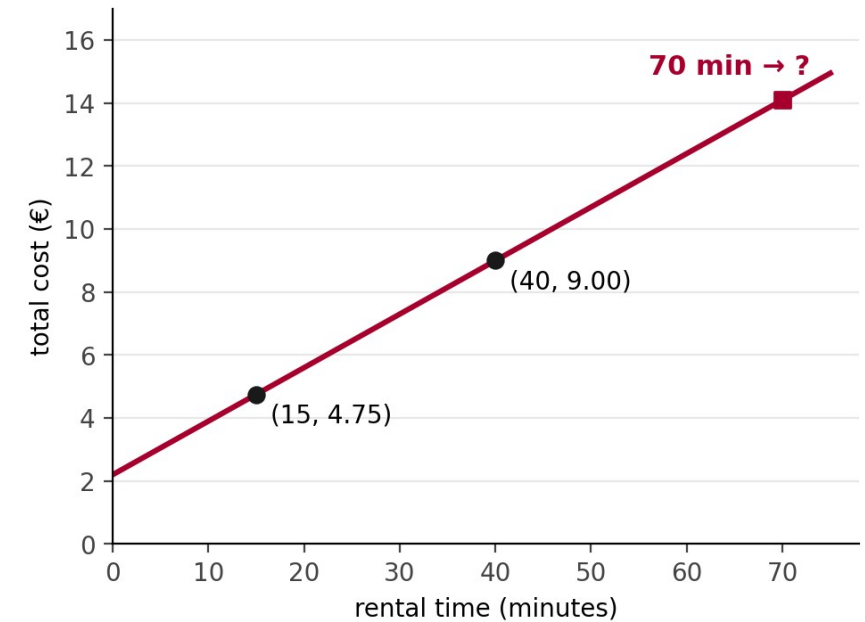
03

Unit Conversions & Rates

Dimensional analysis, rates, and converting area and volume units.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



Key Terminology

Term (English)	Ελληνικά
Unit conversion	Μετατροπή μονάδων
Conversion factor	Συντελεστής μετατροπής
Rate	Ρυθμός
Dimensional analysis	Ανάλυση διαστάσεων (μονάδων)

The big picture

- Every conversion is multiplication by a form of 1.
- Track units like algebra — they must cancel.
- Area and volume conversions square or cube the factor.

Theory Explained (1/3)

Conversion Factors

multiply by (new unit / old unit)

WHAT IT MEANS

- Write the factor so the old unit cancels.
- Chain several factors in one line.
- Check: is the answer bigger or smaller, as expected?



Example

$72 \text{ km/h} \times (1000 \text{ m}/1 \text{ km}) \times (1 \text{ h}/3600 \text{ s}) = 20 \text{ m/s}.$

Rates

rate = amount / time amount = rate $\times$ time

WHAT IT MEANS

- Average speed = total distance $\div$ total time.
- Convert mixed times first: 4 h 10 min = $25/6$ h.
- Seconds $\rightarrow$ minutes: $\div 60$, never $\div 100$.



Example

500 km in $25/6$ h = 120 km/h.

Theory Explained (2/3)

Area & Volume Units

area: factor^2 volume: factor^3

WHAT IT MEANS

- $1 \text{ m} = 100 \text{ cm} \rightarrow 1 \text{ m}^2 = 10,000 \text{ cm}^2$.
- $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$.
- Convert lengths BEFORE computing the volume if easier.



Example

$$2.4 \text{ m}^3 \times (1.09361)^3 \approx 3.1 \text{ yd}^3.$$

Inverting Rates

$\text{L}/100 \text{ km} \rightarrow \text{km}/\text{L} \rightarrow \text{mi}/\text{gal}$

WHAT IT MEANS

- Consumption (L per km) is the reciprocal of efficiency (km per L).
- Invert first, then convert.
- Label every step with its units.



Example

$$6.5 \text{ L}/100 \text{ km} \rightarrow 15.38 \text{ km}/\text{L} \rightarrow \approx 36.2 \text{ mpg}.$$

Theory Explained (3/3)

Time Arithmetic

$$\text{decimal hours} \times 60 = \text{minutes}$$

WHAT IT MEANS

- 5 min 15 s = 5.25 min (15/60), not 5.15.
- Split decimals: 3.692 h = 3 h + 0.692 × 60 min.
- Round only at the very end.



Example

$$5.25 \times 42.195 = 221.52 \text{ min} = 3 \text{ h } 41 \text{ min } 31 \text{ s.}$$

Watch Out



Base 60, not base 10

- Minutes and seconds convert by 60.
- $0.5 \text{ h} = 30 \text{ min}$, not 50.



Square and cube the factor

- Area units need factor², volume units factor³.
- Using the linear factor is the classic trap.

Problem 1

Very Hard

A speed camera near Lisbon reports a car's speed in two different units at the same instant: feet per second and miles per hour. The technician notices the feet-per-second reading is exactly 14 more than the miles-per-hour reading. Given that 1 mile = 5280 feet and 1 hour = 3600 seconds, what is the car's speed in miles per hour?

Formula / Rule

Convert one variable's units using the exact conversion fraction, then solve the resulting linear equation.



First move

When a problem relates the same speed expressed in two different units, convert everything into one variable using the exact conversion factor as a fraction (don't round it to a decimal), then solve algebraically.

Solution

1. Let v = the car's speed in miles per hour.
2. Convert v to feet per second: $v \text{ mph} = v \times (5280 \text{ ft} / 3600 \text{ s}) = (22/15)v \text{ ft/s}$.
3. Translate the given relationship into an equation: the ft/s reading equals the mph reading plus 14, so $(22/15)v = v + 14$.
4. Clear the fraction by multiplying every term by 15: $22v = 15v + 210$.
5. Subtract $15v$ from both sides: $7v = 210$.
6. Divide by 7: $v = 30$.
7. Check: $30 \text{ mph} = 30 \times 22/15 = 44 \text{ ft/s}$, and $44 - 30 = 14$, which matches the given condition.

**Answer: 30 mph**



Problem 1 — Strategy & Elimination



Tip

When a problem relates the same speed expressed in two different units, convert everything into one variable using the exact conversion factor as a fraction (don't round it to a decimal), then solve algebraically.



Shortcut

Keep the conversion factor as the exact fraction $\frac{22}{15}$ and immediately clear denominators by multiplying the whole equation by 15 — this turns it into $22v = 15v + 210$ with no decimals or rounding anywhere.



Trap

A student who rounds $\frac{5280}{3600}$ to a decimal too early, or who writes the equation backwards as $v - (\frac{22}{15})v = 14$ (assuming the mph reading is the larger one), will get an incorrect or nonsensical value instead of correctly treating the ft/s reading as the larger one.



Desmos Tip

Graph $y = (\frac{22}{15})x$ and $y = x + 14$ and find their intersection point — the x-coordinate (30) is the mph speed and the y-coordinate (44) confirms the ft/s reading.



Process of Elimination

Correct: 30 mph

Tempting wrong answer	Exactly what went wrong
28 mph	Rounded the conversion factor $\frac{5280}{3600}$ to 1.5 instead of using the exact fraction $\frac{22}{15}$, then solved $1.5v - v = 14$ to get $v = 28$.
44 mph	Correctly solved for the ft/s value (44) but reported that number as the final mph answer instead of continuing to identify v itself.
≈6.5 mph	When isolating v from $7v = 210$ (or the equivalent $(\frac{7}{15})v = 14$), multiplied both sides by $\frac{7}{15}$ instead of dividing by it (multiplying by the reciprocal $\frac{15}{7}$), inverting the fraction incorrectly.

Problem 2

Hard

A delicatessen in Thessaloniki sells graviera cheese at €16.80 per kilogram. A tourist orders 3.5 pounds of cheese to take home, but the shop's scale only measures in 50-gram increments and always rounds a customer's order UP to the next increment before weighing. Using 1 kilogram = 2.20462 pounds, how much will the tourist pay, rounded to the nearest cent?

Formula / Rule

Unit conversion (pounds to kilograms) combined with a per-unit rate (price × quantity), plus a stated rounding rule.



First move

When a real-world purchase combines a unit conversion with a price rate and a rounding rule, convert first, apply the rounding rule exactly as stated, and only then multiply by the rate.

Solution

1. Convert 3.5 pounds to kilograms: $3.5 \div 2.20462 \approx 1.5876$ kg.
2. The scale reads only in 0.05 kg (50 g) increments; the increments bracketing 1.5876 are 1.55 kg and 1.60 kg.
3. Since the shop always rounds up, the weighed amount used for billing is 1.60 kg.
4. Multiply the rounded weight by the price per kilogram: $1.60 \text{ kg} \times €16.80/\text{kg} = €26.88$.
5. €26.88 already has exactly two decimal places, so no further rounding is needed.

**Answer: €26.88**



Problem 2 — Strategy & Elimination



Tip

When a real-world purchase combines a unit conversion with a price rate and a rounding rule, convert first, apply the rounding rule exactly as stated, and only then multiply by the rate.



Trap

A student who converts pounds to kilograms correctly but forgets the shop's 'round up to the nearest 50 g' rule will multiply the unrounded 1.5876 kg directly by the price instead of the correct rounded 1.60 kg.



Shortcut

You can estimate quickly with $1\text{ lb} \approx 0.4536\text{ kg}$ ($3.5 \times 0.4536 \approx 1.588\text{ kg}$) to sanity-check the division, but the rounding-to-50g step and final multiplication still have to be done by hand.



Process of Elimination

Correct: €26.88

Tempting wrong answer	Exactly what went wrong
€26.67	Skipped the rounding-up-to-nearest-50g step entirely and multiplied the unrounded 1.5876 kg directly by €16.80.
€130.20	Multiplied pounds by the conversion factor instead of dividing ($3.5 \times 2.20462 = 7.716\text{ kg}$ instead of $3.5 \div 2.20462$), then rounded up to 7.75 kg and paid for nearly five times too much cheese.
€58.80	Never converted pounds to kilograms at all — treated the 3.5 lb quantity as if it were already 3.5 kg and multiplied directly by the per-kilogram price.

Problem 3

Hard

A juice bar in Ljubljana blends its signature drink using orange, beet, and ginger juice according to a rule the owner describes this way: 'for every 2 liters of orange juice, we use 750 milliliters of beet juice, and the ginger juice is always one third the volume of the beet juice.' A catering order calls for 10 liters of orange juice. How many milliliters of ginger juice are needed?

Formula / Rule

Set up a proportion between two of the three linked quantities described in words, then apply the stated fractional relationship to the third.

First move

When a ratio is written as a sentence linking three quantities, first convert everything to the same unit, then isolate the two-quantity proportion you actually need before bringing in the third relationship.

Solution

1. Convert the base ratio to one unit: 2 liters of orange juice = 2000 mL, paired with 750 mL of beet juice.
2. Set up a proportion for 10 liters (10,000 mL) of orange juice: $2000/750 = 10000/x$, where x is the beet juice needed.
3. Cross-multiply: $2000x = 750 \times 10000 = 7,500,000$.
4. Solve for x : $x = 7,500,000 \div 2000 = 3750$ mL of beet juice.
5. Apply the stated relationship for ginger juice, which is always one third the volume of beet juice: $\text{ginger} = 3750 \div 3 = 1250$ mL.

 **Answer: 1250 mL**



Problem 3 — Strategy & Elimination



Tip

When a ratio is written as a sentence linking three quantities, first convert everything to the same unit, then isolate the two-quantity proportion you actually need before bringing in the third relationship.



Shortcut

Since orange:beet simplifies to 8:3, you can skip the milliliter setup and compute directly in liters: $10\text{ L} \times \frac{3}{8} = 3.75\text{ L}$ of beet juice, then divide by 3 for the ginger juice.



Trap

A rushed student applies 'ginger is one third of beet' to the wrong quantity — computing one third of the 10,000 mL of orange juice instead of one third of the correctly scaled 3750 mL of beet juice.



Desmos Tip

Type $y = (750/2000) \cdot (1/3) \cdot x$ into the calculator and evaluate at $x = 10000$ to get the ginger juice volume (1250) in one expression.



Process of Elimination

Correct: 1250 mL

Tempting wrong answer	Exactly what went wrong
≈3333 mL	Applied 'ginger is one third of beet' directly to the orange juice quantity instead of the scaled beet quantity, computing $10000/3$ instead of $3750/3$.
3750 mL	Correctly scaled the beet juice to 3750 mL but stopped there, reporting the beet juice volume instead of taking one third of it for the ginger juice.
11250 mL	Inverted the ginger-to-beet relationship, tripling the beet volume (3750×3) instead of taking one third of it.

Problem 4

Hard

The high-speed rail line between Athens and Thessaloniki covers 500 kilometers. A particular train completes the trip in 4 hours and 10 minutes, while carrying 312 passengers and using electricity equivalent to 18,000 liters of diesel. What is the train's average speed in meters per second?

Formula / Rule

Average speed = distance/time, then convert km/h to m/s using the factor 5/18.



First move

Strip out any data not needed for the actual question first — Digital SAT unit problems often bury the two numbers you need in a sentence full of irrelevant details like passenger counts or energy use.

Solution

1. Identify the data actually needed: distance = 500 km and time = 4 hours 10 minutes; the passenger count and energy figure are irrelevant to a speed calculation.
2. Convert the time to hours: 10 minutes = $10/60$ hour = $1/6$ hour, so total time = $4 + 1/6 = 25/6$ hours.
3. Compute average speed in km/h: $500 \div (25/6) = 500 \times 6/25 = 120$ km/h.
4. Convert km/h to m/s: $120 \text{ km/h} \times (1000 \text{ m/km}) \div (3600 \text{ s/hr}) = 120,000/3600 \text{ m/s}$.
5. Simplify using the shortcut factor 5/18: $120 \times 5/18 = 600/18 = 100/3 \approx 33.3 \text{ m/s}$.



Answer: $100/3 \text{ m/s} \approx 33.3 \text{ m/s}$



Problem 4 — Strategy & Elimination



Tip

Strip out any data not needed for the actual question first — Digital SAT unit problems often bury the two numbers you need in a sentence full of irrelevant details like passenger counts or energy use.



Shortcut

Memorize that km/h $\rightarrow$ m/s is always $\times(5/18)$ (and m/s $\rightarrow$ km/h is $\times 18/5$) — this skips writing out the full 1000/3600 conversion every time.



Trap

A student who tries to work the diesel/electricity figure into the calculation (assuming every number given must be used) wastes time or misapplies it; the speed calculation only needs distance and time.



Desmos Tip

Type $500/(25/6)$ into the calculator to get 120 km/h instantly, then multiply by $5/18$ to confirm $100/3$ m/s without doing the arithmetic by hand.



Process of Elimination

Correct: $100/3$ m/s ≈ 33.3 m/s

Tempting wrong answer	Exactly what went wrong
120 m/s	Computed the correct average speed in km/h (120) but forgot to convert to m/s at all, reporting the km/h value as if it were already meters per second.
≈ 33.9 m/s	Misread '10 minutes' as 0.10 hours instead of $1/6$ hour, using a total time of 4.10 hours, which shifts the final answer slightly off from the correct 33.3.
432 m/s	Multiplied by the inverse conversion factor ($18/5$) instead of $5/18$ when converting km/h to m/s.

Problem 5

Very Hard

A hiking app records elevation gain in a proprietary unit called 'notches,' where each notch equals an unknown but constant number of meters. A trail with 840 meters of elevation gain shows as 24 notches on the app. Using this same conversion, a second trail shows 63 notches. How many kilometers of elevation gain is that?

Formula / Rule

Find the unknown unit rate (constant) from the first data pair, then apply that same rate to a second quantity.



First move

When a problem defines a made-up or proprietary unit, your first move is always to solve for the conversion constant from the given example before touching the actual question.

Solution

1. Use the given pair to find the conversion constant: 24 notches = 840 meters, so 1 notch = $840 \div 24 = 35$ meters.
2. Sanity-check the direction: more notches should mean more elevation, and 35 meters/notch is a reasonable rate, so the constant is meters per notch, not the reverse.
3. Apply the constant to the second trail: 63 notches $\times$ 35 meters/notch = 2205 meters.
4. Convert the result to kilometers, as the question asks: $2205 \div 1000 = 2.205$ km.
5. Sanity check: 63 notches is 2.625 times the original 24, so the elevation should be about 2.625 times 840 m $\approx$ 2205 m — consistent.



Answer: 2.205 km



Problem 5 — Strategy & Elimination



Tip

When a problem defines a made-up or proprietary unit, your first move is always to solve for the conversion constant from the given example before touching the actual question.



Shortcut

Skip solving for the constant explicitly and set up a direct proportion instead: $840/24 = x/63$, cross-multiplying to solve for x (meters) in one line.



Trap

A student who inverts the proportion — treating 24 as meters per notch and 840 as the number of notches — gets a wildly wrong constant and a wrong answer for the second trail.



Desmos Tip

Graph $y = 35x$ (notches to meters) and read the y -value at $x = 63$ to confirm 2205 meters before converting to kilometers.



Process of Elimination

Correct: **2.205 km**

Tempting wrong answer	Exactly what went wrong
2205 km	Correctly found the constant (35 m/notch) and multiplied to get 2205 meters, but forgot to convert to kilometers and reported the meter value with the wrong unit label.
≈0.0018 km	Solved for the constant upside down ($24 \div 840$ instead of $840 \div 24$), producing about 0.0286 meters per notch instead of 35, then used that on the second trail.
2.142 km	Made a division slip computing $840 \div 24$, getting 34 instead of the correct 35 meters per notch, and carried that error through the final multiplication and conversion.

Problem 6

Hard

During the Berlin Marathon, a runner maintains an exact pace of 5 minutes and 15 seconds per kilometer for the entire 42.195-kilometer race. If she keeps this pace the whole way, how long will it take her to finish, in hours, minutes, and seconds, rounded to the nearest second?

Formula / Rule

Convert a compound pace (min/km) entirely into one unit (seconds per km), multiply by total distance, then convert the total seconds back into h:m:s.



First move

Translate a 'per kilometer' pace into a single unit (seconds per km) first, multiply by total distance to get total seconds, then convert that one number back into h:m:s at the very end.

Solution

1. Convert the pace to seconds per kilometer: 5 minutes 15 seconds = $5 \times 60 + 15 = 315$ seconds per km.
2. Multiply by the total distance: $315 \text{ s/km} \times 42.195 \text{ km} = 13,291.425$ seconds.
3. Convert seconds to whole hours: $13,291.425 \div 3600 \approx 3.692$ hours, so 3 full hours = 10,800 seconds.
4. Find the remaining seconds after those 3 hours: $13,291.425 - 10,800 = 2491.425$ seconds.
5. Convert the remainder to whole minutes: $2491.425 \div 60 \approx 41.52$ minutes, so 41 full minutes = 2460 seconds.
6. Find the final remaining seconds: $2491.425 - 2460 = 31.425$, which rounds to 31 seconds.



Answer: 3 hours, 41 minutes, 31 seconds



Problem 6 — Strategy & Elimination



Tip

Translate a 'per kilometer' pace into a single unit (seconds per km) first, multiply by total distance to get total seconds, then convert that one number back into h:m:s at the very end.



Shortcut

Convert the pace straight to seconds per km (315) and multiply by the exact distance in one line ($315 \times 42.195 = 13,291.425$ s) rather than handling minutes and seconds separately — this avoids the decimal-minute trap entirely.



Trap

The most common trap is converting '5 minutes 15 seconds' to decimal minutes as 5.15 instead of 5.25 (treating the 15 seconds as 0.15 minutes rather than the correct $15/60 = 0.25$), which throws off every step that follows.



Desmos Tip

Type $5 + 15/60$ into the calculator to get the exact decimal pace 5.25 min/km before multiplying, avoiding the 5.15 decimal-conversion trap by hand.



Process of Elimination

Correct: **3 hours, 41 minutes, 31 seconds**

Tempting wrong answer	Exactly what went wrong
3 hours, 37 minutes, 18 seconds	Converted the pace to decimal minutes incorrectly as 5.15 instead of 5.25 (treating 15 seconds as 0.15 minutes rather than $15/60 = 0.25$ minutes), understating the total time.
3 hours, 69 minutes, 12 seconds	Converted the decimal remainder of hours (0.692) into minutes by multiplying by 100 instead of 60, treating time as base-10 instead of base-60.
3 hours, 40 minutes, 30 seconds	Used 42 km instead of the exact marathon distance 42.195 km, dropping the extra 195 meters from the calculation.

Problem 7

Very Hard

A car rented in Germany has a fuel-efficiency rating of 6.5 liters per 100 kilometers. Given that 1 gallon = 3.78541 liters and 1 mile = 1.60934 kilometers, what is the car's fuel efficiency in miles per gallon, rounded to the nearest tenth?

Formula / Rule

Multi-step dimensional analysis: invert L/100km into km/L, then convert km/L → km/gal → mi/gal.



First move

For any multi-hop conversion, decide which conversions multiply and which divide before calculating, and never round between intermediate steps.

Solution

1. Recognize that 6.5 L/100km is a consumption rate, not an efficiency rate — invert it first: km per liter = $100 \div 6.5 \approx 15.3846$ km/L.
2. Convert km/L to km per gallon by multiplying by the liters-in-a-gallon factor: $15.3846 \times 3.78541 \approx 58.2371$ km/gal.
3. Convert km/gal to miles per gallon by dividing by the km-in-a-mile factor: $58.2371 \div 1.60934 \approx 36.184$ mi/gal.
4. Round to the nearest tenth as requested: 36.2 mpg.
5. Sanity check using the shortcut constant: $235.215 \div 6.5 \approx 36.19$, matching the full chain (small difference is just rounding).



Answer: ≈ 36.2 mpg



Problem 7 — Strategy & Elimination



Tip

For any multi-hop conversion, decide which conversions multiply and which divide before calculating, and never round between intermediate steps.



Trap

Because 'liters per 100 km' is a fuel-consumption rate (lower = more efficient), a student who treats it like an ordinary rate and multiplies 6.5 straight through the gallon and mile conversions — instead of first inverting it to km-per-liter — lands on a badly wrong, too-small mpg figure.



Shortcut

There's a precomputed constant for exactly this conversion: $\text{mpg} \approx 235.215 \div (\text{L}/100\text{km})$ — plugging in $235.215/6.5 \approx 36.19$ matches the full chain almost instantly.



Desmos Tip

Type $100/6.5 \cdot 3.78541/1.60934$ directly into the calculator as one expression to get 36.18 without tracking any intermediate rounding.



Process of Elimination

Correct: $\approx 36.2 \text{ mpg}$

Tempting wrong answer	Exactly what went wrong
$\approx 15.3 \text{ mpg}$	Multiplied the consumption rate (6.5 L/100km) straight through the gallon and mile conversions without first inverting it to km-per-liter, so the result never represents an actual efficiency figure.
$\approx 58.2 \text{ mpg}$	Correctly converted to kilometers per gallon (58.2) but forgot the final step of converting kilometers to miles, reporting the km/gal figure as if it were already mi/gal.
$\approx 93.7 \text{ mpg}$	Multiplied by the km-to-mile factor (1.60934) instead of dividing by it, inflating the distance instead of converting it down into the smaller mile count.

Problem 8

Very Hard

A landscaping company in Zurich needs to cover a rectangular garden bed measuring 4 meters by 6 meters with a 10-centimeter-deep layer of mulch. The mulch is sold by the cubic yard. Given that 1 meter = 1.09361 yards, how many cubic yards of mulch are needed, rounded to the nearest tenth?

Formula / Rule

Volume conversions scale with the CUBE of the linear conversion factor, not the factor itself.

First move

Whenever a conversion involves area or volume, never apply the linear conversion factor directly — square it for area or cube it for volume before multiplying.

Solution

1. Convert the mulch depth to meters, matching the other dimensions:
 $10 \text{ cm} = 0.10 \text{ m}$.
2. Compute the volume in cubic meters: $4 \text{ m} \times 6 \text{ m} \times 0.10 \text{ m} = 2.4 \text{ cubic meters}$.
3. Convert cubic meters to cubic yards using the CUBE of the linear factor, since volume scales with length cubed: $1 \text{ m}^3 = (1.09361 \text{ yd})^3 \approx 1.30794 \text{ yd}^3$.
4. Multiply: $2.4 \text{ m}^3 \times 1.30794 \text{ yd}^3/\text{m}^3 \approx 3.1391 \text{ yd}^3$.
5. Round to the nearest tenth: $\approx 3.1 \text{ cubic yards}$.

 **Answer:** $\approx 3.1 \text{ cubic yards}$



Problem 8 — Strategy & Elimination



Tip

Whenever a conversion involves area or volume, never apply the linear conversion factor directly — square it for area or cube it for volume before multiplying.



Trap

The most common mistake is converting 2.4 cubic meters to cubic yards by multiplying by 1.09361 only once, treating a volume conversion as if it were a simple length conversion, which gives a badly wrong result.



Shortcut

Memorize that $1 \text{ m}^3 \approx 1.308 \text{ yd}^3$ (the cube of 1.09361) as a ready-made volume conversion constant, so you can multiply straight from cubic meters to cubic yards without recomputing the cube each time.



Desmos Tip

Type $4 \cdot 6 \cdot 0.1 \cdot 1.09361^3$ directly into the calculator as one expression to get the answer in a single step without a separate cubing calculation.



Process of Elimination

Correct: **≈ 3.1 cubic yards**

Tempting wrong answer	Exactly what went wrong
≈ 2.6 cubic yards	Multiplied the volume by the linear factor 1.09361 only once instead of cubing it, treating a volume conversion as if it were a length conversion.
≈ 313.9 cubic yards	Forgot to convert the mulch depth from 10 centimeters to 0.10 meters before computing the volume, using 10 directly as meters and inflating the volume 100-fold.
≈ 2.9 cubic yards	Squared the linear conversion factor (as if converting an area) instead of cubing it, which is required for a volume conversion.

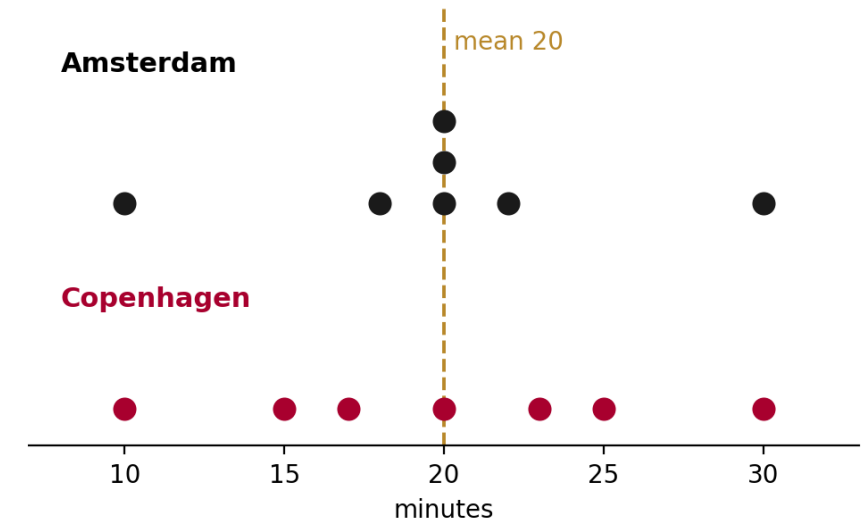
04

Statistics — Center & Spread

Mean, median, weighted averages, outliers, and standard deviation.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

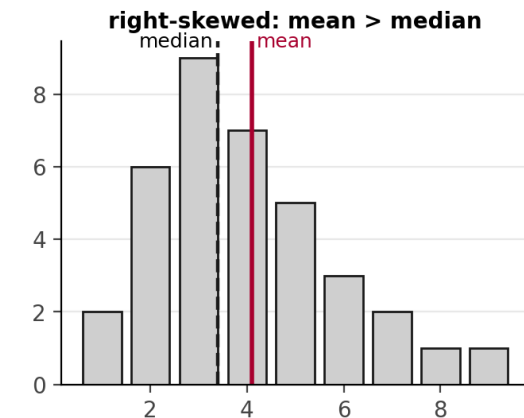


Key Terminology

Term (English)	Ελληνικά
Mean (average)	Μέση τιμή
Median	Διάμεσος
Mode	Επικρατούσα τιμή
Standard deviation	Τυπική απόκλιση
Range	Εύρος
Outlier	Ακραία τιμή

A ㄱ The big picture

- Center: mean and median. Spread: range, IQR, SD.
- Many questions work backwards from a mean.
- Outliers move the mean far more than the median.



Theory Explained (1/3)

Mean as a Total

$$\text{mean} = \text{sum} / n \quad \text{sum} = \text{mean} \times n$$

WHAT IT MEANS

- Convert every mean into a total first.
- Combined groups: add totals, add counts.
- Removing a value: subtract from the total, reduce n.



Example

12 at 1180, n at 1340, overall 1244 $\rightarrow$ n = 8.

Median

middle value of the SORTED list

WHAT IT MEANS

- Odd n: the middle value.
- Even n: average the two middle values.
- Always sort before finding it.



Example

62, 74, 81, 85, 90, 96, x $\rightarrow$ median = 85.

Theory Explained (2/3)

Weighted Average

$$\Sigma(\text{weight} \times \text{value}) / \Sigma \text{ weights}$$

WHAT IT MEANS

- Weights can be counts or percentages.
- Heavier weight → more influence.
- Solve for a missing value by isolating it.



Example

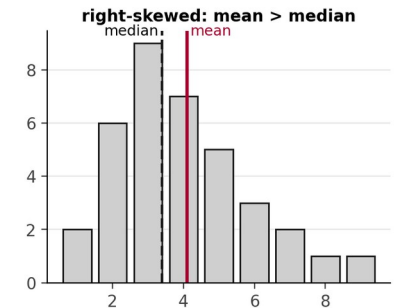
$$0.25(82) + 0.35(78) + 0.40F = 85 \rightarrow F = 93.$$

Outliers & Skew

mean chases the tail; median resists

WHAT IT MEANS

- Right skew (long tail right): mean > median.
- Removing a high outlier drops the mean a lot.
- Median changes little — often by one position.



Example

Remove €1,400: mean -€100, median -€20.

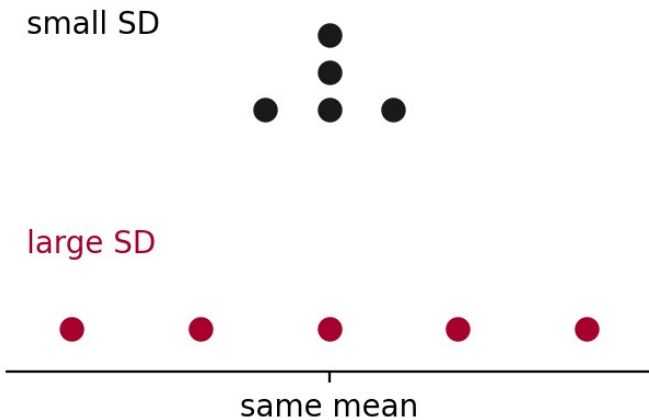
Theory Explained (3/3)

Standard Deviation

SD = typical distance from the mean

WHAT IT MEANS

- No formula needed on the SAT — compare shapes.
- Values clustered near the mean → smaller SD.
- Same range does NOT mean same SD.



Example

10, 20, 20, 20, 30 has smaller SD than 10, 15, 20, 25, 30.

Watch Out



Sort before the median

- Data given in random order is a deliberate trap.
- Count positions carefully.



Range $\neq$ standard deviation

- Range uses only two values.
- SD depends on every value's distance from the mean.

Problem 1

Hard

A CLC test-prep program in Thessaloniki runs two practice groups. Group A has 12 students with a mean practice-test score of 1180. Group B has an unknown number of students with a mean score of 1340. When all students from both groups are combined, the overall mean score is 1244. How many students are in Group B?

Formula / Rule

Weighted average: total sum = (count1 x mean1) + (count2 x mean2), divided by total count.



First move

Convert each group's average into a total by multiplying by its count before combining — never average two averages directly unless the groups are the same size.

Solution

1. Let n = the number of students in Group B.
2. Group A's total score = $12 \times 1180 = 14160$.
3. Group B's total score = $n \times 1340$.
4. The combined mean equation is: $(14160 + 1340n) / (12 + n) = 1244$.
5. Multiply both sides by $(12 + n)$: $14160 + 1340n = 1244(12 + n) = 14928 + 1244n$.
6. Subtract $1244n$ from both sides and subtract 14160 from both sides: $1340n - 1244n = 14928 - 14160$, so $96n = 768$.
7. Solve for n : $n = 768 / 96 = 8$.



Answer: 8 students



Problem 1 — Strategy & Elimination



Tip

Convert each group's average into a total by multiplying by its count before combining — never average two averages directly unless the groups are the same size.



Shortcut

Use the alligation shortcut: the ratio of group sizes equals the inverse ratio of how far each group's mean is from the combined mean — $n/12 = (1244-1180)/(1340-1244) = 64/96 = 2/3$, so $n = 8$ immediately.



Trap

It's tempting to pair each average with whichever group size sits nearest it out of habit; here that means matching 1340 with 12 and 1180 with n , which silently reverses the whole equation and produces $n = 18$ instead of the correct $n = 8$.



Desmos Tip

Graph $y = (14160 + 1340x)/(12 + x)$ and $y = 1244$ on the built-in calculator and read the x -value where they intersect to confirm $x = 8$.



Process of Elimination

Correct: 8 students

Tempting wrong answer	Exactly what went wrong
18	Swapped which average belongs to which group size, solving $12(1340) + n(1180) = 1244(12+n)$ instead of $12(1180) + n(1340) = 1244(12+n)$.
20	Correctly solved for the total number of students across both groups combined ($12 + 8 = 20$) but reported that total instead of isolating just Group B's count.
6	Reached the correct equation $96n = 768$ but divided 768 by 128 instead of 96 (a copying slip on the coefficient), getting $n = 6$.

Problem 2

Very Hard

A math club in Vienna gives a 150-point placement test. Six students have already been scored, from lowest to highest: 62, 74, 81, 85, 90, 96. A seventh student, whose score is higher than all six of these, is added to the group. Once the seventh score is included, the mean of all seven scores equals the median of all seven scores plus 2. What is the seventh student's score?

Formula / Rule

Median of an odd-sized list is the single middle value once sorted; mean = sum / count.



First move

When a data set has an odd number of values and a new value is added only at the very top or bottom, find the median first — it often doesn't move at all, which simplifies the rest of the problem.

Solution

1. Since the new score is higher than all six existing scores, the sorted list of seven scores is 62, 74, 81, 85, 90, 96, x .
2. With 7 values, the median is the 4th value in the sorted list, which is 85 — this stays 85 no matter how large x is, as long as x remains the largest.
3. Set up the mean: $(62+74+81+85+90+96+x)/7 = (488+x)/7$.
4. The condition says mean = median + 2, so $(488+x)/7 = 85 + 2 = 87$.
5. Multiply both sides by 7: $488 + x = 609$.
6. Solve for x : $x = 609 - 488 = 121$.

**Answer: 121 points**



Problem 2 — Strategy & Elimination



Tip

When a data set has an odd number of values and a new value is added only at the very top or bottom, find the median first — it often doesn't move at all, which simplifies the rest of the problem.



Shortcut

Since the median (85) is fixed and unaffected by x , you can jump straight to $\text{mean} = 87$ and solve the one-line equation $488 + x = 609$ without extra setup.



Trap

Treating the 7-score set like an even-sized list and averaging the 3rd and 4th values (81 and 85) instead of taking the single 4th value (85) is exactly the mistake this problem is built to catch.



Desmos Tip

Type `mean({62,74,81,85,90,96,121})` into the calculator to confirm it equals 87, and `median(...)` of the same list to confirm 85.



Process of Elimination

Correct: **121 points**

Tempting wrong answer	Exactly what went wrong
107	Treated the 7-value set as if it were even-sized, averaging the 3rd and 4th values (81 and 85) to get a 'median' of 83, then solved $(488 + x)/7 = 85$ instead of 87.
34	Used the correct median (85) and required mean (87), but divided by 6 instead of 7, forgetting the new score makes this a seven-value set, not six.
609	Correctly computed $7 \times 87 = 609$ but reported this total sum as the final answer instead of subtracting 488 to isolate x .

Problem 3

Hard

At a vineyard on Santorini, two harvesting teams pick grapes. Team A has 4 workers, and Team B has 2 workers. The total kilograms picked by Team A is triple the total kilograms picked by Team B. Across all 6 workers combined, the mean amount picked per worker is 12 kg. What is the mean amount picked per worker on Team A?

Formula / Rule

Mean = total sum / count; translate a 'triple the sum' sentence into an equation between the two totals before dividing.



First move

Whenever a ratio is described between two group TOTALS (not group means), define variables for the totals first and translate the sentence into an equation — don't average anything until the totals are set up.

Solution

1. Total kilograms picked by all 6 workers = $6 \times 12 = 72$ kg.
2. Let S_b = Team B's total kilograms (2 workers). Team A's total is triple that: $S_a = 3 \times S_b$.
3. Since $S_a + S_b = 72$, substitute: $3S_b + S_b = 72$, so $4S_b = 72$.
4. Solve: $S_b = 18$ kg (Team B's total).
5. Then $S_a = 3 \times 18 = 54$ kg (Team A's total).
6. Mean per worker on Team A = $S_a / 4 = 54/4 = 13.5$ kg.

**Answer: 13.5 kg**



Problem 3 — Strategy & Elimination



Tip

Whenever a ratio is described between two group TOTALS (not group means), define variables for the totals first and translate the sentence into an equation — don't average anything until the totals are set up.



Trap

It's easy to misread 'the sum of four numbers is triple the sum of two others' backwards, setting Team B's total as triple Team A's total instead of the other way around.



Shortcut

Since total = 4 x Sb (because Sa is 3Sb and Sa+Sb=4Sb), solve $S_b = 72/4 = 18$ in one line without ever writing $S_a+S_b=72$ as a separate step.



Desmos Tip

This is a two-line arithmetic problem — just type $72/4$ and then $3 \cdot 18$ directly into the calculator as a fast check rather than graphing anything.



Process of Elimination

Correct: 13.5 kg

Tempting wrong answer	Exactly what went wrong
4.5 kg	Flipped which group's total is triple the other, setting $S_b = 3 \times S_a$ instead of $S_a = 3 \times S_b$, which makes Team A's total come out as only 18 kg instead of 54 kg.
9 kg	Correctly found Team A's total (54 kg) but divided by 6 (all workers) instead of 4 (just Team A's workers) to get the mean.
11.25 kg	Correctly solved $S_b = 18$, but then miscalculated $S_a = 3 \times 18$ as 45 instead of 54, giving a final mean of $45/4 = 11.25$ kg.

Problem 4

Hard

A real-estate site lists the monthly rent (in euros) for 7 available apartments in different Barcelona neighborhoods, along with each apartment's floor number: Floor 6, €800; Floor 3, €600; Floor 12, €1400 (a renovated penthouse); Floor 8, €720; Floor 1, €760; Floor 5, €640; Floor 2, €680. The agency removes the €1400 penthouse listing as an outlier before analyzing typical rent in the area. By how many more euros does the mean change than the median when this listing is removed?

Formula / Rule

Mean = sum/count; median = middle value (or average of the two middle values); outliers pull the mean far more than the median.



First move

Recompute both the mean and median separately after removing a data point — don't assume they shift by the same amount, and remember that going from an odd count to an even count turns the median into an average of two values.

Solution

- Sort the 7 rents from least to greatest (floor numbers are extra information and don't affect any calculation): 600, 640, 680, 720, 760, 800, 1400.
- Mean of all 7: $\text{sum} = 600 + 640 + 680 + 720 + 760 + 800 + 1400 = 5600$, so $\text{mean} = 5600/7 = 800$.
- Median of all 7 (the 4th value in the sorted list) = 720.
- Remove the outlier, 1400: the remaining 6 sorted rents are 600, 640, 680, 720, 760, 800.
- New mean: $\text{sum} = 4200$, $\text{mean} = 4200/6 = 700$. The mean dropped from 800 to 700, a change of 100.
- New median: average of the 3rd and 4th values, $(680 + 720)/2 = 700$. The median dropped from 720 to 700, a change of only 20.
- Difference between how much each measure changed: $100 - 20 = 80$.



Answer: €80 (the mean drops by €100 while the median drops by only €20)



Problem 4 — Strategy & Elimination



Tip

Recompute both the mean and median separately after removing a data point — don't assume they shift by the same amount, and remember that going from an odd count to an even count turns the median into an average of two values.



Shortcut

You don't need to recompute the mean from scratch — new mean = (old mean x n - removed value)/(n-1) = (800x7-1400)/6 = 4200/6 = 700 directly.



Trap

Assuming the median won't change at all when an outlier is removed is the single biggest trap here — the median almost always shifts slightly, just far less than the mean does.



Desmos Tip

Type both six- and seven-value lists into the calculator (e.g. L1={600,640,680,720,760,800}) and use mean(L1) and median(L1) to instantly confirm both new values.



Process of Elimination

Correct: €80 (the mean drops by €100 while the median drops by only €20)

Tempting wrong answer	Exactly what went wrong
€100	Computed only the change in the mean and reported that, ignoring that the question asks for the DIFFERENCE between the mean's change and the median's change.
€60	Mis-computed the new median of the six-value list as a single middle value (the 3rd value, 680) instead of averaging the two middle values, giving a wrong median change of 40 instead of 20 (100-40=60).
€180	Forgot to reduce the denominator when computing the new mean, dividing 4200 by 7 instead of 6, getting a wrong mean of 600 and an inflated mean-change of 200 (200-20=180).

Problem 5

Very Hard

A startup in Berlin has 9 employees, and their mean salary is €52,000. The founder's salary, €148,000, is far above everyone else's and is treated as an outlier. If the founder's salary is excluded, what is the mean salary of the remaining 8 employees?

Formula / Rule

Mean = sum/count; removing a value changes both the total AND the count, so recompute both.

First move

Turn the mean back into a total first (mean $\times$ count), do the subtraction on the total, and only divide by the new count at the very end.

Solution

1. Convert the mean into a total: total payroll for 9 employees = $9 \times 52,000 = €468,000$.
2. Identify the founder's salary as the value to be removed: €148,000.
3. Subtract the founder's salary from the total: $468,000 - 148,000 = €320,000$ (this is now the total for the remaining employees).
4. Update the count: with the founder excluded, only 8 employees remain, not 9.
5. Divide the new total by the new count: $320,000 / 8 = €40,000$.
6. Sanity check: since the founder's salary was far above the original mean, removing it should pull the mean down substantially — €40,000 < €52,000 confirms the direction makes sense.

 **Answer:** €40,000



Problem 5 — Strategy & Elimination



Tip

Turn the mean back into a total first (mean x count), do the subtraction on the total, and only divide by the new count at the very end.



Shortcut

new mean = (old mean x n - removed value)/(n-1) = (52,000x9 - 148,000)/8 = 320,000/8 = 40,000 in one continuous calculation, skipping the separate 'find the old total' step.



Trap

The single biggest trap: after removing one employee's salary from the total, it's easy to forget the denominator also shrinks from 9 to 8, and re-divide by 9 out of habit.



Desmos Tip

Type (52000·9-148000)/8 directly into the built-in calculator to get 40000 in one line, confirming the arithmetic without writing out each step by hand.



Process of Elimination

Correct: €40,000

Tempting wrong answer	Exactly what went wrong
€35,555.56 (approximately €35,556)	Correctly subtracted the founder's salary from the total (320,000) but then divided by 9 instead of 8, forgetting the founder is no longer part of the group being averaged.
-€12,000	Subtracted the founder's salary directly from the mean instead of from the total sum: (52,000-148,000)/8 = -96,000/8 = -12,000.
€320,000	Correctly found the new total payroll but stopped there without dividing by the new count, reporting the total instead of the mean.

Problem 6

Hard

A retail chain tracks one week's sales (in thousands of euros) at 6 branches across Europe. When sorted from least to greatest, the figures form the list r , $r+9$, $r+18$, $r+27$, $r+36$, and s , where r is the smallest branch's sales and s is the largest branch's sales. The range of the six figures is 45 (thousand euros). Separately, the finance report notes that s is also equal to 3 more than double r . What is the mean weekly sales, in thousands of euros, across all 6 branches?

Formula / Rule

Range = largest - smallest; combine with a second equation for the same unknown, then compute mean = sum/count.



First move

When two separate statements both describe the same unknown value (here, the largest branch s), set the two expressions equal to each other immediately instead of solving them one at a time.

Solution

1. By the definition of range, $s - r = 45$, so $s = r + 45$.
2. The finance report gives a second equation for the same value: $s = 2r + 3$.
3. Set the two expressions for s equal to each other: $r + 45 = 2r + 3$.
4. Solve for r : $45 - 3 = 2r - r$, so $r = 42$.
5. Find s using either equation: $s = 42 + 45 = 87$ (check: $2(42)+3 = 87$, confirmed).
6. List all six branch sales figures: $r=42$, $r+9=51$, $r+18=60$, $r+27=69$, $r+36=78$, and $s=87$.
7. Sum the six figures: $42+51+60+69+78+87 = 387$.
8. Mean = $387/6 = 64.5$.



Answer: 64.5 thousand euros (€64,500)



Problem 6 — Strategy & Elimination



Tip

When two separate statements both describe the same unknown value (here, the largest branch s), set the two expressions equal to each other immediately instead of solving them one at a time.



Shortcut

The five evenly-spaced smaller terms ($r, r+9, \dots, r+36$) sum to $5r+90$; plug in $r=42$ to get 300 instantly, then just add the largest term $s=87$ for a total of 387 — no need to add six separate numbers one at a time.



Trap

Because finding r takes several steps on its own, it's tempting to treat $r=42$ as the final answer — but the question never asked for r , it asked for the mean of the full six-branch list.



Desmos Tip

Enter the list $L1=\{42,51,60,69,78,87\}$ into the calculator and use `mean(L1)` to instantly confirm 64.5.



Process of Elimination

Correct: **64.5 thousand euros (€64,500)**

Tempting wrong answer	Exactly what went wrong
42 (thousand euros)	Solved correctly for r but stopped there, submitting the smallest branch's sales as if it were the final mean instead of continuing to build the list and average it.
-40.5 (thousand euros)	Misapplied the range definition with the difference reversed, using $s = r - 45$ instead of $s = r + 45$, which flips the sign of every downstream value (r becomes -48, s becomes -93).
61.8 (thousand euros)	Correctly found $r=42$ and $s=87$ but dropped one middle term ($r+36=78$) when listing the six branches, averaging only 5 values (42,51,60,69,87) instead of all 6.

Problem 7

Very Hard

At a Lyceum in Patras, a student's overall course grade is weighted: coursework counts 25%, the midterm counts 35%, and the final exam counts 40%. She has an 82 in coursework and a 78 on the midterm. What score does she need on the final exam to earn an overall grade of exactly 85?

Formula / Rule

Weighted average: sum of (decimal weight x component score) = target overall grade.

First move

Convert every percent weight to a decimal and multiply each by its matching score before adding anything — treat the unknown final term as the last piece to isolate.

Solution

1. Convert each percentage weight to a decimal: coursework 25% = 0.25, midterm 35% = 0.35, final exam 40% = 0.40.
2. Set up the weighted average equation: $0.25(82) + 0.35(78) + 0.40(F) = 85$, where F is the unknown final exam score.
3. Compute the known terms: $0.25(82) = 20.5$ and $0.35(78) = 27.3$.
4. Add the known terms: $20.5 + 27.3 = 47.8$.
5. Substitute back into the equation: $47.8 + 0.40F = 85$.
6. Isolate the term with F: $0.40F = 85 - 47.8 = 37.2$.
7. Solve for F: $F = 37.2 / 0.40 = 93$.

 **Answer: 93**



Problem 7 — Strategy & Elimination



Tip

Convert every percent weight to a decimal and multiply each by its matching score before adding anything — treat the unknown final term as the last piece to isolate.



Shortcut

Since coursework and midterm together already lock in 47.8 fixed points toward the 85 target, immediately compute the 'points still needed,' $85 - 47.8 = 37.2$, and divide by the final's weight — no need to write the full three-term equation symbolically first.



Trap

Seeing three numbers (82, 78, and the target 85) listed side by side tempts a plain three-way average; the 25/35/40 weights must be used as multipliers, not ignored.



Desmos Tip

Graph $y = 0.25(82) + 0.35(78) + 0.40x$ and $y = 85$ on the built-in calculator and find the x-coordinate of their intersection to confirm $F = 93$.



Process of Elimination

Correct: 93

Tempting wrong answer	Exactly what went wrong
95	Treated the three grade components as equally weighted, computing a plain average $(82+78+F)/3=85$ instead of using the given 25/35/40 percent weights.
93.75	Rounded 0.35×78 to 27 instead of keeping the exact value 27.3, so solved $20.5+27+0.40F=85$, giving $0.40F=37.5$ and $F=93.75$.
14.88	Correctly reached $0.40F=37.2$, but then multiplied by 0.40 instead of dividing, computing $F=37.2 \times 0.40=14.88$.

Problem 8

Very Hard

Two coding bootcamps each timed 7 students solving the same practice problem set, in minutes. Amsterdam's times: 10, 18, 20, 20, 20, 22, 30. Copenhagen's times: 10, 15, 17, 20, 23, 25, 30. Both groups have the same mean (20 minutes) and the same range (20 minutes). Without calculating the exact standard deviation, which group's completion times have the greater standard deviation?

Formula / Rule

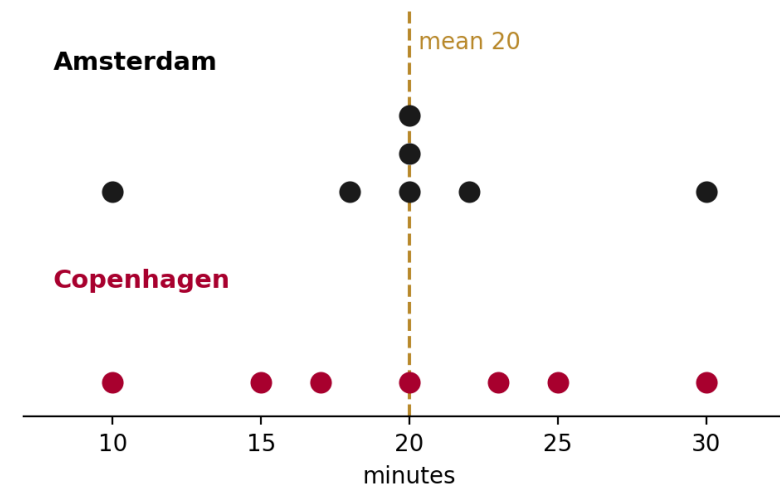
Standard deviation measures the typical distance of ALL data points from the mean — not just the distance between the minimum and maximum (the range).

Solution

1. Confirm both means are equal: Amsterdam's sum = $10+18+20+20+20+22+30 = 140$, mean = $140/7 = 20$. Copenhagen's sum = $10+15+17+20+23+25+30 = 140$, mean = $140/7 = 20$. Both means are 20 minutes.
2. Confirm both ranges are equal: Amsterdam's range = $30-10 = 20$. Copenhagen's range = $30-10 = 20$. Since the ranges match, range alone cannot distinguish the two groups' spread.
3. List each value's deviation from the mean (20) for Amsterdam: -10, -2, 0, 0, 0, 2, 10.
4. List each value's deviation from the mean (20) for Copenhagen: -10, -5, -3, 0, 3, 5, 10.
5. Compare the two lists: five of Amsterdam's seven values sit exactly at, or within 2 minutes of, the mean, with only the two extremes (10 and 30) far away — while Copenhagen's values are spread more evenly across the entire range, with several (15, 17, 23, 25) sitting well away from the mean.

✓ **Answer: Copenhagen**

7. Check with the sum of squared deviations (proportional to variance): Amsterdam: $100+4+0+0+0+4+100 = 208$. Copenhagen: $100+25+9+0+9+25+100=268$. Since $268 > 208$, this confirms Copenhagen's spread is greater.





Problem 8 — Strategy & Elimination



Tip

When two data sets share the same mean and range, look past the two extreme values and compare how closely the remaining (interior) values cluster around the mean.



Trap

Assuming equal range means equal spread is the exact trap here — Amsterdam's tight cluster of five values right at the mean actually makes it LESS spread out overall than Copenhagen's more evenly distributed values, despite both having the same min and max.



Shortcut

No numeric shortcut beats comparing the deviations directly — since both sets share the same mean, listing each value's distance from 20 and comparing which set has more/larger interior deviations is already the fastest legitimate method.



Process of Elimination

Correct: **Copenhagen**

Tempting wrong answer	Exactly what went wrong
Amsterdam	Reasoned that having two dramatic outliers (10 and 30) sitting alone must create more spread than Copenhagen's smoother distribution, without actually comparing how far the typical (non-extreme) values sit from the mean.
They have equal standard deviation	Concluded that because both groups share the same mean AND the same range, their spread must be identical, incorrectly treating range as if it were the same thing as standard deviation.
Cannot be determined without computing the exact standard deviation	Assumed that comparing spread requires the full standard deviation formula, when comparing how tightly the interior values cluster around the mean already settles the comparison.

05

Data Displays — Histograms, Dot Plots & Box Plots

Reading frequency displays and five-number summaries.

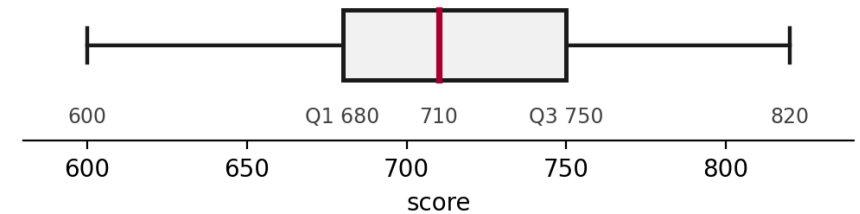
In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

Center A (dot plot)



Center B (box plot)

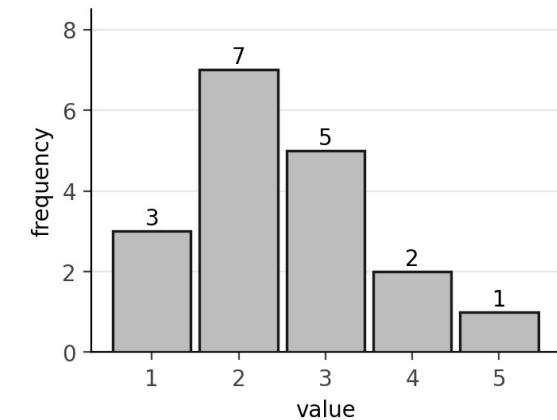


Key Terminology

Term (English)	Ελληνικά
Histogram	Ιστόγραμμα
Dot plot	Διάγραμμα κουκκίδων
Box plot	Διάγραμμα κουτιού (θηκόγραμμα)
Five-number summary	Σύνοψη πέντε αριθμών
Frequency	Συχνότητα

A **ἄ** The big picture

- Each display hides the raw list in a different way.
- Frequency = how many times a value occurs.
- Box plots show position, not individual values.



Theory Explained (1/3)

Dot Plots & Frequency Tables

$$\text{sum} = \Sigma(\text{value} \times \text{frequency}) \quad n = \Sigma \text{frequency}$$

WHAT IT MEANS

- Each dot = one data point.
- Multiply each value by its count before adding.
- An unknown frequency changes BOTH the sum and n.



Example

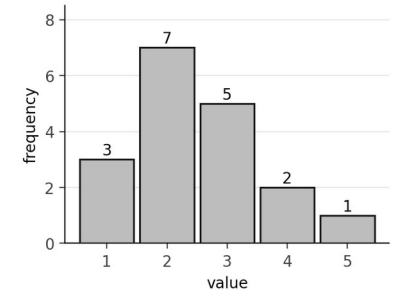
$$(0 \cdot 6 + 10 \cdot 8 + 20n + 30 \cdot 3 + 40 \cdot 2) / (19 + n) = 14.8 \rightarrow n = 6.$$

Histograms

$$\text{bar height} = \text{frequency}$$

WHAT IT MEANS

- The median is found by counting, not by looking.
- Use cumulative totals to reach the middle position.
- Bars can cover ranges — values inside are unknown.



Example

20 values → average the 10th and 11th.

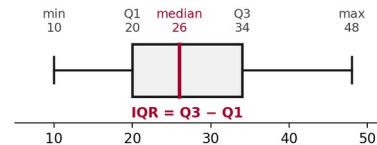
Theory Explained (2/3)

Box Plots

min · Q1 · median · Q3 · max

WHAT IT MEANS

- Each section holds about 25% of the data.
- $IQR = Q3 - Q1$ (spread of the middle half).
- $Range = max - min$.



Example

$Q1 = 68, Q3 = 84 \rightarrow IQR = 16$; 25% scored above 84.

Quartiles by Hand

split around the median (exclude it when n is odd)

WHAT IT MEANS

- Sort the data first.
- $Q1$ = median of the lower half.
- $Q3$ = median of the upper half.



Example

11 values: $Q1 = 3rd$ value, $Q3 = 9th$ value.

Theory Explained (3/3)

Outlier Fences

below $Q1 - 1.5 \cdot IQR$ or above $Q3 + 1.5 \cdot IQR$

WHAT IT MEANS

- Compute IQR first.
- Compare the value to the fence, not to the old minimum.
- A new low value is not automatically an outlier.



Example

$Q1 = 71$, $IQR = 17 \rightarrow \text{fence} = 45.5 \rightarrow 47$ is not an outlier.

Watch Out



Height is not position

- The tallest stack of dots is the mode, not the maximum.
- Read values from the horizontal axis.



Box plots hide counts

- You know percentages (25% per section).
- You don't know individual values inside the box.

Problem 1

Hard

A dot plot shows the number of books read last month by 9 students in a reading club in Thessaloniki: 1 book (2 students), 2 books (3 students), 3 books (2 students), 4 books (1 student), and x books (1 student). If the mean number of books read by the group is 3, what is the value of x ?

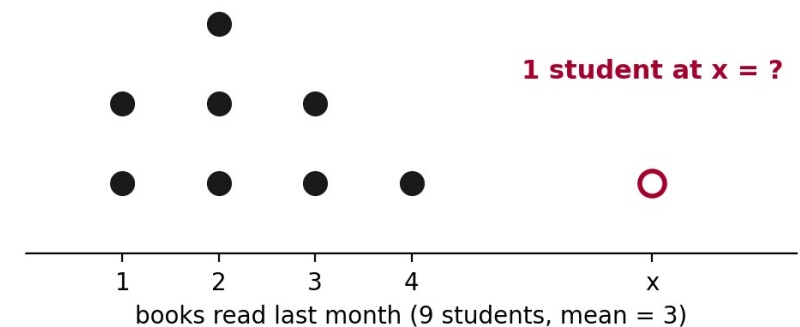
Formula / Rule

Mean = (sum of all values)/(number of values); solve for a missing data value.

Solution

- Count the total number of students represented: $2 + 3 + 2 + 1 + 1 = 9$ students.
- Multiply each known value by its frequency and add: $1(2) + 2(3) + 3(2) + 4(1) = 2 + 6 + 6 + 4 = 18$ (this is the sum of every student's books except the one who read x).
- Add the unknown student's contribution to get the full total: total books = $18 + x$.
- Set up the mean equation using all 9 students: $(18 + x)/9 = 3$.
- Multiply both sides by 9: $18 + x = 27$.
- Solve for x : $x = 27 - 18 = 9$.

✓ **Answer:** $x = 9$





Problem 1 — Strategy & Elimination



Tip

When a mean is given and one data value is missing, translate the mean formula into $\text{sum} = \text{mean} \times n$ first, then subtract the sum of everything you already know.



Shortcut

Skip writing out the frequency table twice — just compute the target total ($\text{mean} \times n = 27$) and subtract the sum of the 8 known dots (18) in one line: $x = 27 - 18 = 9$.



Trap

A rushed student forgets that the student who read x books is still one of the 9 people in the mean, and instead divides by 8 (the count of students whose value is already known), which produces the wrong x .



Desmos Tip

Type $(18+x)/9=3$ into the calculator's equation solver to confirm $x = 9$ in one step.



Process of Elimination

Correct: **$x = 9$**

Tempting wrong answer	Exactly what went wrong
$x = 6$	Forgot that the student with x books counts toward n , used $n = 8$ instead of 9, and solved $(18+x)/8 = 3$.
$x = 27$	Set x equal to $\text{mean} \times n$ ($3 \times 9 = 27$) directly, forgetting to subtract the sum of the 8 known values (18) first.
$x = 17$	Added only the four distinct dot-plot values ($1+2+3+4 = 10$) instead of value $\times$ frequency (18), then solved $x = 27 - 10 = 17$.

Problem 2

Hard

A histogram shows the number of goals scored by a football (soccer) club in each of its 20 matches this season: bars of height 4 over 0 goals, height 6 over 1 goal, height 5 over 2 goals, height 3 over 3 goals, and height 2 over 4 goals. What is the median number of goals scored per match?

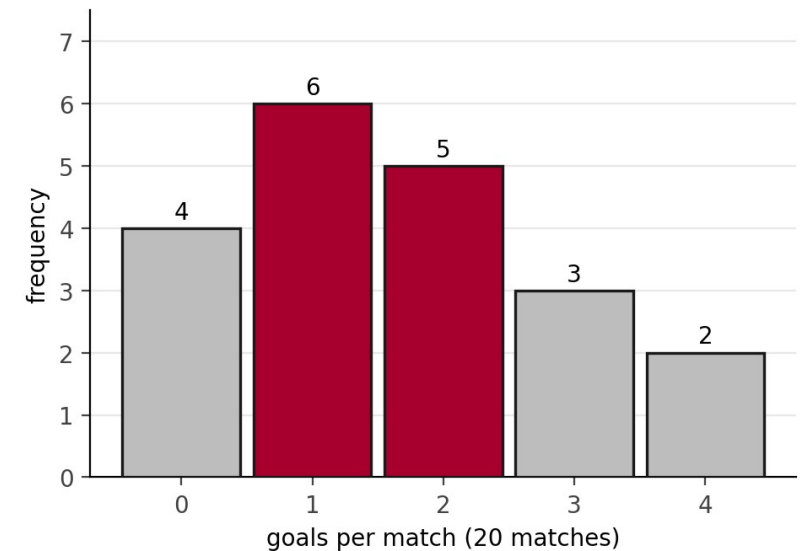
Formula / Rule

Median from a frequency table = average of the two middle values, located via cumulative frequency (when n is even).

Solution

1. Total matches: $4 + 6 + 5 + 3 + 2 = 20$, so n is even and the median is the average of the 10th and 11th values once ordered.
2. Build cumulative frequency: 0 goals covers positions 1-4; 1 goal covers positions 5-10 ($4+6=10$); 2 goals covers positions 11-15 ($10+5=15$); 3 goals covers positions 16-18; 4 goals covers positions 19-20.
3. The 10th position is the last position in the '1 goal' group, so the 10th value = 1.
4. The 11th position is the first position in the '2 goals' group, so the 11th value = 2.
5. Median = $(10\text{th value} + 11\text{th value})/2 = (1+2)/2 = 1.5$.

✓ **Answer: Median = 1.5 goals**





Problem 2 — Strategy & Elimination



Tip

When a display groups data by value instead of listing raw data, find the median by tracking cumulative frequency to the exact middle position(s) — never just read off whichever bar looks centrally located.



Trap

Because '2 goals' is the visually central bar, it's tempting to call it the median directly — but cumulative counting shows the true middle falls between the '1 goal' and '2 goals' groups, giving 1.5, not 2.



Shortcut

Since $n = 20$ is even, you only need to track the running cumulative total until it first passes 10 — once it does, you've identified both middle positions without listing all 20 data points individually.



Process of Elimination

Correct: **Median = 1.5 goals**

Tempting wrong answer	Exactly what went wrong
2	Picked the bar that looks centrally located among the five categories instead of counting cumulative frequency to the actual 10th and 11th positions.
1	Correctly found the 10th value (1) but stopped there, forgetting that $n = 20$ is even and the median requires averaging the 10th AND 11th values.
1.65	Computed the mean $(0(4)+1(6)+2(5)+3(3)+4(2))/20 = 33/20 = 1.65$ instead of the median.

Problem 3

Very Hard

A box plot summarizes the exam scores (out of 100) of 31 students in a Digital SAT prep class in Vienna: minimum = 58, Q1 = 71, median = 79, Q3 = 88, maximum = 96. A 32nd student transfers in with a score of 47. Using the standard rule that a value is an outlier if it falls more than $1.5 \times \text{IQR}$ below Q1 or above Q3, find the lower fence and determine whether the new student's score is an outlier.

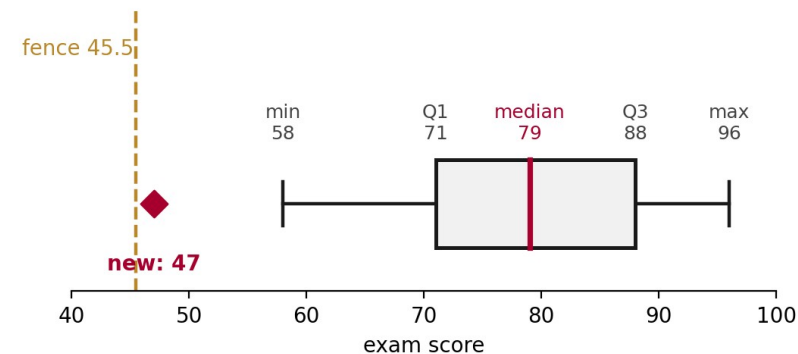
Formula / Rule

$\text{IQR} = Q3 - Q1$; outlier fences = $Q1 - 1.5(\text{IQR})$ and $Q3 + 1.5(\text{IQR})$.

Solution

- Find the IQR from the box plot: $\text{IQR} = Q3 - Q1 = 88 - 71 = 17$.
- Multiply by 1.5: $1.5 \times 17 = 25.5$.
- Find the lower fence: lower fence = $Q1 - 25.5 = 71 - 25.5 = 45.5$.
- Compare the new student's score to the lower fence: 47 vs. 45.5.
- Since $47 > 45.5$, the score of 47 lies inside the fence, not beyond it.
- Conclude: the new student's score of 47 is NOT an outlier by the $1.5 \times \text{IQR}$ rule, even though it is well below the original minimum of 58.

✓ **Answer:** Lower fence = 45.5; 47 is NOT an outlier ($47 > 45.5$).





Problem 3 — Strategy & Elimination



Tip

To test for an outlier, always compute the actual fence ($Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) before comparing — never judge 'extreme-ness' just by comparing to the box plot's own minimum or maximum.



Trap

Because 47 is lower than the original minimum of 58, it looks obviously extreme — but the fence calculation shows 47 still falls inside the allowed range, so the intuitive answer (outlier) is wrong.



Shortcut

Once you know Q1 and IQR, compute the fence in one line — $Q1 - 1.5 \times IQR$ — and just check whether 47 clears it; no need to re-sort or re-derive the whole distribution.



Process of Elimination

Correct: Lower fence = 45.5; 47 is NOT an outlier ($47 > 45.5$).

Tempting wrong answer	Exactly what went wrong
Outlier (fence = 54)	Computed the fence as $Q1 - IQR$ ($71 - 17 = 54$) instead of $Q1 - 1.5 \times IQR$, forgetting the 1.5 multiplier entirely.
Outlier, since 47 is below the original minimum of 58	Assumed any new value below the group's previous minimum must automatically be an outlier, without computing a fence at all.
Outlier (fence = 53.5)	Used the median (79) instead of Q1 (71) as the base for the lower fence, computing $79 - 25.5 = 53.5$ and comparing 47 to that.

Problem 4

Very Hard

Two Digital SAT prep centers in Athens gave the same practice test. Center A's results are shown as a dot plot: 620 (1 student), 660 (2 students), 700 (1 student), 740 (4 students), and 780 (2 students). Center B's results are shown as a box plot based on 10 students: minimum = 600, Q1 = 680, median = 710, Q3 = 750, maximum = 820. Which center has the greater median score, and which center has the greater range?

Formula / Rule

Median and range from a dot plot (order the data by frequency) versus reading the five-number summary directly from a box plot.

Solution

1. List Center A's 10 dot-plot values in increasing order: 620, 660, 660, 700, 740, 740, 740, 740, 780, 780.
2. Since $n = 10$ is even, Center A's median = average of the 5th and 6th values = $(740 + 740) / 2 = 740$.
3. Center A's range = $\text{max} - \text{min} = 780 - 620 = 160$.
4. Center B's statistics are read directly from the box plot: median = 710, and range = $\text{max} - \text{min} = 820 - 600 = 220$.
5. Compare medians: 740 (Center A) > 710 (Center B), so Center A has the greater median.
6. Compare ranges: 220 (Center B) > 160 (Center A), so Center B has the greater range.

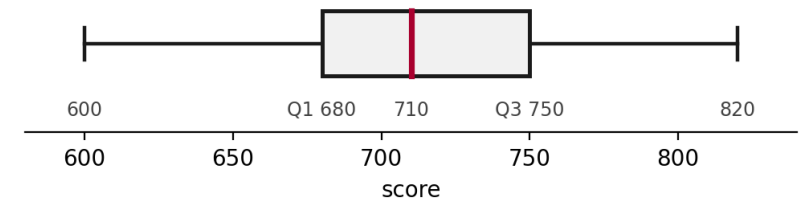


Answer: Center A has the greater median (740 vs. 710); Center B has the greater range (220 vs. 160).

Center A (dot plot)



Center B (box plot)





Problem 4 — Strategy & Elimination



Tip

When comparing distributions shown in different display types, extract the SAME statistic for each display using its own reading rules before comparing — never assume one group dominates on every measure just because of one eye-catching value.



Trap

Center B's box plot has the single highest score of the two groups (820), which tempts students into assuming Center B must also have the higher median — but Center A's dot plot is heavily weighted toward its high end (four students at 740, two at 780), which actually gives Center A the higher median despite its lower maximum.



Shortcut

Center B's five-number summary is already given directly from the box plot, so only Center A needs to be built from the raw dot-plot data by hand.



Desmos Tip

Enter Center A's 10 scores as a list $L1=\{620,660,660,700,740,740,740,740,780,780\}$ in the calculator and use $\text{median}(L1)$ and $\text{max}(L1)-\text{min}(L1)$ to instantly check the median (740) and range (160) against your hand sort.



Process of Elimination

Correct: **Center A has the greater median (740 vs. 710); Center B has the greater range (220 vs. 160).**

Tempting wrong answer	Exactly what went wrong
Center B has the greater median and the greater range.	Assumed the group with the higher maximum score (Center B, 820) automatically has the higher median too, without actually sorting and averaging Center A's dot-plot values.
Center A's median = 700	Read the middle of the five DISTINCT dot-plot values (620, 660, 700, 740, 780) instead of sorting all 10 individual data points and averaging the true 5th and 6th values.
Center A's median = 720	Averaged the 4th and 5th data values (700 and 740) instead of the 5th and 6th (740 and 740).

Problem 5

Very Hard

A ferry company tracked delay times (rounded to the nearest 10 minutes) for a number of Piraeus–Santorini crossings last month, summarized in this frequency table: 0 min delay: 6 crossings; 10 min: 8 crossings; 20 min: n crossings; 30 min: 3 crossings; 40 min: 2 crossings. If the mean delay across all crossings was 14.8 minutes, what is the value of n ? Then, using that value of n , find the median delay.

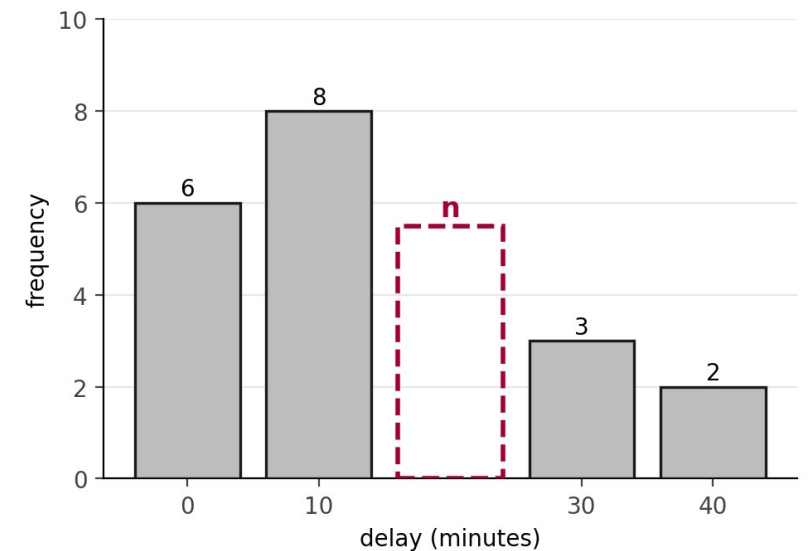
Formula / Rule

Weighted mean = sum of (value $\times$ frequency)/sum of frequencies; median via cumulative frequency.

Solution

- Let n be the unknown frequency for the 20-minute row. Total crossings = $6+8+n+3+2 = 19+n$.
- Total delay-minutes = $0(6)+10(8)+20n+30(3)+40(2) = 0+80+20n+90+80 = 250+20n$.
- Set up the mean equation: $(250+20n)/(19+n) = 14.8$.
- Cross-multiply: $250+20n = 14.8(19+n) = 281.2+14.8n$.
- Collect the n terms on one side: $20n-14.8n = 281.2-250$, so $5.2n = 31.2$.
- Solve: $n = 31.2/5.2 = 6$.
- Total crossings = $19+6 = 25$, which is odd, so the median is the 13th crossing once sorted.
- Cumulative frequency: 0 min covers positions 1–6; 10 min covers positions 7–14 ($6+8=14$) — the 13th position falls inside the 10–

✓ **Answer:** $n = 6$; median delay = 10 minutes.





Problem 5 — Strategy & Elimination



Tip

When a frequency is unknown, let the TOTAL count also be an expression in terms of that unknown (here, $19+n$) — don't assume the total is fixed unless the problem states it directly.



Shortcut

No faster path than setting up the mean equation directly — but if solving ever produces a fractional or negative n , that's a built-in red flag that a setup mistake was made, since a frequency must be a positive whole number.



Trap

A rushed student fixes the denominator at 19 (the sum of the known frequencies only) instead of $19+n$, forgetting that the missing frequency also adds to the total number of crossings.



Desmos Tip

Graph $y=(250+20x)/(19+x)$ and $y=14.8$ in the calculator and read the intersection's x -value, or use the equation solver on $(250+20n)/(19+n)=14.8$, to confirm $n=6$.



Process of Elimination

Correct: $n = 6$; median delay = 10 minutes.

Tempting wrong answer	Exactly what went wrong
$n \approx 1.56$	Kept the denominator fixed at 19 instead of $19+n$, solving $(250+20n)/19 = 14.8$, which ignores that the unknown frequency also adds to the total crossing count.
$n = 6$, median = 20 minutes	Correctly solved for n but then found the median by picking the value in the middle ROW of the 5-row table (the 3rd of 5 delay categories, 20 min) instead of counting cumulative frequency to the true 13th crossing.
$n \approx -2.26$	Wrote the total delay-minutes as $250+n$ instead of $250+20n$, forgetting to multiply the 20-minute row's delay by its own frequency n before adding it into the sum.

Problem 6

Hard

At a Model United Nations conference in Brussels, the scores of 28 delegates on a 100-point policy memo (graded on a 40-item rubric) are shown in a box plot: minimum = 54, Q1 = 68, median = 76, Q3 = 84, maximum = 91. Delegates spent an average of 3 hours writing the memo. What is the interquartile range of the scores, and how many delegates scored above 84?

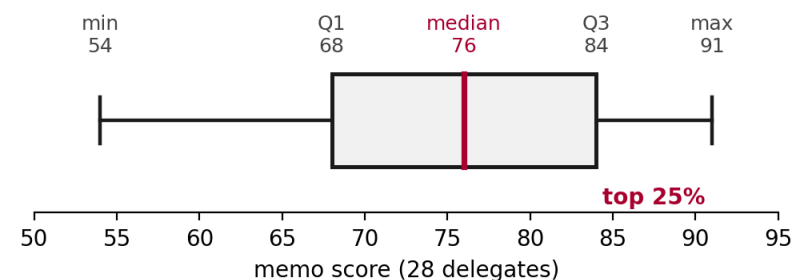
Formula / Rule

$IQR = Q3 - Q1$; each quartile of a box plot represents 25% of the data set.

Solution

1. Identify the irrelevant details: the memo's 100-point scale, the 40-item rubric, and the 3-hour average writing time don't affect the box plot's summary statistics — set them aside.
2. Find Q1 and Q3 directly from the box plot: $Q1 = 68$, $Q3 = 84$.
3. Compute the interquartile range: $IQR = Q3 - Q1 = 84 - 68 = 16$.
4. Recall the quartile definition: each quarter of a box plot (below Q1, Q1-to-median, median-to-Q3, above Q3) always contains exactly 25% of the data, regardless of the actual score values.
5. Since 'above 84' means above Q3, that region is exactly the top 25% of delegates.
6. Convert to a count: 25% of 28 delegates = $0.25 \times 28 = 7$ delegates.

✓ **Answer:** $IQR = 16$; 25% of delegates (7 delegates) scored above 84.





Problem 6 — Strategy & Elimination



Tip

Before touching any numbers in a box-plot word problem, identify which facts actually belong to the five-number summary and discard details that only describe the scenario.



Trap

The '100-point, 40-item' memo details tempt students into computing a percentage relative to the point scale (like $84/100 = 84\%$) instead of recognizing that 'above Q3' is always exactly 25% of the data by definition, independent of the score scale.



Shortcut

No calculation is needed for the percentage part — by definition each quartile always holds 25% of the data, so 'above Q3' is always 25% regardless of the actual scores; just multiply 0.25 by the total count.



Process of Elimination

Correct: **IQR = 16; 25% of delegates (7 delegates) scored above 84.**

Tempting wrong answer	Exactly what went wrong
IQR = 37	Computed the full range ($\text{max} - \text{min} = 91 - 54 = 37$) instead of the interquartile range ($Q3 - Q1$).
84% of delegates scored above 84	Confused the raw score value at Q3 (84 out of 100 points) with the percentage of delegates above that score, treating the score itself as if it were a percentile.
14 delegates (50%) scored above 84	Assumed 'above Q3' means the entire upper half of the box plot (above the median) rather than just the top quarter above Q3, doubling the correct percentage.

Problem 7

Hard

In a chess tournament in Zagreb, the number of moves in each of 11 completed games was recorded in the order the games finished: 38, 31, 44, 28, 52, 36, 47, 31, 41, 34, 38. What is the interquartile range of the number of moves?

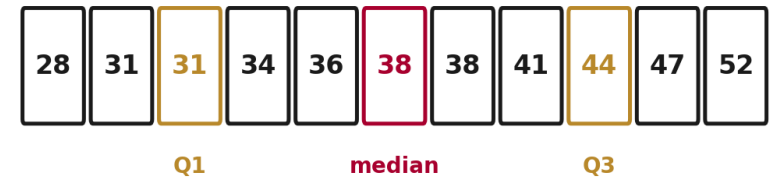
Formula / Rule

$IQR = Q3 - Q1$, found by splitting the ordered data at the median (excluding the median itself when n is odd).

Solution

- Sort the 11 move counts from least to greatest: 28, 31, 31, 34, 36, 38, 38, 41, 44, 47, 52.
- With $n = 11$ (odd), the overall median is the 6th value: 38.
- To find $Q1$, take the lower half of the data, EXCLUDING the median itself: 28, 31, 31, 34, 36 (5 values).
- $Q1$ is the median of this lower half, the 3rd value: $Q1 = 31$.
- To find $Q3$, take the upper half of the data, again EXCLUDING the median: 38, 41, 44, 47, 52 (5 values).
- $Q3$ is the median of this upper half, the 3rd value: $Q3 = 44$.
- $IQR = Q3 - Q1 = 44 - 31 = 13$.

✓ **Answer:** $IQR = 13$ ($Q1 = 31$, $Q3 = 44$)





Problem 7 — Strategy & Elimination



Tip

For odd n , always exclude the actual median value from BOTH the lower and upper halves before finding $Q1$ and $Q3$ — including it shifts both quartiles inward and changes the IQR.



Shortcut

Once sorted, you don't need to write out the two halves as separate lists — just count in 3 positions from each end (since each half has 5 values) to read off $Q1$ and $Q3$ directly.



Trap

The list is given in the order the games finished, not sorted — using position-in-the-original-list instead of position-in-sorted-order, or accidentally including the median in one or both halves, are the two most common ways students corrupt $Q1$ and $Q3$ here.



Desmos Tip

Enter the moves as a list $L1=\{28,31,31,34,36,38,38,41,44,47,52\}$ in the calculator and use `quartile(L1,1)` and `quartile(L1,3)` to instantly check $Q1$ and $Q3$ against your hand calculation.



Process of Elimination

Correct: **IQR = 13 ($Q1 = 31$, $Q3 = 44$)**

Tempting wrong answer	Exactly what went wrong
IQR = 10	Included the overall median (38) in both the lower and upper halves when splitting the data, instead of excluding it, which shifts $Q1$ up to 32.5 and $Q3$ down to 42.5.
IQR = 24	Computed the range ($\text{max} - \text{min} = 52 - 28 = 24$) instead of the interquartile range.
IQR = 11.5	Correctly excluded the median from the lower half ($Q1 = 31$) but accidentally included it in the upper half, computing $Q3$ as the average of 41 and 44 (42.5) instead of the correct 44.

Problem 8

Very Hard

A logistics company with distribution hubs in Rotterdam, Hamburg, and Gdańsk tracked the number of containers delayed at customs each day over 24 operating days. The results are shown in this frequency table: 0 delays: 9 days; 1 delay: 6 days; 2 delays: d days; 3 delays: 2 days; 5 delays: 1 day. The company's insurance policy waives its deductible for the month only if the mean number of daily delays over all 24 days is less than 1. First find the value of d , then determine whether the deductible is waived.

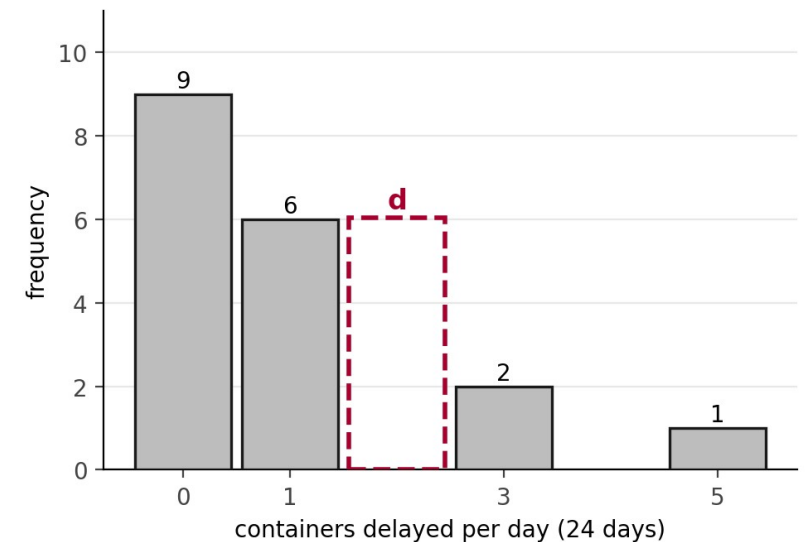
Formula / Rule

Weighted mean from a frequency table, used to test a real-world threshold condition.

Solution

- Find the missing frequency d using the total day count: $9+6+d+2+1 = 24$.
- Solve: $18+d = 24$, so $d = 6$.
- Compute the total delay-count sum: $0(9)+1(6)+2(6)+3(2)+5(1) = 0+6+12+6+5 = 29$.
- Compute the mean: $\text{mean} = 29/24 \approx 1.21$ delays per day.
- Compare to the policy threshold: the deductible is waived only if the mean is LESS THAN 1.
- Since 1.21 is not less than 1, the deductible is NOT waived.

✓ **Answer:** $d = 6$; mean ≈ 1.21 delays/day, so the deductible is NOT waived.





Problem 8 — Strategy & Elimination



Tip

Always solve for a missing frequency using the total-count equation FIRST (frequencies must sum to the stated total), then plug that value into the weighted-sum formula for the mean — don't try to solve both at once.



Shortcut

Since three known frequencies plus d must sum to 24, find d from one quick equation ($18+d=24$) before ever touching the weighted-mean formula.



Trap

It's tempting to add d directly into the delay-total sum instead of $2d$, since the '2 delays' row is easy to skim past — this understates the mean just enough to flip the real-world conclusion.



Desmos Tip

Once $d=6$ is known, type $(0\cdot9+1\cdot6+2\cdot6+3\cdot2+5\cdot1)/24$ directly into the calculator to get the mean in one line and check it against 1 before answering the waiver question.



Process of Elimination

Correct: $d = 6$; mean ≈ 1.21 delays/day, so the deductible is NOT waived.

Tempting wrong answer	Exactly what went wrong
Deductible IS waived (mean ≈ 0.96)	Added d directly into the total delay sum instead of $2d$ (forgot to multiply the '2 delays' row by its frequency d before summing), understating the mean just enough to drop it below the threshold of 1.
$d = 7$	Miscounted the known frequencies as 17 instead of 18 (e.g., dropped the single day with 5 delays), then solved $17+d=24$ to get $d=7$ instead of $d=6$.
Deductible IS waived (mean ≈ 0.40)	Correctly computed the mean as $29/24 \approx 1.21$ but then divided again by the 3 distribution hubs mentioned in the setup, wrongly treating the threshold as a per-hub average instead of the overall daily mean.

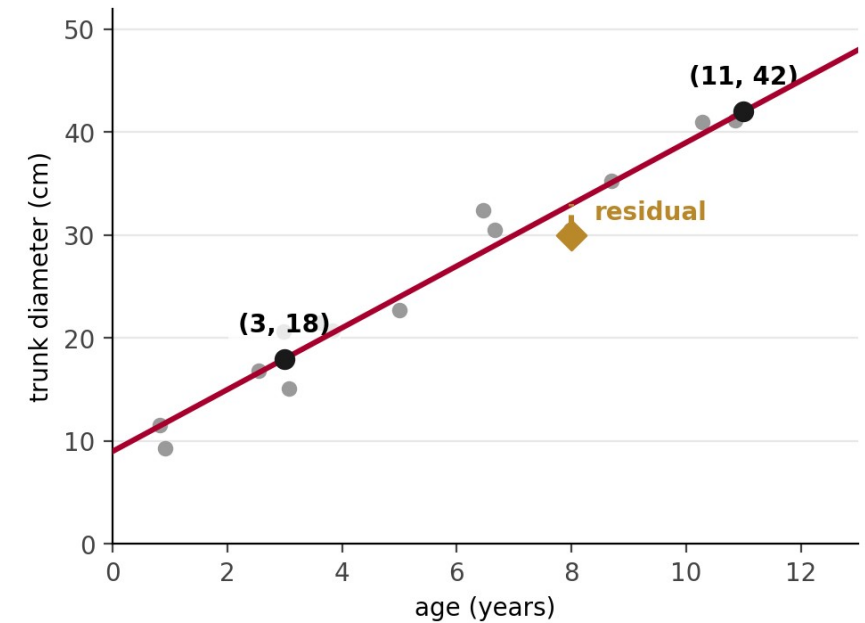
06

Scatterplots

Lines of best fit, residuals, model types, and extrapolation.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)

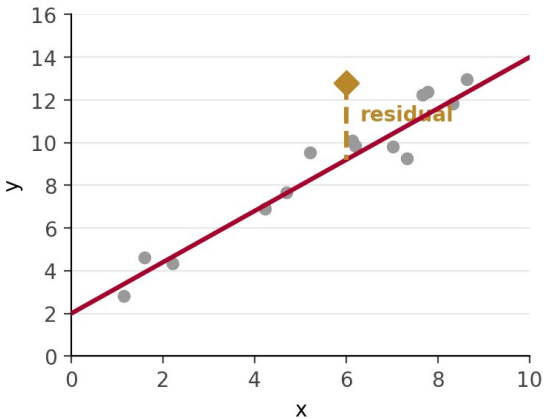


Key Terminology

Term (English)	Ελληνικά
Scatterplot	Διάγραμμα διασποράς
Line of best fit	Ευθεία βέλτιστης προσαρμογής
Residual	Υπόλοιπο (residual)
Positive/negative association	Θετική/αρνητική συσχέτιση
Nonlinear model	Μη γραμμικό μοντέλο

The big picture

- A scatterplot shows how two variables move together.
- The line of best fit turns data into predictions.
- Residuals measure how far reality is from the model.



Theory Explained (1/3)

Line of Best Fit

$$\hat{y} = mx + b \text{ from two points on the line}$$

WHAT IT MEANS

- Slope = predicted change in y per 1 unit of x.
- Intercept = predicted y when x = 0.
- Check whether the intercept makes sense in context.



Example

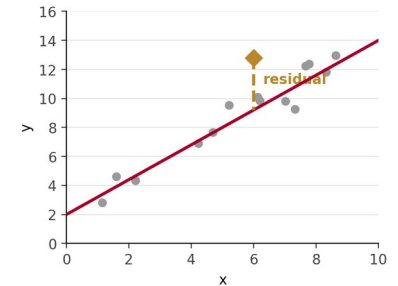
(100, 22), (500, 118) $\rightarrow m = 0.24$, $b = -2$.

Residuals

$$\text{residual} = \text{actual} - \text{predicted}$$

WHAT IT MEANS

- Positive: the point is ABOVE the line (underestimate).
- Negative: BELOW the line (overestimate).
- Zero: the point lies exactly on the line.



Example

Actual 30, predicted 33 $\rightarrow$ residual $-3 \rightarrow$ overestimate.

Theory Explained (2/3)

Choosing a Model

equal differences → linear; equal ratios → exponential

WHAT IT MEANS

- Check differences between consecutive y-values.
- If differences grow, check ratios.
- Constant ratio $b \rightarrow y = a \cdot b^x$.



Example

256, 384, 576, 864 → ratio 1.5 → exponential.

Units & Context

match x to the model's units before substituting

WHAT IT MEANS

- 'x in hundreds' → 650 members means $x = 6.5$.
- Ignore extra facts not in the model.
- State predictions with units.



Example

$y = 34(6.5) + 120 = 341$ trips.

Theory Explained (3/3)

Interpolation vs Extrapolation

inside the data range → reliable; outside → risky

WHAT IT MEANS

- Predictions far outside the data are unreliable.
- Remove flagged faulty points before fitting.
- Correlation $\neq$ causation.



Example

Data from 4–7 hours; predicting 10 hours = extrapolation.

Watch Out



Residual direction

- Always actual – predicted.
- Reversing it flips over/under.



Intercepts in context

- A negative count or trips with zero members is unrealistic.
- The SAT asks you to judge this.

Problem 1

Very Hard

A mobile-game studio in Rotterdam ran a monthly analysis linking its social-media advertising spending (in euros) to the number of new app downloads that month. Studying 14 months of data, analysts noted that a month with 100 euros of ad spending averaged 22 new downloads, while a month with 500 euros of ad spending averaged 118 new downloads. Using these two typical months as points on the line of best fit, (a) find the predicted number of new downloads for a month with 350 euros of spending, and (b) determine whether the model's y-intercept is realistic in context.

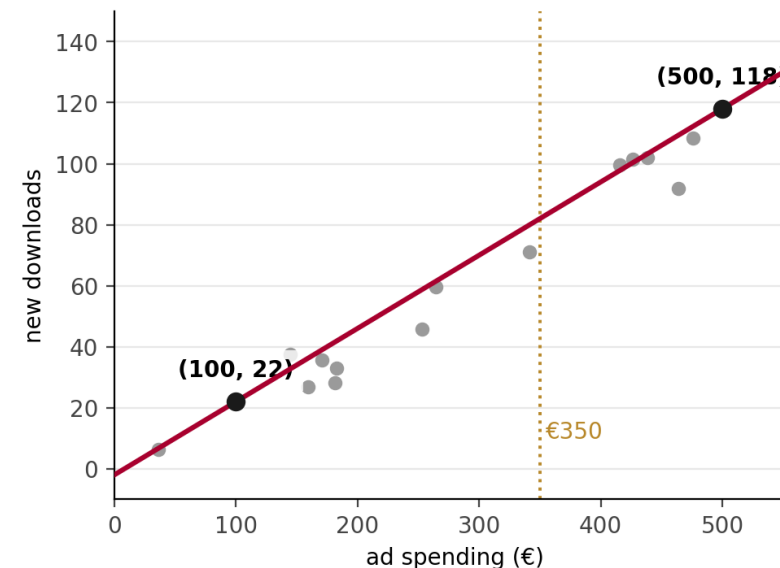
Formula / Rule

Slope-intercept equation from two data points; realism check on the y-intercept.

Solution

1. Translate the two described months into ordered pairs: (100, 22) and (500, 118), where x = euros spent and y = new downloads.
2. Find the slope: $m = (118 - 22)/(500 - 100) = 96/400 = 0.24$ downloads per euro.
3. Use point-slope form with (100, 22): $22 = 0.24(100) + b$, so $22 = 24 + b$.
4. Solve for b : $b = 22 - 24 = -2$.
5. Write the line of best fit: $y = 0.24x - 2$.
6. Predict downloads for $x = 350$: $y = 0.24(350) - 2 = 84 - 2 = 82$ downloads.
7. Check the y-intercept: at $x = 0$ the model predicts $y = -2$ downloads, which is impossible since a download count cannot be negative, so the intercept is not realistic — it is just an artifact of extending the line past the observed data.

✓ **Answer:** 82 downloads predicted at 350 euros; the y-intercept of -2 is not realistic because download counts cannot be negative.





Problem 1 — Strategy & Elimination



Tip

When a word problem describes two 'typical' months or cases instead of giving a table, translate them into ordered pairs first, then treat the problem exactly like finding a line through two points.



Trap

A rushed student might treat 22 and 118 as slope values instead of y-coordinates paired with their given x-values, or accept -2 downloads as a valid prediction without questioning whether a negative count makes sense.



Shortcut

Since the two x-values are 400 apart, the slope $96/400 = 0.24$ can be found in one line without formally writing point-slope form first.



Desmos Tip

Plot (100, 22) and (500, 118) in Desmos and use its built-in linear regression on the two points (or graph $y = 0.24x - 2$) to confirm the line crosses below $y = 0$ near $x = 0$.



Process of Elimination

Correct: 82 downloads predicted at 350 euros; the y-intercept of -2 is not realistic because download counts cannot be negative.

Tempting wrong answer	Exactly what went wrong
≈1064 downloads predicted	Flipped the slope ratio, computing $(500-100)/(118-22) = 400/96$ instead of $96/400$, giving an inflated slope and a wildly overestimated prediction at $x = 350$.
106 downloads predicted	Set $b = 22$ directly from the point (100, 22) without solving $22 = 24 + b$ for b , effectively treating the y-coordinate itself as the intercept and using $y = 0.24x + 22$.
82 downloads predicted, with the y-intercept of -2 accepted as realistic	Computed the prediction correctly but didn't check whether a negative predicted download count at $x = 0$ makes sense, accepting -2 as plausible instead of recognizing counts can't be negative.

Problem 2

Hard

Researchers studying tree growth in a park outside Ljubljana plotted age since replanting (years, x) against trunk diameter (cm, y) for a grove of poplar trees. The line of best fit for the grove passes approximately through the points $(3, 18)$ and $(11, 42)$. One particular tree in the grove is 8 years old and has an actual trunk diameter of 30 cm. Find the residual for this tree, and state whether the line overestimated or underestimated its diameter.

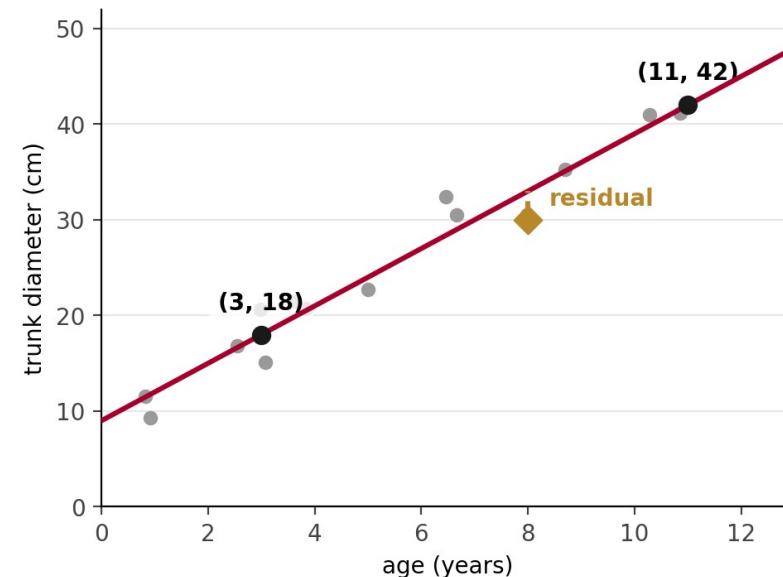
Formula / Rule

Line of best fit from two points, then residual = actual - predicted.

Solution

1. Find the slope of the line of best fit: $m = (42 - 18)/(11 - 3) = 24/8 = 3$ cm per year.
2. Use point-slope form with $(3, 18)$: $18 = 3(3) + b$, so $18 = 9 + b$.
3. Solve for b : $b = 9$.
4. Write the equation of the line of best fit: $y = 3x + 9$.
5. Find the predicted diameter for an 8-year-old tree: $y = 3(8) + 9 = 24 + 9 = 33$ cm.
6. Compute the residual: residual = actual - predicted = $30 - 33 = -3$.
7. Since the residual is negative, the actual diameter (30 cm) is below the predicted diameter (33 cm), so the line of best fit overestimated this tree's diameter.

✓ **Answer:** Residual = -3 cm; the line overestimated the tree's diameter.





Problem 2 — Strategy & Elimination



Tip

When only two approximate points on the line are given, build the full equation first (slope, then intercept), then substitute the specific x-value asked about before computing any residual.



Shortcut

There is no way to skip building the full equation here, since age 8 is not one of the two given points — the two-step process (build the line, then predict) is already the fastest reliable path.



Trap

A student might compute predicted - actual ($33 - 30 = +3$) instead of actual - predicted, flipping the sign and concluding the tree was underestimated when it was actually overestimated.



Desmos Tip

Enter $y = 3x + 9$ in Desmos and use the table feature at $x = 8$ to instantly confirm the predicted value of 33 before comparing it to the actual 30.



Process of Elimination

Correct: **Residual = -3 cm; the line overestimated the tree's diameter.**

Tempting wrong answer	Exactly what went wrong
Residual = 3; line underestimated	Computed predicted - actual ($33 - 30 = 3$) instead of actual - predicted, flipping the sign and reversing the over/under conclusion.
Residual = -21 (using $y = 3x + 27$)	Solved $18 = 9 + b$ by adding instead of subtracting, getting $b = 27$ instead of $b = 9$, which produced a badly inflated predicted diameter of 51 cm.
Residual = 6; line underestimated	Forgot to add the y-intercept when computing the predicted diameter, using predicted = $3(8) = 24$ instead of $3(8) + 9 = 33$.

Problem 3

Very Hard

A microbiology lab in Ghent records the number of bacteria in a colony at the start of each hour: 256 at hour 0, 384 at hour 1, 576 at hour 2, and 864 at hour 3. A lab assistant must decide whether a linear, quadratic, or exponential model best describes this growth before writing a report. (a) Determine which type of model fits the pattern and justify the choice using the data. (b) Using a model of the form $y = a \cdot b^x$, predict the bacteria count at hour 5.

Formula / Rule

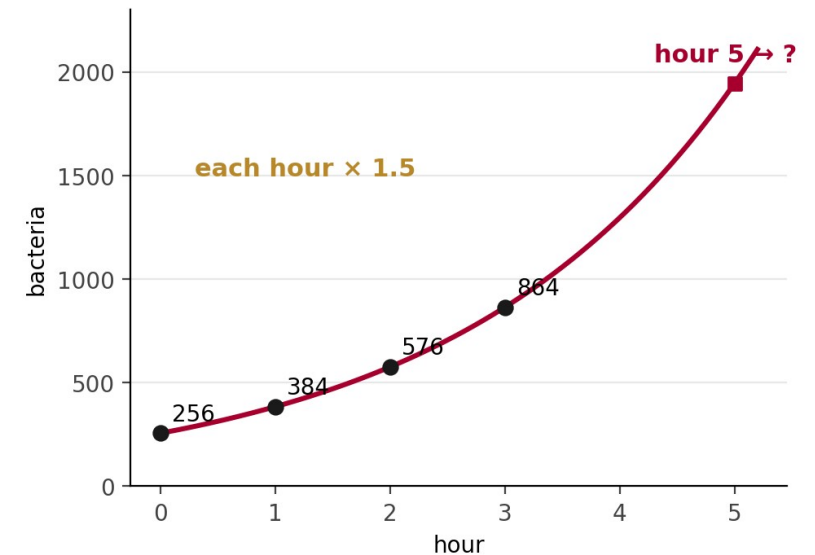
Constant-difference test vs. constant-ratio test to identify the model; exponential equation $y = a \cdot b^x$.

Solution

1. Test for a constant difference (the signature of a linear model): $384 - 256 = 128$, $576 - 384 = 192$, $864 - 576 = 288$ — these differences are not constant, so the data is not linear.
2. Test for a constant ratio (the signature of an exponential model): $384/256 = 1.5$, $576/384 = 1.5$, $864/576 = 1.5$ — the ratio between consecutive terms is constant at 1.5, which confirms an exponential model rather than quadratic (a quadratic model would show constant second differences, not a constant ratio).
3. Since $y = a \cdot b^x$ and the constant multiplier per hour is 1.5, the growth factor is $b = 1.5$.
4. Find a using hour 0: at $x = 0$, $y = a \cdot b^0 = a$, and the data gives $y = 256$ at hour 0, so $a = 256$.
5. Write the model: $y = 256 \cdot (1.5)^x$.
6. Predict hour 5: $y = 256 \cdot (1.5)^5 = 256 \cdot 7.59375 = 1944$ bacteria.



Answer: Exponential model, $y = 256 \cdot (1.5)^x$; predicted count at hour 5 is 1944 bacteria.





Problem 3 — Strategy & Elimination



Tip

Test consecutive differences first for a constant amount (linear); if that fails, test the ratio of consecutive y-values for a constant multiplier (exponential) before ever assuming quadratic.



Shortcut

Once a constant ratio is confirmed, there is no need to also test second differences for a quadratic fit — a constant ratio alone guarantees exponential and automatically rules out both linear and quadratic.



Trap

A student might see the increasing differences (128, 192, 288) and wrongly conclude the data is quadratic because growth is 'speeding up,' without checking that the ratios are constant — which is the actual signature of exponential growth, not just any acceleration.



Desmos Tip

Graph $y = 256 \cdot 1.5^x$ in Desmos and check it passes through (0,256), (1,384), (2,576), (3,864) exactly, then read the y-value at $x = 5$ directly from the table.



Process of Elimination

Correct: **Exponential model, $y = 256 \cdot (1.5)^x$; predicted count at hour 5 is 1944 bacteria.**

Tempting wrong answer	Exactly what went wrong
≈1269 bacteria at hour 5	Never tested for a constant ratio, instead extrapolated linearly using the average difference between consecutive terms (about 202.67 per hour), badly underestimating the true exponential growth.
1920 bacteria at hour 5	Used the correct model $y = 256 \cdot 1.5^x$ but computed 1.5^5 by multiplying $1.5 \times 5 = 7.5$ instead of repeated multiplication ($1.5^5 = 7.59375$), giving $256 \cdot 7.5 = 1920$.
2916 bacteria at hour 5	Used the hour-1 value (384) as the initial value instead of the hour-0 value (256), shifting the entire exponential model up by one time step: $384 \cdot 1.5^5 = 2916$.

Problem 4

Hard

In a scatterplot relating cumulative SAT prep hours (x) to a student's practice-test section score out of 800 (y) for a tutoring cohort, the line of best fit is $y = 5x + 400$. One student in the cohort had an actual practice-test score of 740, and the residual for that student was -10. How many hours of SAT prep had that student completed?

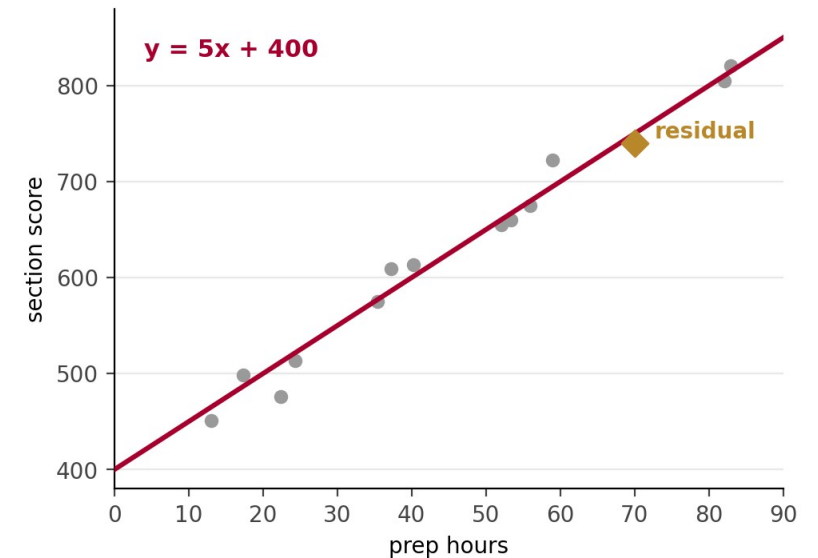
Formula / Rule

Residual = actual - predicted, solved in reverse for the input variable x .

Solution

1. Recall the residual formula: residual = actual - predicted.
2. Rearrange to solve for the predicted value: predicted = actual - residual.
3. Substitute the known values: predicted = $740 - (-10) = 740 + 10 = 750$.
4. Set this predicted score equal to the line of best fit expression: $750 = 5x + 400$.
5. Subtract 400 from both sides: $5x = 750 - 400 = 350$.
6. Divide by 5: $x = 350/5 = 70$.
7. The student had completed 70 hours of SAT prep.

✓ **Answer:** 70 hours of SAT prep.





Problem 4 — Strategy & Elimination



Tip

When a residual and one of the two values (actual or predicted) are given, first isolate whichever value is missing using $\text{predicted} = \text{actual} - \text{residual}$ (or $\text{actual} = \text{predicted} + \text{residual}$), then substitute into the line equation to solve for the remaining unknown.



Shortcut

Once $\text{predicted} = 750$ is found, solving $750 = 5x + 400$ is a single two-step linear equation — there is no faster path since x is embedded inside the line's formula.



Trap

A student might compute $\text{predicted} = \text{actual} + \text{residual}$ ($740 + (-10) = 730$) instead of $\text{actual} - \text{residual}$, since the direction of the residual formula is easy to misremember, leading to $5x = 330$ and an incorrect $x = 66$.



Desmos Tip

Graph $y = 5x + 400$ in Desmos and trace or use the table to find where $y = 750$, which confirms $x = 70$ without redoing the algebra by hand.



Process of Elimination

Correct: **70 hours of SAT prep.**

Tempting wrong answer	Exactly what went wrong
x = 66 hours	Computed $\text{predicted} = \text{actual} + \text{residual}$ ($740 + (-10) = 730$) instead of $\text{actual} - \text{residual}$, misremembering the direction of the residual formula, then solved $730 = 5x + 400$ to get $x = 66$.
x = 150 hours	Correctly found $\text{predicted} = 750$ but forgot to subtract 400 from both sides before dividing, solving $5x = 750$ directly to get $x = 150$ instead of $x = 70$.
750 hours	Correctly computed the predicted score (750) but stopped there and reported it as the final answer, instead of continuing to substitute into $y = 5x + 400$ and solve for x .

Problem 5

Hard

A researcher studying study hours (x) versus quiz score out of 50 (y) for a class models the relationship with the equation $2x - y = -14$ as the line of best fit, not yet solved for y . A particular student studied 12 hours and scored an actual 44 on the quiz. Find the residual for this student, and state whether the line overestimated or underestimated the score.

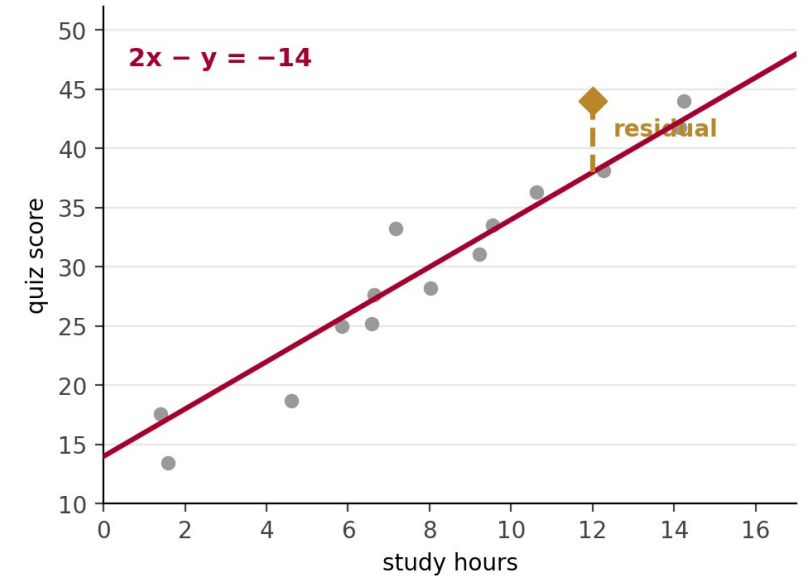
Formula / Rule

Rearrange standard form to slope-intercept form, then $\text{residual} = \text{actual} - \text{predicted}$.

Solution

1. Recognize that $2x - y = -14$ is not solved for y , so it cannot be used directly as a prediction rule.
2. Rearrange: $2x - y = -14$, so $-y = -14 - 2x$, and multiplying both sides by -1 gives $y = 2x + 14$.
3. Substitute the student's study hours, $x = 12$, into the corrected equation: predicted $y = 2(12) + 14$.
4. Compute: $2(12) = 24$, so predicted $y = 24 + 14 = 38$.
5. Recall $\text{residual} = \text{actual} - \text{predicted}$.
6. Compute the residual: $44 - 38 = 6$.
7. Since the residual is positive, the actual score (44) is above the predicted score (38), so the line of best fit underestimated this student's quiz score.

✓ **Answer:** Residual = 6; the line underestimated the student's score.





Problem 5 — Strategy & Elimination



Tip

Whenever a line of best fit is given in standard form ($Ax + By = C$) or any form not solved for y , isolate y first — never substitute x into the unsolved equation and call the result the predicted value.



Shortcut

Rearranging once to $y = 2x + 14$ gives a reusable prediction rule for the whole dataset, so it's worth doing before plugging in any x -value rather than manipulating the standard-form equation freshly each time.



Trap

A student might make a sign error while isolating y (getting $y = 2x - 14$ instead of $y = 2x + 14$), predicting $y = 2(12) - 14 = 10$ and computing a badly wrong residual of $44 - 10 = 34$.



Desmos Tip

Type $2x - y = -14$ directly into Desmos, which graphs implicit equations, and trace to $x = 12$ to confirm the predicted $y = 38$ before computing the residual by hand.



Process of Elimination

Correct: **Residual = 6**; the line underestimated the student's score.

Tempting wrong answer	Exactly what went wrong
Residual = 34; line underestimated	Made a sign error isolating y from $2x - y = -14$, getting $y = 2x - 14$ instead of $y = 2x + 14$, which produced a predicted score of only 10 instead of 38.
Residual = -6; line overestimated	Computed predicted - actual ($38 - 44 = -6$) instead of actual - predicted, flipping the sign and reversing the over/underestimate conclusion even though the magnitude was right.
Residual = 8; line underestimated	Made an arithmetic slip computing $2(12)$, getting 22 instead of 24, which shifted the predicted score to 36 and the residual to 8.

Problem 6

Hard

A city bike-share program in Utrecht tracked, over 18 months, the number of registered members (x , in hundreds) against the number of bike trips taken per day (y). Analysts also recorded the average trip duration each month, which turned out to be irrelevant to this particular model. The line of best fit relating members to daily trips is $y = 34x + 120$, and the program currently operates 45 docking stations across the city. Using the model, predict the number of daily bike trips when there are 650 registered members, and state whether the y -intercept is realistic in context.

Formula / Rule

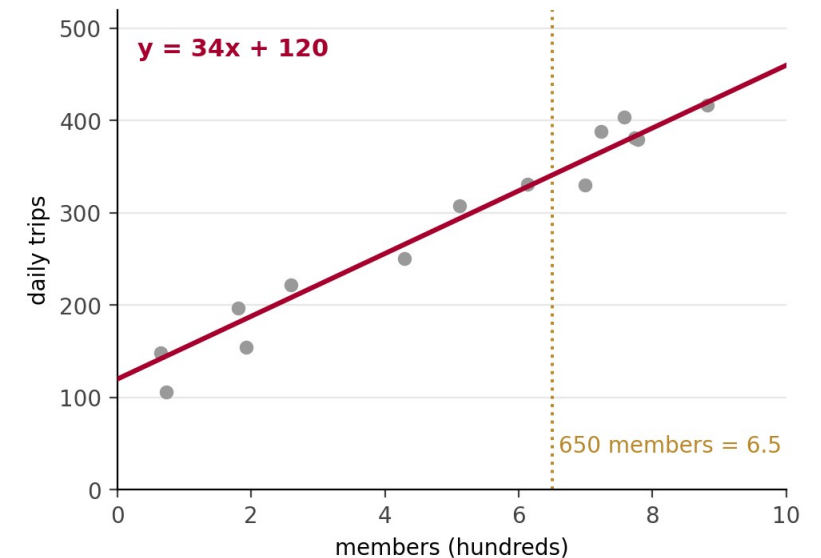
Slope/intercept interpretation with unit conversion before substitution.

Solution

1. Note that x is defined in hundreds of members, so 650 registered members corresponds to $x = 650/100 = 6.5$.
2. Substitute $x = 6.5$ into the line of best fit: $y = 34(6.5) + 120$.
3. Compute $34(6.5) = 221$.
4. Add: $y = 221 + 120 = 341$ predicted daily trips.
5. Identify that the average trip duration and the 45 docking stations are not part of this linear model and are not needed for the prediction.
6. Interpret the y -intercept (120): at $x = 0$ (zero registered members), the model predicts 120 trips per day, which is not fully realistic taken literally, since zero registered members should produce close to zero member trips — the intercept would only make sense if it reflects trips by non-member/casual riders.



Answer: Predicted trips are about 341 per day; the y -intercept of 120 is not realistic taken literally, since it implies daily trips even with zero registered members.





Problem 6 — Strategy & Elimination



Tip

Before substituting, double-check what each variable's units actually are (e.g., 'x in hundreds of members') and ignore any recorded facts that never appear in the equation itself.



Trap

A student might substitute $x = 650$ directly into $y = 34x + 120$ (ignoring that x is measured in hundreds of members), or try to work the 45 docking stations into the slope or intercept interpretation, both of which are irrelevant to the actual linear model given.



Shortcut

Convert the member count to the model's units ($650/100 = 6.5$) in one step before plugging in, rather than rescaling the whole equation to raw member counts.



Desmos Tip

Graph $y = 34x + 120$ in Desmos and use the table feature to evaluate at $x = 6.5$, instantly confirming $y = 341$ without arithmetic errors.



Process of Elimination

Correct: Predicted trips are about 341 per day; the y-intercept of 120 is not realistic taken literally, since it implies daily trips even with zero registered members.

Tempting wrong answer	Exactly what went wrong
≈22,220 predicted trips	Substituted $x = 650$ directly into $y = 34x + 120$ without converting to the model's units (hundreds of members), since x should have been $650/100 = 6.5$, not 650.
386 predicted trips	Mistakenly added the 45 docking stations into the calculation ($341 + 45 = 386$), treating the irrelevant docking-station count as part of the linear model instead of recognizing it plays no role in $y = 34x + 120$.
341 predicted trips, with the y-intercept of 120 accepted as realistic	Computed the prediction correctly but didn't examine what the y-intercept means in context, accepting 120 trips at zero members as reasonable instead of questioning it.

Problem 7

Very Hard

A sports dietician in Thessaloniki collects data on daily protein intake (grams, x) versus lean muscle mass gained in a month (kg, y) for her clients, and models the relationship with the line of best fit $y = mx + 0.5$, where m has not yet been determined. One client consumed 120 grams of protein per day, and her data point had a residual of exactly 0 — her actual muscle mass gain was 2.9 kg. (a) Find the value of m . (b) Using this equation, predict the muscle mass gain, rounded to the nearest tenth of a kg, for a client who consumes 180 grams of protein per day.

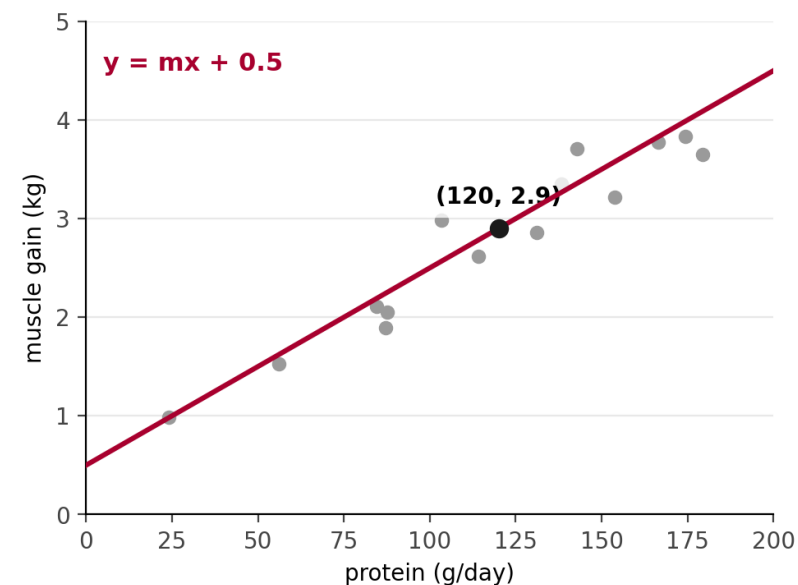
Formula / Rule

A residual of 0 means the point lies exactly on the line of best fit; solve for the unknown slope, then predict.

Solution

1. Recognize that a residual of 0 means actual = predicted for this client, so her point (120, 2.9) satisfies the line exactly.
2. Substitute (120, 2.9) into $y = mx + 0.5$: $2.9 = m(120) + 0.5$.
3. Subtract 0.5 from both sides: $120m = 2.9 - 0.5 = 2.4$.
4. Solve for m : $m = 2.4/120 = 0.02$.
5. Write the completed line of best fit: $y = 0.02x + 0.5$.
6. Predict the gain for $x = 180$ grams: $y = 0.02(180) + 0.5$.
7. Compute: $0.02(180) = 3.6$, so $y = 3.6 + 0.5 = 4.1$ kg.

 **Answer:** $m = 0.02$; predicted muscle mass gain at 180 g protein/day is 4.1 kg.





Problem 7 — Strategy & Elimination



Tip

When a data point is described as having a residual of 0, treat it as if it were just handed to you as a point that satisfies the equation exactly, and substitute it in to solve for the missing constant before doing anything else.



Shortcut

Since the intercept 0.5 is already fixed, the only unknown is m — subtract 0.5 and divide in one line ($120m = 2.4$, $m = 0.02$) without ever writing a full point-slope setup.



Trap

A student in a hurry sets $mx = 2.9$ directly (ignoring the +0.5 intercept already in the equation) instead of subtracting 0.5 first, since it's tempting to treat the actual value itself as equal to only the mx term.



Desmos Tip

Once $m = 0.02$ is found, graph $y = 0.02x + 0.5$ in Desmos and use the table feature at $x = 180$ to confirm the predicted 4.1 kg.



Process of Elimination

Correct: $m = 0.02$; predicted muscle mass gain at 180 g protein/day is 4.1 kg.

Tempting wrong answer	Exactly what went wrong
$m \approx 0.024$, predicted gain ≈ 4.85 kg	Set $120m = 2.9$ directly instead of subtracting the 0.5 intercept first, skipping $2.9 - 0.5 = 2.4$ and computing $m = 2.9/120 \approx 0.0242$ instead of $m = 0.02$.
predicted gain = 3.6 kg	Correctly solved for $m = 0.02$ but then computed only $0.02(180) = 3.6$ for the final prediction and forgot to add back the y-intercept of 0.5.
$m \approx 0.028$, predicted gain ≈ 5.6 kg	Made a sign error rearranging $2.9 = 120m + 0.5$, adding 0.5 instead of subtracting it ($120m = 2.9 + 0.5 = 3.4$), giving $m \approx 0.0283$ instead of 0.02.

Problem 8

Very Hard

A sleep researcher in Krakow records five swimmers' hours of sleep the night before a meet (x) against their reaction time off the blocks in milliseconds (y): (4, 340), (5, 310), (6, 280), (7, 250), and (8, 100). The coach explains that the swimmer at (8, 100) had a faulty starting sensor that night, so that data point should be excluded from the model. (a) Using the line of best fit built from two of the remaining reliable points, predict the reaction time for a swimmer who gets 10 hours of sleep. (b) State whether this prediction is a reliable interpolation or a risky extrapolation, and why.

Formula / Rule

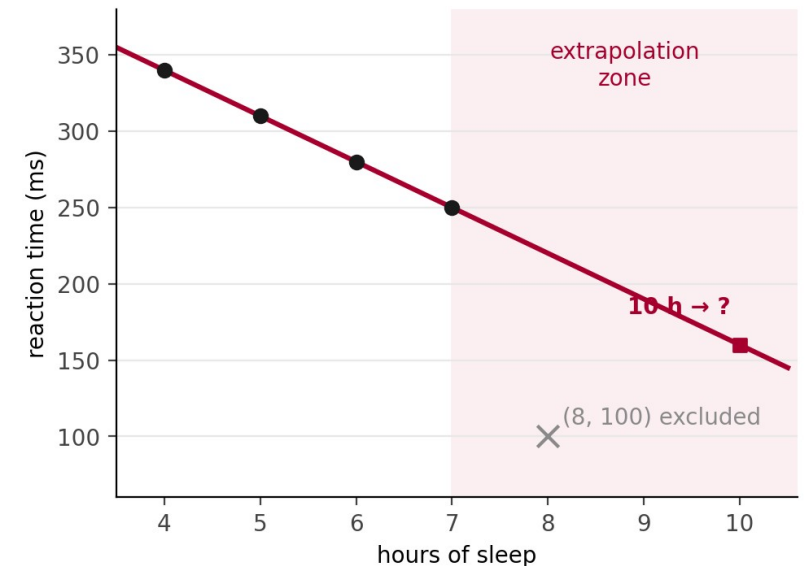
Identify and exclude an outlier before building the line of best fit; then judge interpolation vs. extrapolation.

Solution

1. Exclude the flagged outlier (8, 100), leaving the reliable points (4, 340), (5, 310), (6, 280), and (7, 250).
2. Use two of the reliable points, (4, 340) and (7, 250), to find the slope: $m = (250 - 340)/(7 - 4) = -90/3 = -30$ ms per hour of sleep.
3. Use point-slope form with (4, 340): $340 = -30(4) + b$, so $340 = -120 + b$.
4. Solve for b : $b = 340 + 120 = 460$.
5. Write the line of best fit: $y = -30x + 460$.
6. Predict reaction time at $x = 10$: $y = -30(10) + 460 = -300 + 460 = 160$ ms.
7. Compare $x = 10$ to the range of the reliable data (4 to 7 hours): since 10 falls outside that range, this prediction is an



Answer: 160 ms predicted at 10 hours of sleep; this is a risky extrapolation, since $x = 10$ falls outside the 4-to-7 hour range of the reliable data used to build the line.





Problem 8 — Strategy & Elimination



Tip

Before building any line of best fit from a table, scan for a point that clearly breaks the pattern (like a data-entry or equipment error the problem tells you about) and exclude it, then check whether your final x-value actually falls inside or outside the range you used.



Trap

A student might build the line using all five points, including the flagged faulty outlier (8, 100), which drags the slope far steeper than the true underlying trend and produces an unrealistic (even negative) prediction.



Shortcut

Once the outlier is identified, pick the two reliable points that are farthest apart (4 and 7) to compute the slope — this minimizes rounding and matches what a quick eyeball trend line would give.



Desmos Tip

Plot all five points in Desmos, visually confirm (8,100) breaks the downward trend, then graph $y = -30x + 460$ over just the reliable points to see that $x = 10$ sits clearly outside their span.



Process of Elimination

Correct: 160 ms predicted at 10 hours of sleep; this is a risky extrapolation, since $x = 10$ falls outside the 4-to-7 hour range of the reliable data used to build the line.

Tempting wrong answer	Exactly what went wrong
-20 ms predicted	Included the excluded outlier (8, 100) when computing the slope, using (4,340) and (8,100): $m = (100-340)/(8-4) = -60$, giving line $y = -60x + 580$ and an impossible negative prediction at $x = 10$.
520 ms predicted	Made a sign error computing the slope, dropping the negative sign from $(250-340)/(7-4) = -30$ and using $m = 30$ instead, giving line $y = 30x + 220$ and a prediction that rises instead of falls with more sleep.
160 ms, labeled as a reliable interpolation	Correctly computed the predicted reaction time but didn't check that $x = 10$ falls outside the 4-to-7 range used to build the line, incorrectly treating the extrapolated prediction as trustworthy interpolation.

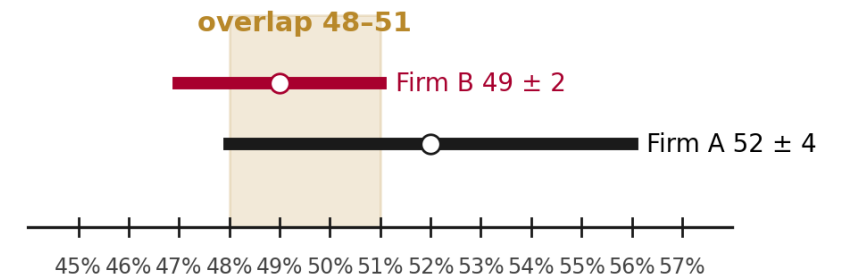
07

Sampling, Bias & Margin of Error

Random samples, bias, generalizing, and confidence intervals.

In this section

- Key terms — English / Ελληνικά
- Explained theory with examples
- 8 worked problems (Hard / Very Hard)



Key Terminology

Term (English)	Ελληνικά
Sample	Δείγμα
Population	Πληθυσμός
Random sampling	Τυχαία δειγματοληψία
Bias	Μεροληψία (σφάλμα)
Margin of error	Περιθώριο σφάλματος

A The big picture

- A sample stands in for a whole population.
- Only a random sample supports generalizing.
- Margin of error turns an estimate into a range.

estimate $\pm$ margin of error



Theory Explained (1/3)

Random Sampling

random selection → valid generalization

WHAT IT MEANS

- Every member must have an equal chance.
- The sample must come from the population described.
- Results apply only to that population.



Example

Random 60 from the full list → valid; first 60 arrivals → not.

Bias

voluntary response · convenience · undercoverage

WHAT IT MEANS

- Voluntary response: people choose to answer.
- Convenience: whoever is easiest to reach.
- A larger biased sample is still biased.



Example

Online 'click to respond' poll → voluntary response bias.

Theory Explained (2/3)

Scaling Up

$$\text{population estimate} = \text{sample proportion} \times \text{population}$$

WHAT IT MEANS

- Find the sample proportion first.
- Multiply by the population size, not the sample size.
- Work backwards with the same proportion.



Example

$$42/60 = 70\% \rightarrow 0.70 \times 1,200 = 840 \text{ students.}$$

Margin of Error

$$\text{estimate} \pm \text{MOE} = \text{confidence interval}$$

WHAT IT MEANS

- The true value is plausibly anywhere in the interval.
- Larger random samples $\rightarrow$ smaller margin of error.
- Convert both bounds when asked for counts.



Example

$$64\% \pm 3\% \text{ of } 30,000 \rightarrow 18,300 \text{ to } 20,100.$$

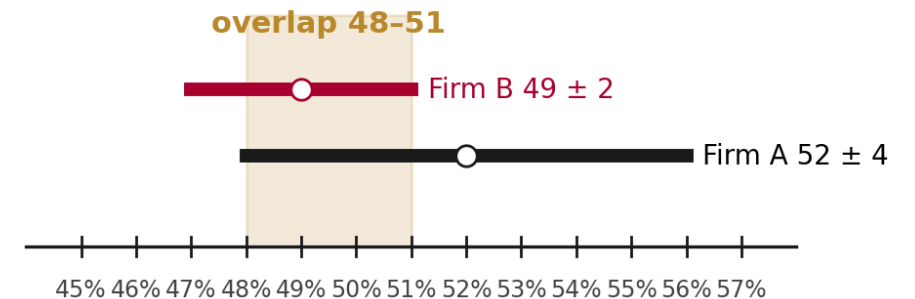
Theory Explained (3/3)

Comparing Estimates

intervals overlap → no clear difference

WHAT IT MEANS

- Compare intervals, not just point estimates.
- Overlap: the two results may be equal.
- Find the overlap: max of lows to min of highs.



Example

48–56% and 47–51% overlap from 48% to 51%.

Watch Out



Sample size can't fix bias

- Bias is about HOW people are chosen.
- More responses only makes a biased estimate more precise.



Use the right base

- Scale percentages to the population, not the sample.
- Report a range when a margin of error is given.

Problem 1

Hard

A tech company in Larnaca, Cyprus has 84,000 registered users of its ride-sharing app. To estimate how many users would pay for a premium subscription, the company selects a simple random sample of registered users and asks them. Based on the sample results, the company estimates that 31,500 of all 84,000 registered users would pay for premium. If the sample contained exactly 189 users who said they would pay for premium, and this sample proportion was used directly to produce the population estimate, how many users were in the sample?

Formula / Rule

Generalizing a sample to a population: the sample proportion equals the population proportion, so set up and solve a proportion equation.

First move

When a problem gives you the final population estimate and asks you to work backward to the sample size, first convert the population estimate into a proportion (population estimate $\div$ total population), then apply that SAME proportion to the sample count.

Solution

1. Find the proportion of the population estimated to want premium: $31500/84000 = 0.375$ (or $3/8$).
2. Since the population estimate was generated by applying the sample proportion, the sample must share this same proportion: $189/n = 0.375$.
3. Solve for n by dividing: $n = 189/0.375$.
4. Compute: $189/0.375 = 504$.
5. So the sample contained 504 users.

 **Answer: 504 users**



Problem 1 — Strategy & Elimination



Tip

When a problem gives you the final population estimate and asks you to work backward to the sample size, first convert the population estimate into a proportion (population estimate ÷ total population), then apply that SAME proportion to the sample count.



Shortcut

Recognize $31,500/84,000$ simplifies to $3/8$; since $189 = 3 \times 63$, the sample size must be $8 \times 63 = 504$, avoiding decimal division entirely.



Trap

It's tempting to treat the population estimate (31,500) as if it were itself the sample size, or to divide the two counts in the wrong order — either move produces a nonsensical or reversed value instead of correctly solving $189/n = 0.375$ for n .



Desmos Tip

Type $189/(31500/84000)$ into the calculator to get 504 directly, skipping manual cross-multiplication.



Process of Elimination

Correct: **504 users**

Tempting wrong answer	Exactly what went wrong
71	Set up the proportion inverted, as $189/n = 84000/31500$ instead of $189/n = 31500/84000$, so dividing by the reciprocal ratio gave 71 instead of 504.
31,500	Stopped after finding the population estimate and reported that population count directly instead of continuing on to solve for the unknown sample size n .
236	Miscalculated $31500/84000$ as 0.8 instead of 0.375 (flipped which number goes on top), then solved $189/0.8 = 236$ instead of $189/0.375 = 504$.

Problem 2

Hard

A researcher surveys a random sample of 300 students at a school in Vienna about their commute to school. She reports that for every 5 students who take public transit, 3 walk, and 2 bike — no student in the sample uses any other method. If the school has 2,400 students total and the sample is representative, how many of all students at the school would be expected to walk to school?

Formula / Rule

Multi-part ratio scaled to a population: convert the ratio sentence into a fraction of the whole, then apply that fraction to the population.

First move

When a ratio is given as a sentence with three (or more) quantities, first add all the parts to get the total, then form the fraction for the specific category the question asks about before scaling to the population.

Solution

1. Add the ratio parts: $5 \text{ (transit)} + 3 \text{ (walk)} + 2 \text{ (bike)} = 10 \text{ parts total}$.
2. Find the fraction of students who walk: $3/10$.
3. Check this fraction against the sample as a sanity check: $300 \times 3/10 = 90$ students walked in the sample of 300.
4. Apply the same fraction to the full school population of 2,400 students: $2400 \times 3/10$.
5. Compute: $2400 \times 3/10 = 720$.
6. So 720 students at the school would be expected to walk.



Answer: 720 students



Problem 2 — Strategy & Elimination



Tip

When a ratio is given as a sentence with three (or more) quantities, first add all the parts to get the total, then form the fraction for the specific category the question asks about before scaling to the population.



Shortcut

No shortcut is needed beyond finding the fraction and multiplying — but you can skip the sample sanity-check (90 students) entirely since it isn't required to answer the population question.



Trap

It's easy to grab the wrong category's number from the sentence since transit, walk, and bike all appear together — always re-read which single category the question is actually asking about before you scale up.



Desmos Tip

Type $2400 \cdot 3/10$ directly into the calculator to confirm 720 without simplifying the fraction by hand first.



Process of Elimination

Correct: 720 students

Tempting wrong answer	Exactly what went wrong
900	Left the bike riders out when adding the ratio parts (used $5+3=8$ instead of $5+3+2=10$), then computed $2400 \times 3/8 = 900$.
1,200	Used the transit fraction ($5/10 = 1/2$) instead of the walk fraction ($3/10$), computing $2400 \times 1/2 = 1200$.
480	Used the bike fraction ($2/10 = 1/5$) instead of the walk fraction, computing $2400 \times 1/5 = 480$.

Problem 3

Very Hard

The city council of Rotterdam wants to know what fraction of all 620,000 residents support a proposed bike-lane expansion. The council posts a link to an online survey on the city's official social media page and invites anyone who sees the post to click and respond. The survey stays open for two weeks and receives 1,850 responses — more than double the roughly 800 responses the council's mailed paper surveys usually generate. Of the 1,850 online respondents, 74% support the expansion. A council member wants to report that '74% of Rotterdam residents support the bike-lane expansion.' What single flaw in the sampling method makes this specific conclusion invalid, regardless of how many responses were collected?

Formula / Rule

Identifying flaws in a study design: check whether participants were randomly selected or self-selected before trusting any generalization to the population.

First move

When a study lets participants decide for themselves whether to respond (a posted link, a call-in poll, a mail-back form), check for self-selection bias first — no sample size, however large, fixes a non-random selection method.

Solution

1. Note that the sample was NOT drawn randomly from all 620,000 residents — whether someone is included depended entirely on whether they saw the post and chose to click it.
2. Consider whether the sample size (1,850) is the issue — it isn't: 1,850 is actually a fairly large sample, and comparing it to the 800 mailed responses is irrelevant to whether the method itself is unbiased.
3. Consider whether the two-week window caused the problem — timing isn't the flaw either; even a random sample collected over two weeks would still be valid.
4. Recognize the real issue: this is a voluntary response sample. People with strong opinions (often more motivated to click and respond) are systematically over-represented compared to indifferent or opposed residents.
5. Because respondents chose themselves rather than being randomly selected, the 74% reflects motivated social-media users, not a representative cross-section of all 620,000 residents.
6. Conclude that the defining flaw is self-selection (voluntary response) bias, not sample size or survey duration.



Answer: Voluntary response (self-selection) bias: only residents who chose to click the online survey are represented, so the 74% cannot be generalized to all Rotterdam residents.



Problem 3 — Strategy & Elimination



Tip

When a study lets participants decide for themselves whether to respond (a posted link, a call-in poll, a mail-back form), check for self-selection bias first — no sample size, however large, fixes a non-random selection method.



Trap

The '1,850 vs. 800' comparison is designed to tempt you into thinking a bigger sample fixed the problem — sample size only affects margin of error, not whether the sample was randomly selected in the first place.



Shortcut

As soon as you see 'anyone who wants to can click and respond,' flag voluntary response bias immediately — there's no need to evaluate sample size, timing, or response rate at all.



Process of Elimination

Correct: **Voluntary response (self-selection) bias: only residents who chose to click the online survey are represented, so the 74% cannot be generalized to all Rotterdam residents.**

Tempting wrong answer	Exactly what went wrong
The sample size (1,850) is too small to be reliable	Confuses margin-of-error concerns (which shrink as n grows) with selection bias; a bigger sample only makes a biased method's estimate more precisely wrong, not more valid.
The two-week survey window is too short to capture opinions	Focuses on an irrelevant detail (duration) instead of recognizing that no survey window length fixes a fundamentally non-random selection process.
The survey undercounts residents without internet access	Names a real but secondary issue (undercoverage) instead of the defining flaw — even if every resident had internet access, letting people self-select into responding would still bias the result.

Problem 4

Very Hard

Two independent research firms use separate random samples to estimate the percentage of a country's residents who support raising the minimum wage. Firm A surveys 400 residents and reports 52% support, with a margin of error of ± 4 percentage points. Firm B surveys 1,600 residents and reports 49% support, with a margin of error of ± 2 percentage points. A news anchor claims, 'Firm A's poll proves support is higher than Firm B's poll shows, since 52% is greater than 49%.' Based on the two margins of error, is the anchor's claim statistically justified, and over what single range of percentages do the two firms' confidence intervals overlap?

Formula / Rule

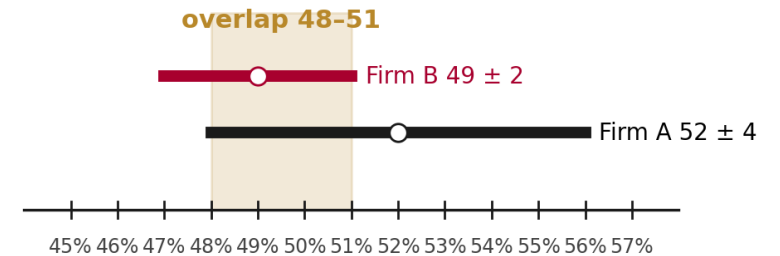
Confidence interval comparison: build estimate $\pm$ margin of error for each result and check whether the intervals overlap before comparing point estimates.

Solution

- Find Firm A's confidence interval: $52\% \pm 4\%$ gives (48%, 56%).
- Find Firm B's confidence interval: $49\% \pm 2\%$ gives (47%, 51%).
- Compare the two intervals for overlap: A runs from 48% to 56%, and B runs from 47% to 51%.
- The overlapping region is where both intervals hold at once: the higher of the two lower bounds (48%) to the lower of the two upper bounds (51%), so 48% to 51%.
- Because the intervals overlap, a true population value like 49% or 50% is consistent with BOTH surveys — the surveys are not actually in conflict.



Answer: Not justified — the confidence intervals overlap from 48% to 51%, so the two results are not significantly different.





Problem 4 — Strategy & Elimination



Tip

To judge whether one poll's estimate is genuinely higher than another's, always build BOTH confidence intervals (estimate $\pm$ margin of error) and check for overlap — never compare the raw point estimates alone.



Trap

The anchor's mistake is comparing only the point estimates (52% vs. 49%) while ignoring that each carries a margin of error; once you build the actual intervals, they clearly overlap, so the claim isn't supported.



Shortcut

Skip building both full intervals and just check: does $|52 - 49| = 3$ exceed the sum of the two margins ($4 + 2 = 6$)? Since $3 < 6$, the intervals must overlap — compute the exact overlap range only if the question specifically asks for it.



Desmos Tip

Plot the two intervals as number-line segments (or type the inequalities $48 \leq x \leq 56$ and $47 \leq x \leq 51$) to visually confirm where they overlap.



Process of Elimination

Correct: **Not justified** — the confidence intervals overlap from 48% to 51%, so the two results are not significantly different.

Tempting wrong answer	Exactly what went wrong
Justified — Firm A's estimate (52%) is greater than Firm B's (49%)	Compares only the point estimates and ignores both margins of error, which is exactly the anchor's flawed reasoning restated rather than actually checked.
Not justified — the intervals overlap from 47% to 56%	Reports the full combined span of both intervals (their union, from the lowest low to the highest high) instead of the overlapping region where both intervals hold simultaneously (their intersection).
Justified, because Firm B has a larger sample and smaller margin of error, so its 49% is more trustworthy	Correctly notes Firm B's estimate is more precise but wrongly treats that as disproving Firm A — a smaller margin of error only narrows B's own interval, it doesn't shrink or invalidate A's overlapping interval.

Problem 5

Hard

At a three-day music festival in Porto with 30,000 total attendees, organizers randomly sample 900 attendees and ask if they plan to return next year. Based on the sample, they estimate that $64\% \pm 3\%$ of all attendees plan to return. Rounded to the nearest hundred, what is the range of the number of attendees who could be expected to return next year?

Formula / Rule

Converting a margin-of-error percentage range into a population count range: find both percentage bounds first, then scale each to the population.

First move

When a margin of error is attached to a percentage, first convert it into a percentage RANGE (estimate minus margin, estimate plus margin), then scale each end of that range to the population separately — never scale just the point estimate if a range is asked for.

Solution

1. Find the lower bound of the estimated proportion: $64\% - 3\% = 61\%$.
2. Find the upper bound of the estimated proportion: $64\% + 3\% = 67\%$.
3. Apply the lower bound to the total attendance: $0.61 \times 30,000 = 18,300$.
4. Apply the upper bound to the total attendance: $0.67 \times 30,000 = 20,100$.
5. Both results already land on whole hundreds, so no further rounding is needed.
6. The expected number of returning attendees ranges from 18,300 to 20,100.

 **Answer:** 18,300 to 20,100 attendees



Problem 5 — Strategy & Elimination



Tip

When a margin of error is attached to a percentage, first convert it into a percentage RANGE (estimate minus margin, estimate plus margin), then scale each end of that range to the population separately — never scale just the point estimate if a range is asked for.



Shortcut

Since $30,000 \times 1\% = 300$, find both bounds fast: the center is $64 \times 300 = 19,200$, then subtract/add $3 \times 300 = 900$ to get 18,300 and 20,100.



Trap

It's tempting to apply the $\pm 3\%$ range to the 900-person sample instead of the 30,000-person population, since the sample size is mentioned right before the percentage — but the question asks about ALL attendees.



Desmos Tip

Type $0.61 \cdot 30000$ and $0.67 \cdot 30000$ into the calculator directly to get both bounds in two steps without hand-converting percentages.



Process of Elimination

Correct: **18,300 to 20,100 attendees**

Tempting wrong answer	Exactly what went wrong
19,200 people	Applied only the point estimate (64%) to the population, ignoring the margin of error entirely and reporting a single value instead of the requested range.
549 to 603	Applied 61%–67% to the 900-person sample instead of the 30,000 attendees.
17,400 to 21,000	Doubled the margin of error to $\pm 6\%$ instead of using $\pm 3\%$, computing bounds of 58% and 70% before scaling to the population.

Problem 6

Very Hard

A high school in Athens has 1,200 students. The student council wants to estimate how many students would support extending the lunch break by 15 minutes. Two methods are proposed: Method 1 — the council president asks the first 60 students who arrive at school on Monday morning. Method 2 — a computer selects 60 students at random from the complete enrollment list. Only Method 2 is carried out, and 42 of the 60 selected students support the extension. Which method would have produced a valid basis for generalizing to the full student body, and, using the results of the valid method, how many of the school's 1,200 students would be expected to support the extension?

Formula / Rule

First determine which sampling method is valid (every student must have an equal chance of selection), then scale that method's sample proportion to the full population.

First move

Before doing any scaling arithmetic, always check whether the sampling method itself was actually random — 'the first people who show up' is a classic convenience sample and invalidates generalizing no matter how clean the numbers look afterward.

Solution

1. Evaluate Method 1: selecting 'the first 60 students to arrive' is not random — early arrivers may differ systematically (bus routes, class schedules) from the full student body, so this is a convenience sample.
2. Evaluate Method 2: a computer randomly selecting 60 students from the complete enrollment list gives every student an equal chance of selection, so this is a valid simple random sample.
3. Since only Method 2 is valid for generalizing to the population, use its results: 42 of the 60 sampled students supported the extension.
4. Find the sample proportion: $42/60 = 0.7$ (or 70%).
5. Apply this proportion to the full school population: 1200×0.7 .
6. Compute: $1200 \times 0.7 = 840$.
7. So Method 2 is the valid method, and 840 students would be expected to support the extension.



Answer: Method 2; 840 students expected to support the extension.



Problem 6 — Strategy & Elimination



Tip

Before doing any scaling arithmetic, always check whether the sampling method itself was actually random — 'the first people who show up' is a classic convenience sample and invalidates generalizing no matter how clean the numbers look afterward.



Shortcut

Once Method 2 is confirmed valid, recognize $42/60$ simplifies to $7/10$, so $1200 \times 7/10 = 840$ can be found with simple mental math instead of long division.



Trap

A student who jumps straight to the arithmetic might try to use or blend in Method 1's results — but Method 1 was never actually carried out, and even if it had been, its non-random selection makes it unusable for the estimate.



Desmos Tip

Type $1200 \cdot (42/60)$ into the calculator to confirm 840 in one step.



Process of Elimination

Correct: **Method 2; 840 students expected to support the extension.**

Tempting wrong answer	Exactly what went wrong
Method 1; 840 students	Computed the population estimate correctly but misidentified Method 1 (arriving early isn't random) as the valid sampling method instead of Method 2.
Method 2; 720 students	Correctly identified Method 2 as valid but mistook the sample size (60) for a percentage, computing 1200×0.6 instead of 1200×0.7 .
Method 2; 42 students	Correctly identified Method 2 but reported the raw sample count (42) instead of scaling it up to the full population of 1,200.

Problem 7

Hard

A pollster surveys a random sample of 800 registered voters out of a district's 25,000 registered voters to estimate support for candidate Nowak. The pollster reports a 95% confidence interval of (46%, 58%) for the true percentage of all 25,000 registered voters who support Nowak. A newspaper wants to convert this into a range for the actual number of votes Nowak could expect, rounded to the nearest hundred votes. What is that range?

Formula / Rule

Applying a confidence interval already stated as a population percentage directly to the population count.

First move

When a confidence interval is already stated as a percentage of the full population, apply it directly to the population size — don't get pulled back into re-scaling from the sample size, which was only relevant to how the pollster built the interval in the first place.

Solution

1. Identify the two ends of the given confidence interval: 46% and 58%.
2. Recognize the interval already describes the full population of registered voters (25,000) — the 800-person sample size was only used to produce the interval and isn't needed again here.
3. Apply the lower bound to the population: $0.46 \times 25,000 = 11,500$.
4. Apply the upper bound to the population: $0.58 \times 25,000 = 14,500$.
5. Both values already land on the nearest hundred, so the range is 11,500 to 14,500 votes.

 **Answer:** 11,500 to 14,500 votes



Problem 7 — Strategy & Elimination



Tip

When a confidence interval is already stated as a percentage of the full population, apply it directly to the population size — don't get pulled back into re-scaling from the sample size, which was only relevant to how the pollster built the interval in the first place.



Shortcut

Since $25,000 \times 1\% = 250$, compute $46 \times 250 = 11,500$ and $58 \times 250 = 14,500$ using mental multiplication instead of decimal multiplication.



Trap

The 800-person sample size is mentioned right before the interval, tempting you to scale the percentages to 800 instead of the actual population of 25,000 that the interval describes.



Desmos Tip

Type $0.46 \cdot 25000$ and $0.58 \cdot 25000$ into the calculator to get both bounds in two quick steps.



Process of Elimination

Correct: **11,500 to 14,500 votes**

Tempting wrong answer	Exactly what went wrong
368 to 464 votes	Applied the 46%–58% interval to the 800-person sample instead of the 25,000-person population, scaling from the irrelevant number mentioned in the problem.
13,000 votes	Averaged the interval down to a single point estimate ($52\% \times 25,000 = 13,000$) instead of reporting the full range the question asked for.
1,150,000 to 1,450,000 votes	Forgot to convert the percentages to decimals before multiplying (used 46 and 58 directly instead of 0.46 and 0.58), inflating both bounds by a factor of 100.

Problem 8

Very Hard

A gym chain in Ljubljana has 18,000 members. A random sample estimates that 10,800 of all 18,000 members (a proportion of 60%) would renew their annual membership, with a reported margin of error of ± 5 percentage points. Reporters convert this into a 95% confidence interval for the actual NUMBER of members (not the percentage) who would renew, and correctly find the lower bound to be 9,900 members. What is the confidence interval's upper bound, in members?

Formula / Rule

A margin-of-error confidence interval is symmetric around its point estimate: the two bounds sit equal distances from the center.

First move

Recognize that a margin-of-error interval is symmetric around the point estimate — once you know the distance from the center to ONE bound, the other bound is the same distance on the opposite side, so you don't need to recompute percentages from scratch.

Solution

1. Confirm the point estimate in raw numbers: 60% of $18,000 = 10,800$ members, matching the given estimate.
2. A confidence interval built from a symmetric margin of error is centered on the point estimate, so the lower and upper bounds sit equal distances from 10,800.
3. Find that distance using the given lower bound: $10,800 - 9,900 = 900$ members.
4. Add the same distance to the point estimate to find the upper bound: $10,800 + 900$.
5. Compute: $10,800 + 900 = 11,700$.
6. Check: this matches $65\% \times 18,000 = 11,700$, confirming the ± 5 percentage-point margin correctly translates to ± 900 members.
7. So the confidence interval's upper bound is 11,700 members.

 **Answer: 11,700 members**



Problem 8 — Strategy & Elimination



Tip

Recognize that a margin-of-error interval is symmetric around the point estimate — once you know the distance from the center to ONE bound, the other bound is the same distance on the opposite side, so you don't need to recompute percentages from scratch.



Shortcut

Since the margin is ± 5 percentage points and $18,000 \times 1\% = 180$, the member-count margin is $5 \times 180 = 900$ directly — skip finding the lower bound's distance by subtraction and just add 900 to 10,800.



Trap

The real trap is treating the interval's full width (1,800 members, lower bound to upper bound) as the amount to add to the point estimate, instead of the 900-member distance from the center to one bound — that mistake produces 12,600 instead of 11,700.



Desmos Tip

Type $10800+900$ and $10800-900$ into the calculator to instantly confirm both interval bounds (11,700 and 9,900) match.



Process of Elimination

Correct: **11,700 members**

Tempting wrong answer	Exactly what went wrong
9,900 members	Misread the question and reported the given lower bound back as the answer instead of computing the upper bound.
10,800 members	Reported the point estimate (the center of the interval) instead of continuing on to find the actual upper bound.
12,600 members	Added the interval's full width (1,800) to the point estimate instead of just the 900-member distance from the center to one bound, computing $10,800 + 1,800 = 12,600$.

Problem-Solving & Data Analysis Cheat Sheet (1/2)

01 Ratios & Proportions

- ▶ $a:b:c \rightarrow ak, bk, ck$
- ▶ $a/b = c/d \rightarrow ad = bc$
- ▶ Chain ratios: match via LCM
- ▶ Direct $y = kx$ · Inverse $xy = k$
- ▶ Area ratio k^2 · Volume k^3

02 Probability

- ▶ $P = \text{favorable} / \text{total}$
- ▶ $P(\text{not } A) = 1 - P(A)$
- ▶ $P(A|B) = P(A \cap B) / P(B)$
- ▶ Independent: $P(A) \cdot P(B)$
- ▶ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- ▶ Odds $a:b \rightarrow a/(a+b)$

03 Unit Conversions & Rates

- ▶ Multiply by (new/old)
- ▶ $d = r \times t$
- ▶ Area: factor^2 · Volume: factor^3
- ▶ Invert consumption rates first
- ▶ Time: $\times 60$ / $\div 60$

04 Statistics — Center & Spread

- ▶ Sum = mean $\times$ n
- ▶ Median: sort, then middle
- ▶ Weighted: $\sum wx / \sum w$
- ▶ Outlier moves mean $\gg$ median
- ▶ Clustered $\rightarrow$ smaller SD

Problem-Solving & Data Analysis Cheat Sheet (2/2)

05 Data Displays — Histograms, Dot Plots & Box Plots

- ▶ $\text{Sum} = \sum \text{value} \times \text{freq}$
- ▶ Median: position $(n+1)/2$
- ▶ $\text{IQR} = Q3 - Q1$
- ▶ Fences: $Q1 - 1.5\text{IQR}$, $Q3 + 1.5\text{IQR}$
- ▶ Each box-plot section $\approx 25\%$

06 Scatterplots

- ▶ $\hat{y} = mx + b$
- ▶ Residual = actual – predicted
- ▶ + above (under-est.) • – below (over-est.)
- ▶ Const. difference $\rightarrow$ linear
- ▶ Const. ratio $\rightarrow$ exponential
- ▶ Outside range = extrapolation

07 Sampling, Bias & Margin of Error

- ▶ Random $\rightarrow$ generalize
- ▶ Voluntary / convenience = bias
- ▶ Estimate = proportion $\times$ population
- ▶ Interval = estimate $\pm$ MOE
- ▶ Bigger random $n \rightarrow$ smaller MOE
- ▶ Overlap $\rightarrow$ not significantly different

Top Traps to Avoid

Skill	The trap	The fix
Ratios & Proportions	Ratio $\neq$ actual count	Always multiply back by k. $4 - 3 = 1$ part, not 1 stamp.
Probability	Pick the correct denominator	'Of all students' vs 'of bike owners' changes everything. Underline the condition in the question.
Unit Conversions & Rates	Base 60, not base 10	Minutes and seconds convert by 60. $0.5 \text{ h} = 30 \text{ min}$, not 50.
Statistics	Sort before the median	Data given in random order is a deliberate trap. Count positions carefully.
Data Displays	Height is not position	The tallest stack of dots is the mode, not the maximum. Read values from the horizontal axis.
Scatterplots	Residual direction	Always actual – predicted. Reversing it flips over/under.
Sampling, Bias & Margin of Error	Sample size can't fix bias	Bias is about HOW people are chosen. More responses only makes a biased estimate more precise.



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